NON-WSQ COURSE · 4 DAYS · 30 HOURS

Google Professional Machine Learning Engineer Training

Build · evaluate · productionize · optimize

C740

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

v2.0 · 12 September 2026



AI-generated course hero · editable typography

COURSE ADMINISTRATION

About the Trainer — Template

Name

1

To be completed by the assigned trainer.

Qualifications

2

Relevant cloud, ML and adult-learning credentials.

Industry evidence

3

Production ML, data engineering, MLOps and governance experience.

Contact

4

Approved course support channel only.

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About the Trainer — Dr Alfred Ang

Role

1

Principal Trainer, Tertiary Infotech Academy Pte Ltd.

Focus

2

Applied AI, machine learning, data engineering and production software systems.

Delivery

3

Hands-on technical courses aligned to certification outcomes.

Provider

4

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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Learner Context Map

Role

1

Data scientist, ML engineer, data engineer, software engineer or cloud architect.

Current stack

2

Python/SQL, data platforms, model frameworks and deployment environment.

Target outcome

3

Design, train, serve and monitor an evidence-backed Google Cloud ML solution.

Constraint

4

Share one real latency, cost, security or governance challenge—without confidential data.

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COURSE ADMINISTRATION

Ground Rules

Participation

1

Question mechanisms, assumptions and metrics; contribute evidence.

Safety

2

Use only the training project and mock data. No production credentials or personal data.

Cost

3

Check estimates, quotas and budgets before paid resources.

Respect

4

One conversation at a time; return punctually and preserve peer confidentiality.

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VISUAL PORTAL FLOW

Download Course Materials

Portal

1

Sign in at lms-tms.tertiaryinfotech.com with your course account.

Course page

2

Open C740 and locate learner slides, Learner Guide and Activities.

Verify

3

Confirm the version shown is v2.0 and filenames match this course.

Use

4

Keep detailed procedures in the LG/activity PDFs; slides are mechanism and evidence maps.

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COURSE ADMINISTRATION

Course Facts

Duration

1

4 days · 30 instructional hours.

Delivery

2

Instructor-led with 16 self-contained technical activities.

Skills Framework

3

Non-WSQ commercial short course.

Certification focus

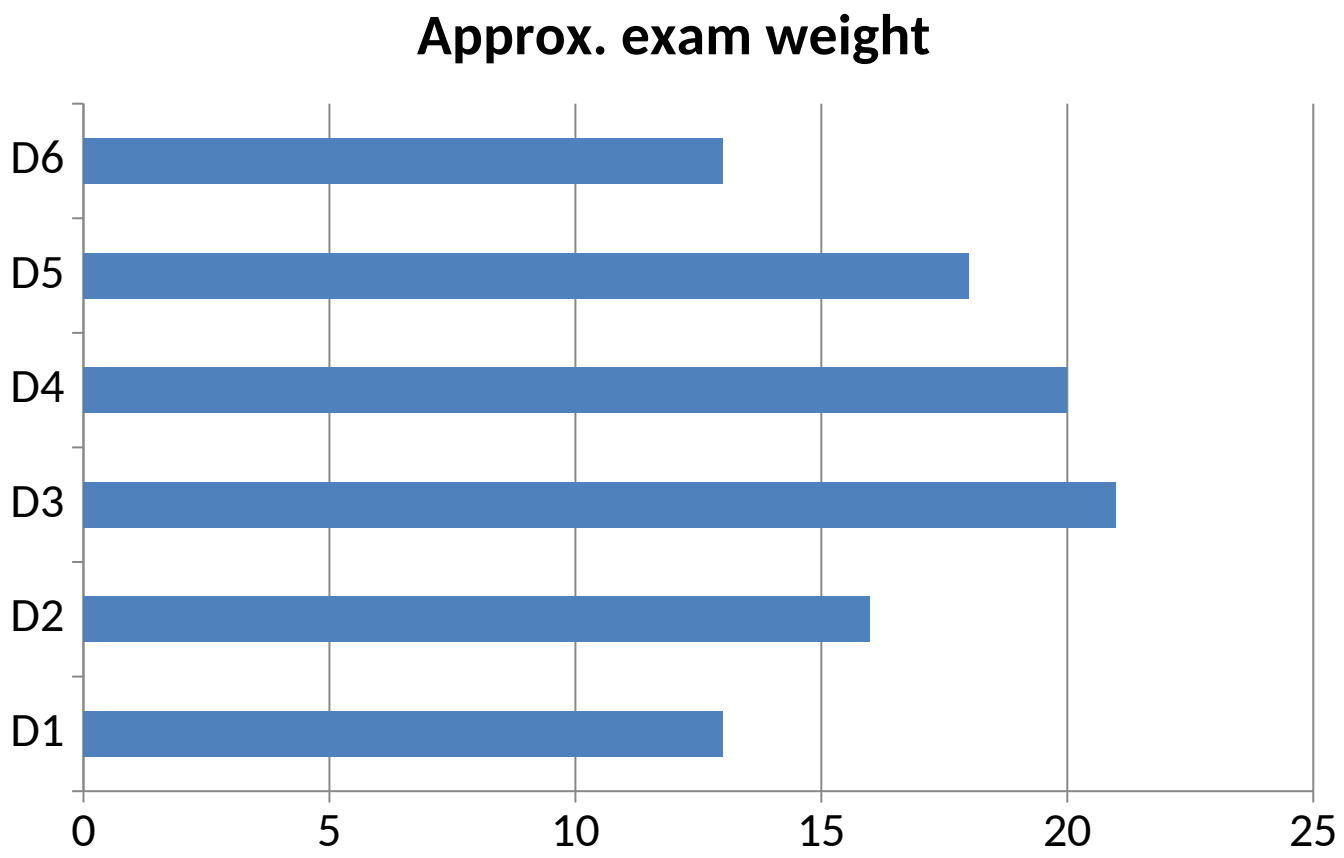
4

Google Cloud Professional Machine Learning Engineer exam guide, current at release date.

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CURRENT OFFICIAL BLUEPRINT

Google PMLE exam domains



Six-domain design

- Architecting low-code AI solutions
- Collaborating to manage data and models
- Scaling prototypes into ML models
- Serving and scaling models
- Automating and orchestrating ML pipelines
- Monitoring AI solutions

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LESSON PLAN

Four-Day Technical Journey

Day 1

1

Low-code AI architecture, Google Cloud setup and data foundations

Day 2

2

Feature engineering, TensorFlow and scalable custom training

Day 3

3

Model serving, deployment strategies and production scaling

Day 4

4

MLOps automation, monitoring, responsible AI and wrap-up

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ACTIVITY 01 · CONCEPT VIEW

Google Cloud Workspace Control Boundary

Identity

1

Dedicated account or approved existing identity; MFA and recovery controls.

Project

2

Isolated training project with explicit region and owner.

Billing

3

Linked billing account, small budget alerts and resource expiry.

Access

4

Least privilege, APIs enabled deliberately, no downloaded service-account keys.

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CERTIFICATION PREPARATION

Practice Exam Portal

Portal

1

https://exams.tertiaryinfotech.com/

Mode

2

Use Practice Mode for feedback and Exam Mode for timed rehearsal.

Boundary

3

Original blueprint-aligned practice questions; never exam dumps.

Study loop

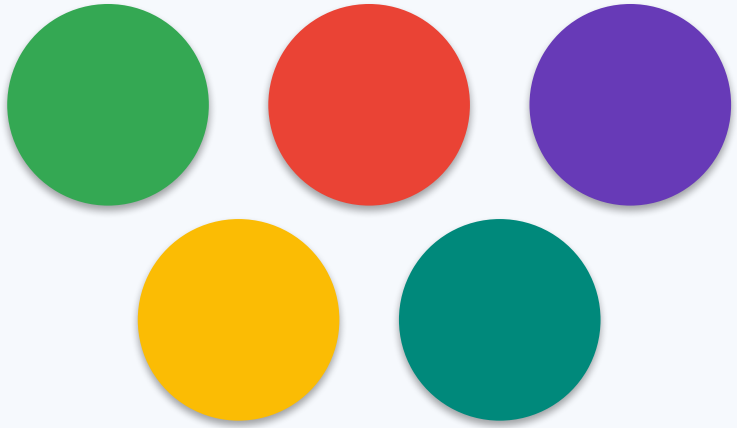
4

Review domain results, revisit activities, and retake under timed conditions.

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DOMAIN 1 · ~13%

Architecting low-code AI solutions

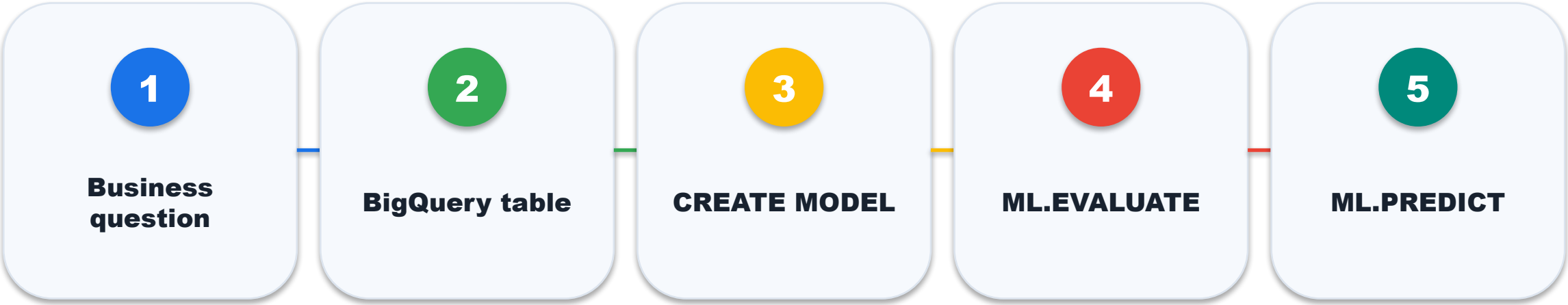


Topics 1–3: GCP fundamentals, Google ML services, ML workflow

8 technical mechanisms · 64 anchored slides

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BigQuery ML model selection — mechanism map



Control point Decision evidence must bind business question to ml.predict through versioned artifacts.

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DOMAIN 1 · DECISION RULE

BigQuery ML model selection — service decision matrix

OPTION 1

Logistic regression

Use when

binary outcome with calibrated probability

OPTION 2

Boosted trees

Use when

non-linear tabular relationships

OPTION 3

ARIMA_PLUS

Use when

forecasting with seasonality

Exam cue: select from workload evidence—not from product familiarity.

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BigQuery ML model selection — runnable configuration

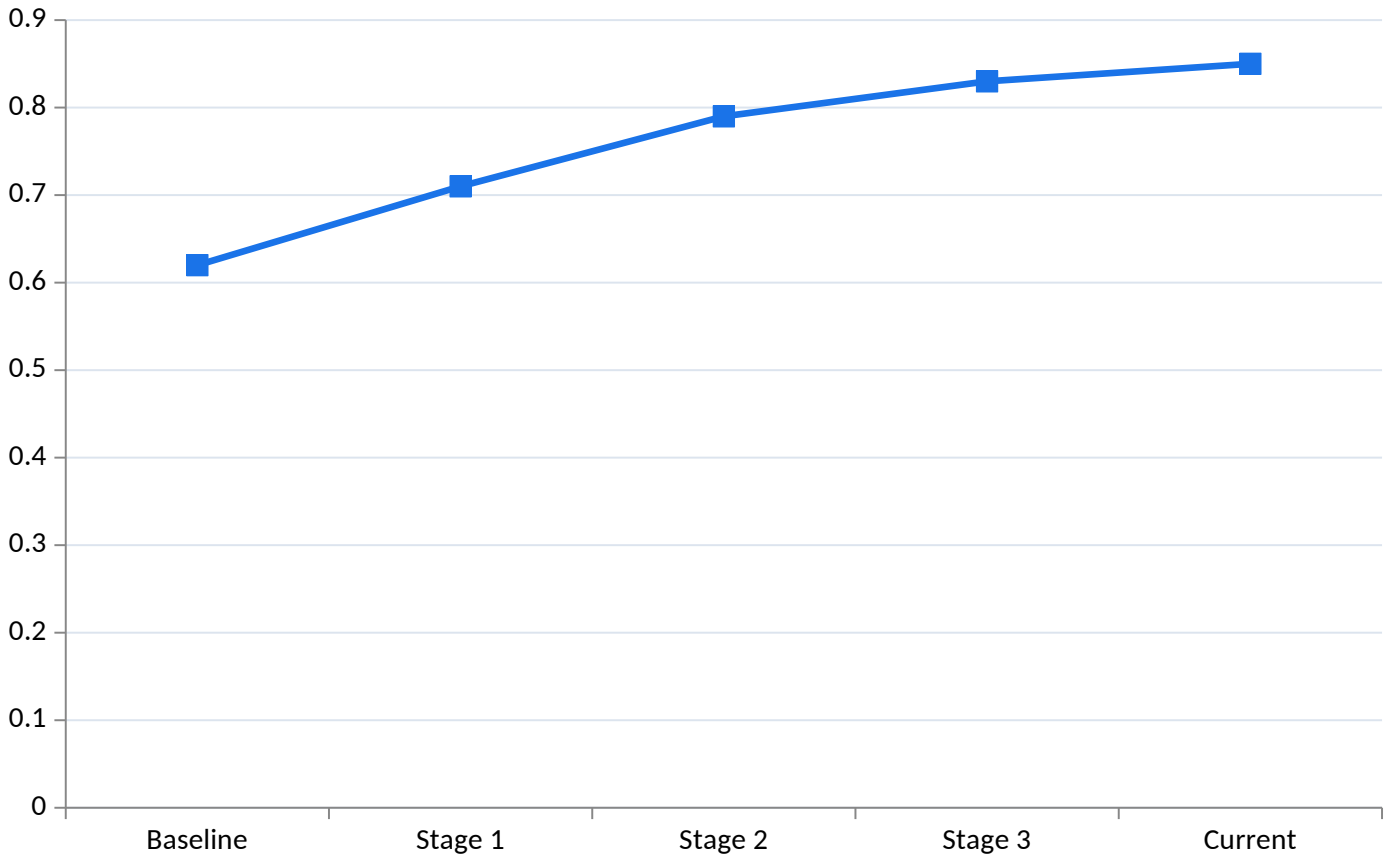
```
CREATE OR REPLACE MODEL `ml.churn`
OPTIONS(model_type='LOGISTIC_REG',
input_label_cols=['churned']) AS
SELECT * EXCEPT(customer_id) FROM `ml.customer_features`;
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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BigQuery ML model selection — metric evidence



Precision–recall trade-off

Formula

```
precision = TP/(TP+FP);
recall = TP/(TP+FN)
```

Interpret trend, threshold and uncertainty before acting.

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BigQuery ML model selection — failure diagnosis

Observable symptom	Likely mechanism	Control or test
High accuracy, poor recall	class imbalance	use PR AUC and class weights
Query cost spike	unbounded scans	partition, cluster and select columns
Leakage	post-outcome feature	point-in-time feature review

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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BigQuery ML model selection — evidence chain



A valid production decision is reproducible from the complete chain.

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BigQuery ML model selection — exam decision

SCENARIO

A retailer needs churn scores from 40 TB already in BigQuery with no custom serving code.

BEST DECISION

Use BigQuery ML; keep training and scoring beside governed data and batch predictions.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 03 evidence — Train and Evaluate a BigQuery ML Churn Model

ACTIVITY 03

Run SQL-defined logistic regression, evaluation and batch prediction on mock customer data.

Evidence: activity artifacts and acceptance checks

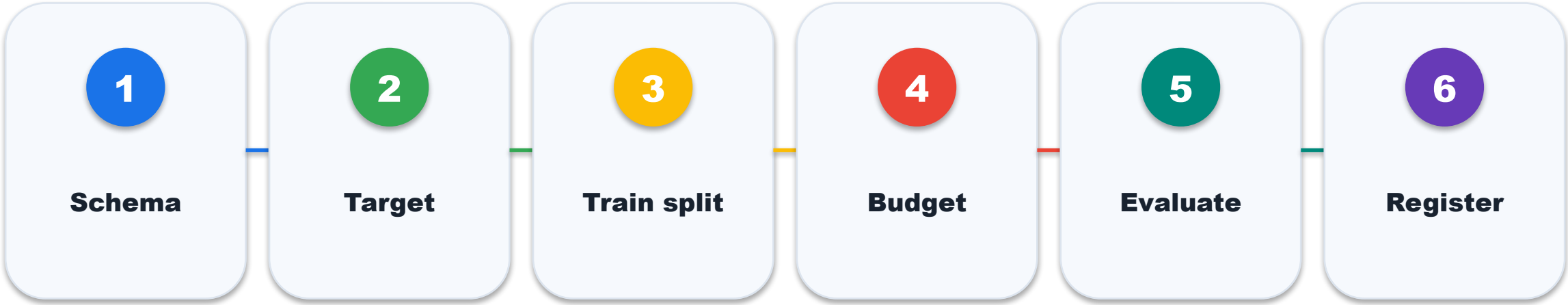
Detailed procedures: Learner Guide + activity README/PDF

- `customers.csv`
- `bigquery_ml.sql`
- `evaluate_metrics.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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AutoML tabular workflow — mechanism map



Control point Decision evidence must bind schema to register through versioned artifacts.

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DOMAIN 1 · DECISION RULE

AutoML tabular workflow — service decision matrix

OPTION 1

AutoML

Use when

strong baseline and limited ML engineering capacity

OPTION 2

Custom training

Use when

custom loss or architecture

OPTION 3

BigQuery ML

Use when

SQL team and warehouse-resident data

Exam cue: select from workload evidence—not from product familiarity.

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AutoML tabular workflow — runnable configuration

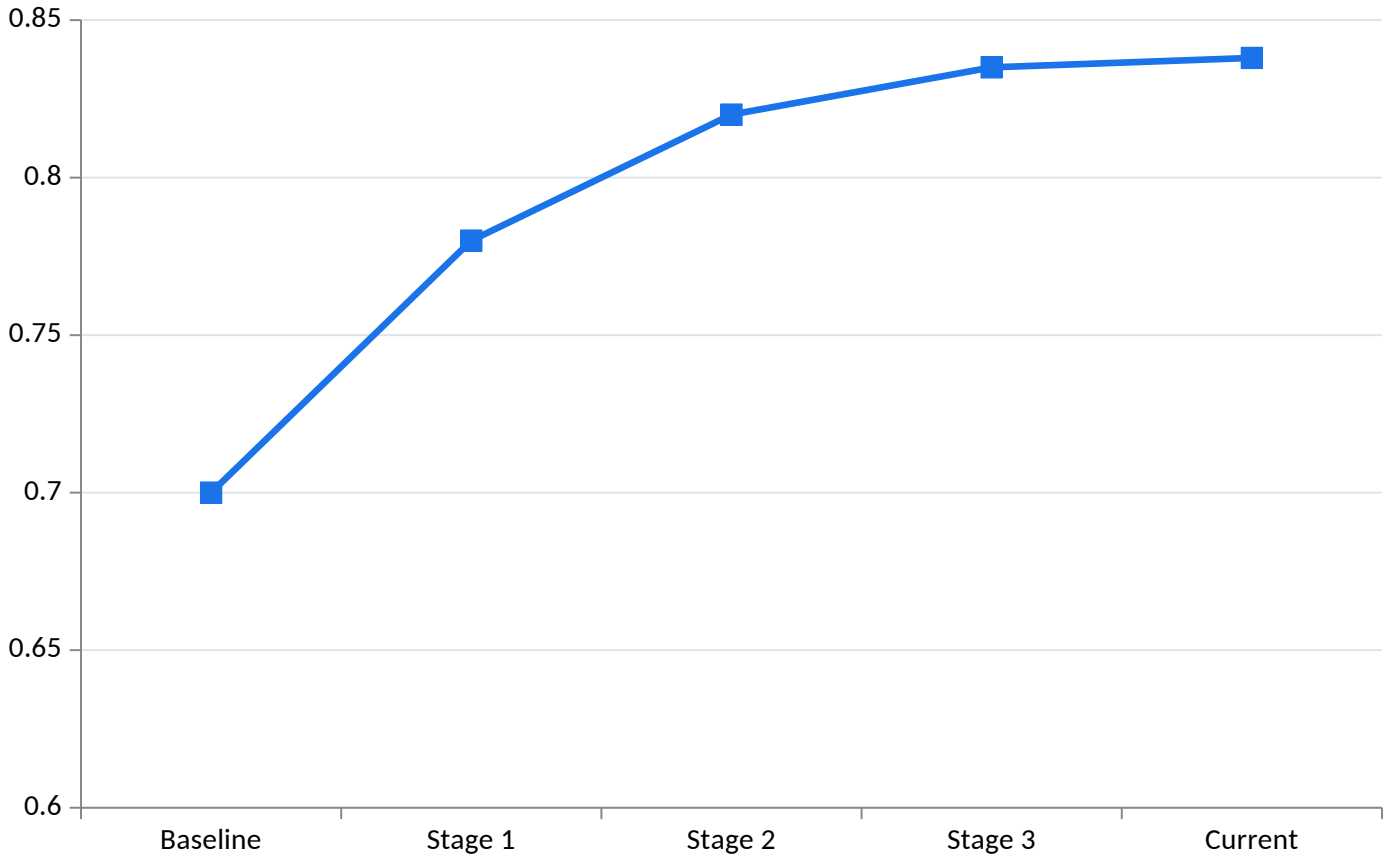
```
gcloud ai models upload --region=$REGION --display-
name=churn-automl \
+  --container-image-uri=$PREDICTION_IMAGE --artifact-
uri=$MODEL_DIR
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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AutoML tabular workflow — metric evidence



Training budget curve

Formula

```
marginal gain =
metric(t)-metric(t-1)
```

Interpret trend, threshold and uncertainty before acting.

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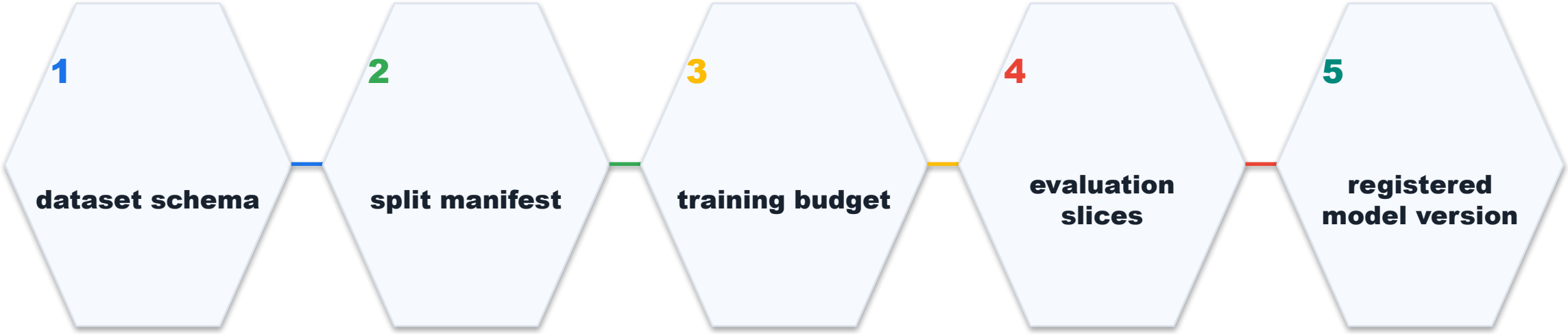
AutoML tabular workflow — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Invalid target	nulls or unsupported label	profile and clean label
Weak validation	random split on temporal data	use time-aware split
Over-budget	no stopping criterion	cap node hours and monitor marginal gain

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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AutoML tabular workflow — evidence chain



A valid production decision is reproducible from the complete chain.

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AutoML tabular workflow — exam decision

SCENARIO

A lending team needs a governed risk baseline quickly, but predictions must be explainable by segment.

BEST DECISION

Start with AutoML tabular, require slice metrics and feature attributions before deployment.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 04 evidence — Design an AutoML Tabular Experiment

ACTIVITY 04

Create a dataset specification, split plan, budget and evaluation slice gate.

Evidence: activity artifacts and acceptance checks

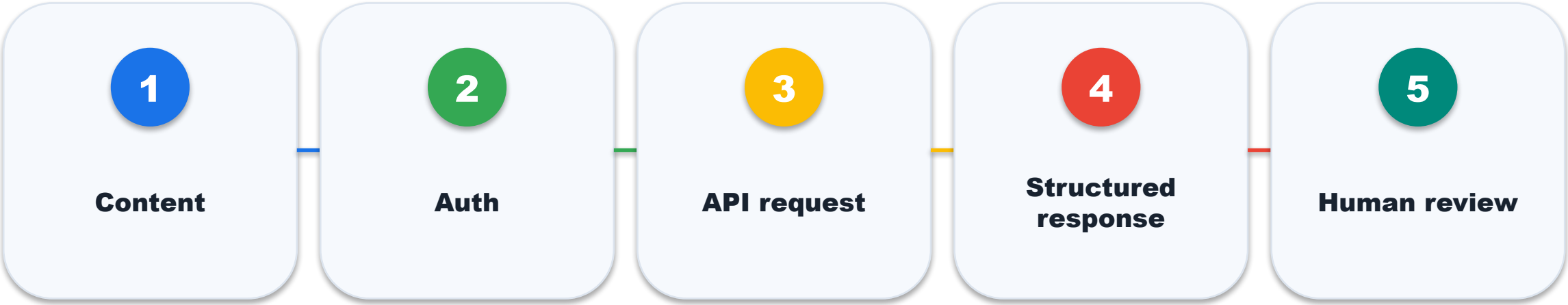
Detailed procedures: Learner Guide + activity README/PDF

- `automl_dataset.csv`
- `automl_config.yaml`
- `validate_splits.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Google Cloud AI APIs — mechanism map



Control point Decision evidence must bind content to human review through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Google Cloud AI APIs — service decision matrix

OPTION 1

Document AI

Use when

forms, invoices and OCR structure

OPTION 2

Vision API

Use when

labels, OCR and image properties

OPTION 3

Translate API

Use when

managed multilingual translation

Exam cue: select from workload evidence—not from product familiarity.

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Google Cloud AI APIs — runnable configuration

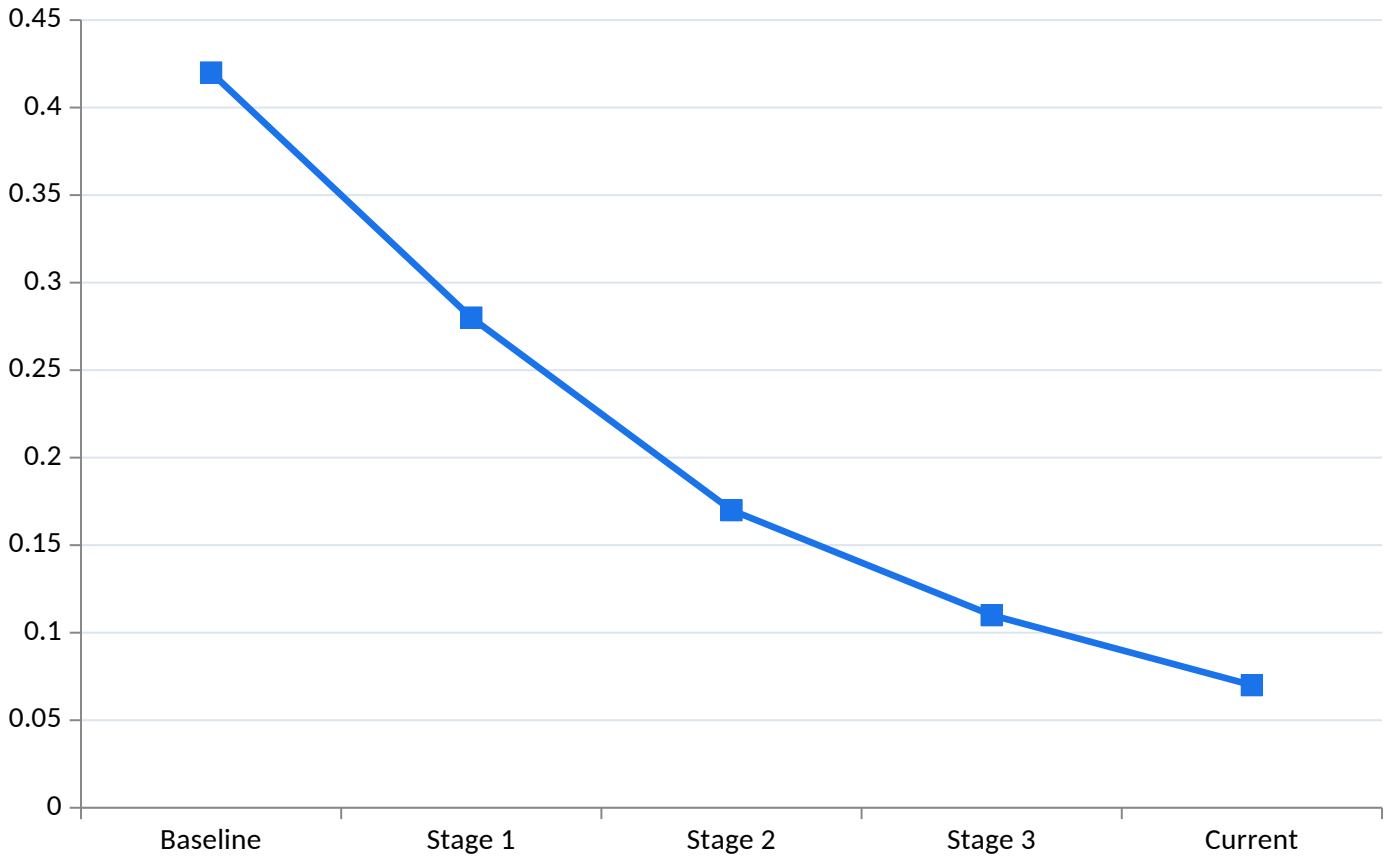
```
curl -s -X POST -H "Authorization: Bearer $(gcloud auth
print-access-token)" \
+  -H 'Content-Type: application/json' \
+  "https://vision.googleapis.com/v1/images:annotate" -d
@request.json
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Google Cloud AI APIs — metric evidence



API confidence threshold

Formula

```
review_rate =
count(score <
threshold)/N
```

Interpret trend, threshold and uncertainty before acting.

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Google Cloud AI APIs — failure diagnosis

Observable symptom	Likely mechanism	Control or test
403	API disabled or IAM missing	enable API; grant narrow role
429	quota exhausted	backoff, batch, request quota
Low confidence	domain mismatch	human review or custom model

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Google Cloud AI APIs — evidence chain



A valid production decision is reproducible from the complete chain.

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Google Cloud AI APIs — exam decision

SCENARIO

An insurer must extract line items from varied claim forms and retain reviewer evidence.

BEST DECISION

Use Document AI processor plus confidence-based human review; do not build OCR from scratch.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 05 evidence — Compare Pretrained Google Cloud AI APIs

ACTIVITY 05

Prepare Vision, Document AI and Natural Language requests and interpret mock responses.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `vision_request.json`
- `mock_vision_response.json`
- `api_selector.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Model Garden and foundation model selection — mechanism map



Control point Decision evidence must bind task to endpoint through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Model Garden and foundation model selection — service decision matrix

OPTION 1

Gemini managed model

Use when

multimodal and managed operations

OPTION 2

Open model

Use when

weight control or special license needs

OPTION 3

Specialized API

Use when

narrow commodity capability

Exam cue: select from workload evidence—not from product familiarity.

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Model Garden and foundation model selection — runnable configuration

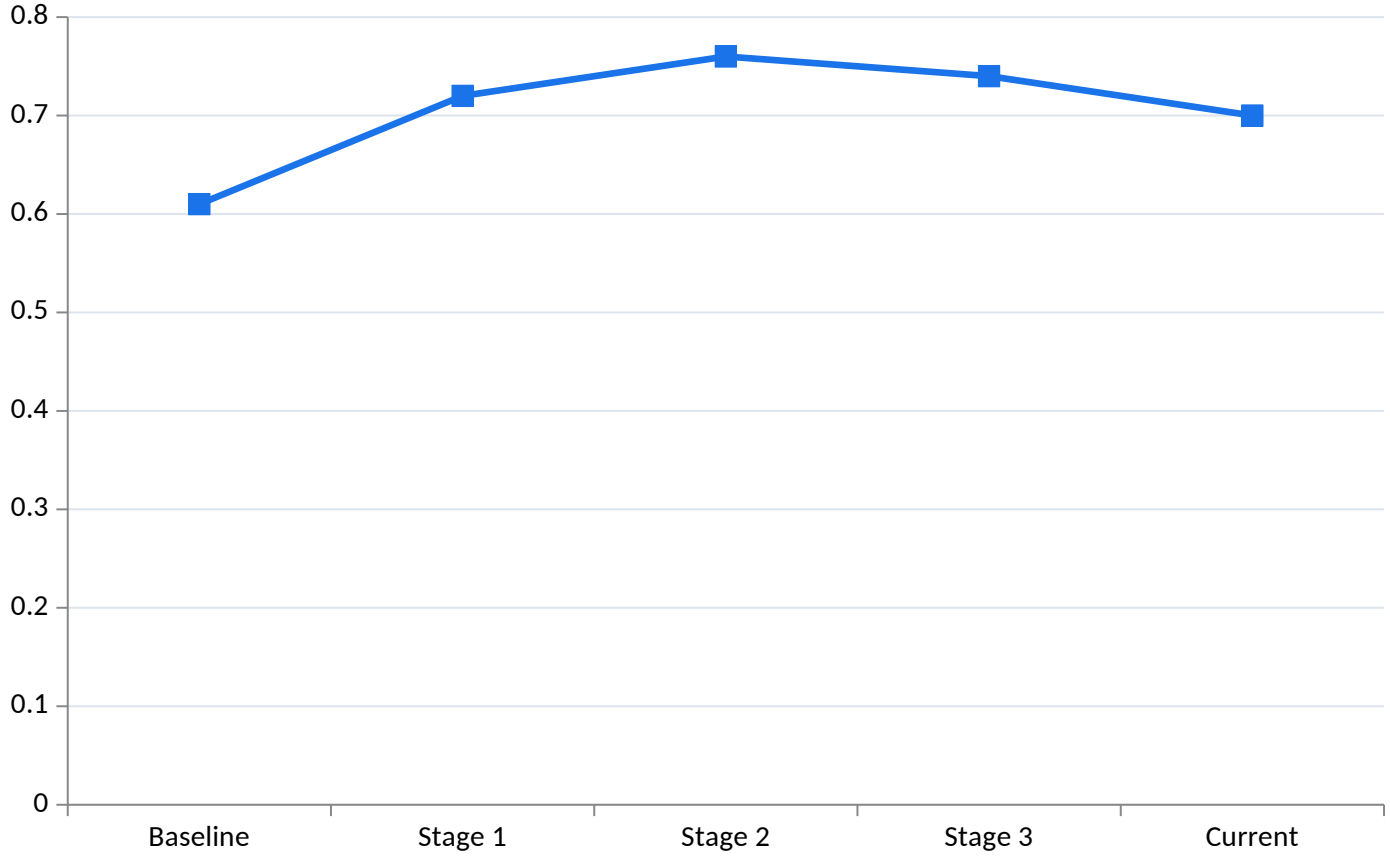
```
MODEL_ID='gemini-2.5-flash'
P95_MS=850
MAX_COST_PER_1K=0.020
SAFETY='BLOCK_MEDIUM_AND_ABOVE'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Model Garden and foundation model selection — metric evidence



Utility score

Formula

```
0.45*quality - 0.25*cost
- 0.20*latency +
0.10*availability
```

Interpret trend, threshold and uncertainty before acting.

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Model Garden and foundation model selection — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Benchmark mismatch	public task differs	use production-like eval set
License risk	terms not reviewed	legal and model-card review
Region gap	model unavailable	confirm region before design

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Model Garden and foundation model selection — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 1 · SCENARIO

Model Garden and foundation model selection — exam decision

SCENARIO

A support assistant must ground answers in policy PDFs with sub-second median response.

BEST DECISION

Evaluate managed Gemini candidates on groundedness, latency and cost; select by measured SLOs.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 02 evidence — Draft an ML Architecture Decision Record

ACTIVITY 02

Convert business volume, latency, security and explainability needs into a service decision.

Evidence: activity artifacts and acceptance checks

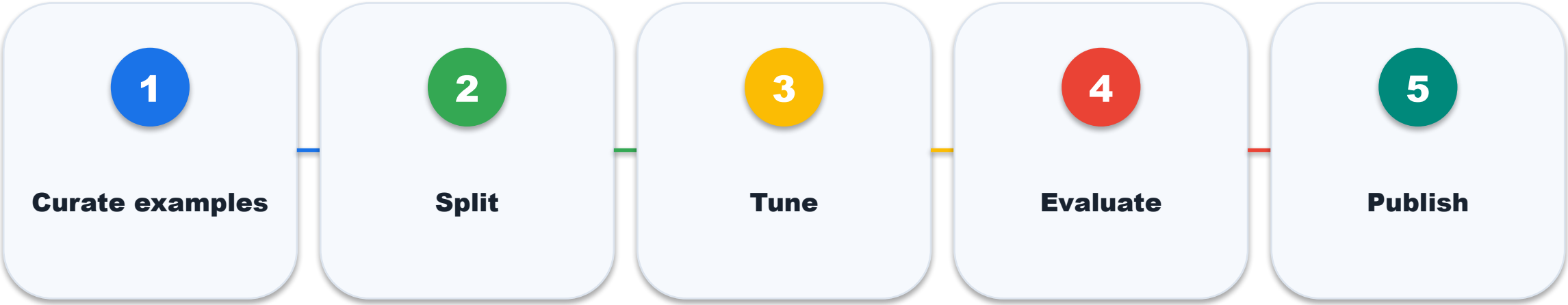
Detailed procedures: Learner Guide + activity README/PDF

- `requirements.csv`
- `architecture_decision.py`
- `expected_decision.json`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Gemini tuning with BigQuery — mechanism map



Control point Decision evidence must bind curate examples to publish through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Gemini tuning with BigQuery — service decision matrix

OPTION 1

Prompt only

Use when

behavior adjustable with instructions

OPTION 2

Supervised tuning

Use when

stable style/task examples

OPTION 3

RAG

Use when

facts change and citations matter

Exam cue: select from workload evidence—not from product familiarity.

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Gemini tuning with BigQuery — runnable configuration

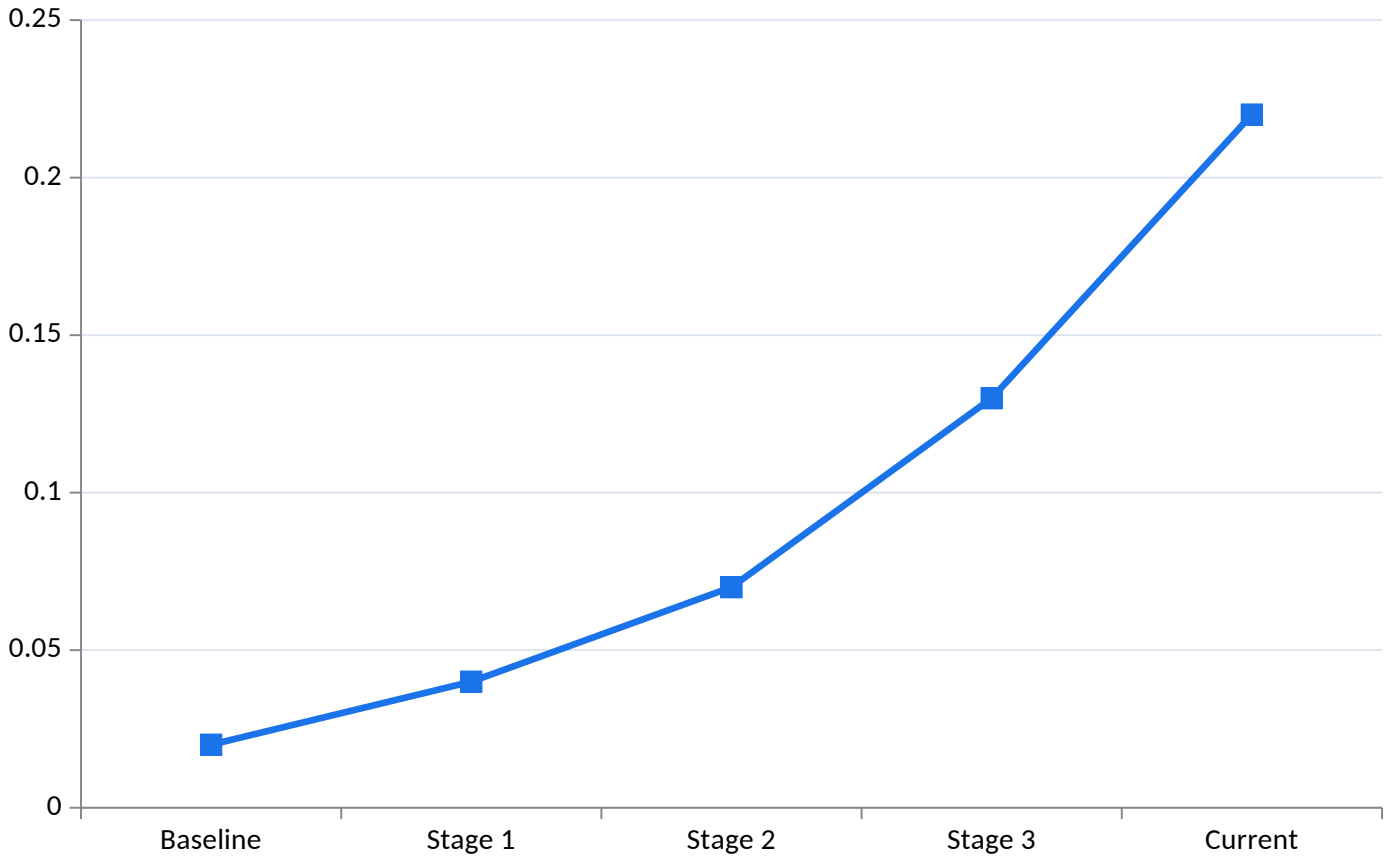
```
SELECT input_text, output_text
FROM `genai.tuning_examples`
WHERE approved = TRUE AND split IN ('TRAIN','VALIDATION');
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Gemini tuning with BigQuery — metric evidence



Tuning generalization gap

Formula

```
gap = train_score - validation_score
```

Interpret trend, threshold and uncertainty before acting.

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Gemini tuning with BigQuery — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Memorization	duplicates or tiny set	deduplicate and widen validation
Unsafe output	bad examples	curation and safety eval
No gain	task is knowledge-bound	use RAG instead

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Gemini tuning with BigQuery — evidence chain



A valid production decision is reproducible from the complete chain.

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Gemini tuning with BigQuery — exam decision

SCENARIO

A legal drafting model must follow a house style, while statute facts remain retrieval-grounded.

BEST DECISION

Tune for stable style and use RAG for changing legal facts; evaluate each separately.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 04 evidence — Design an AutoML Tabular Experiment

ACTIVITY 04

Create a dataset specification, split plan, budget and evaluation slice gate.

Evidence: activity artifacts and acceptance checks

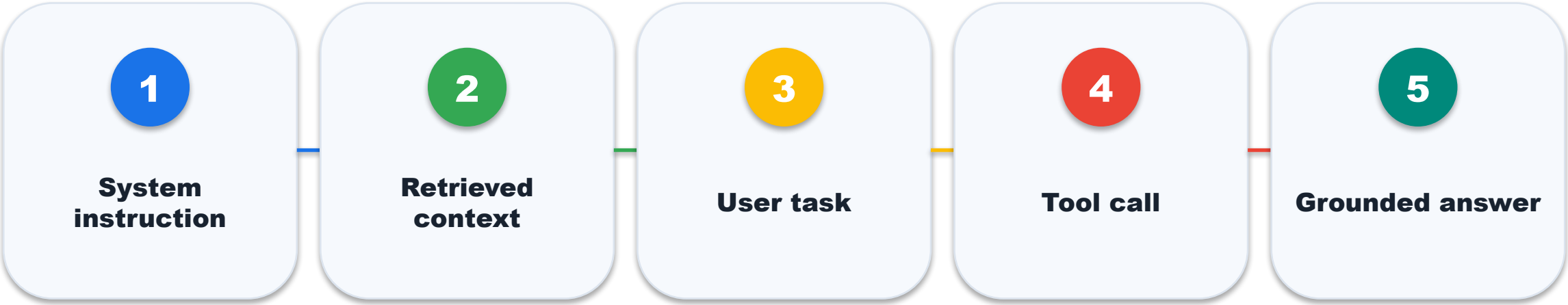
Detailed procedures: Learner Guide + activity README/PDF

- `automl_dataset.csv`
- `automl_config.yaml`
- `validate_splits.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Prompt and context engineering — mechanism map



Control point Decision evidence must bind system instruction to grounded answer through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Prompt and context engineering — service decision matrix

OPTION 1

Few-shot

Use when

stable pattern with representative examples

OPTION 2

RAG

Use when

current private knowledge

OPTION 3

Tool use

Use when

actions or authoritative calculations

Exam cue: select from workload evidence—not from product familiarity.

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Prompt and context engineering — runnable configuration

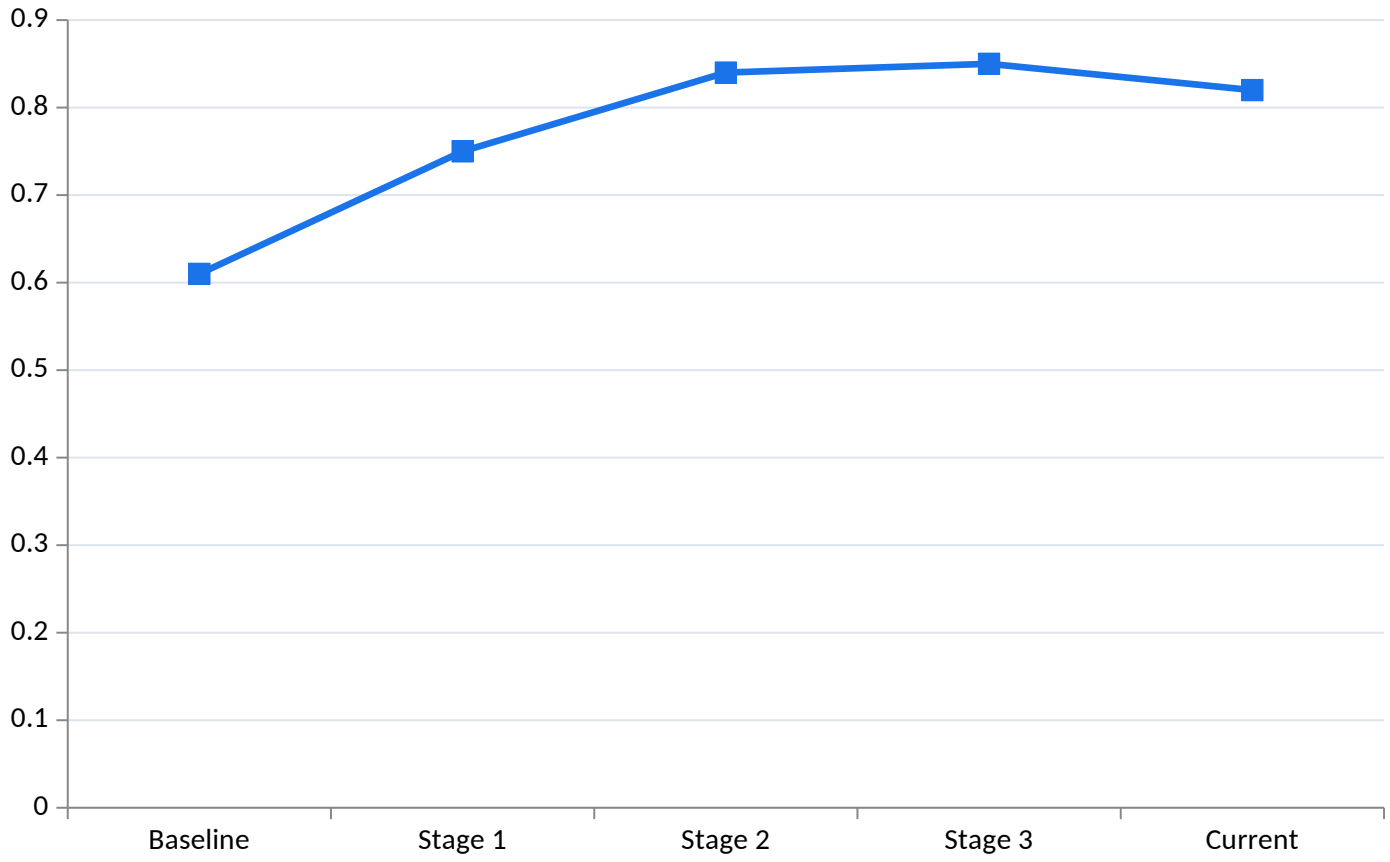
```
SYSTEM='Use only supplied context; cite source_id.'
CONTEXT_TOP_K=5
TEMPERATURE=0.1
MAX_OUTPUT_TOKENS=800
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Prompt and context engineering — metric evidence



Groundedness by retrieval depth

Formula

```
grounded =
supported_claims /
total_claims
```

Interpret trend, threshold and uncertainty before acting.

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Prompt and context engineering — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Hallucination	context missing or ignored	abstain and require citations
Prompt injection	untrusted retrieved text	isolate instructions from data
Context bloat	too many chunks	rerank and cap tokens

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Prompt and context engineering — evidence chain



A valid production decision is reproducible from the complete chain.

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Prompt and context engineering — exam decision

SCENARIO

An internal HR assistant must answer only from approved policies and refuse missing-policy questions.

BEST DECISION

Use RAG with instruction/data separation, citation checks, and a tested abstention path.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 02 evidence — Draft an ML Architecture Decision Record

ACTIVITY 02

Convert business volume, latency, security and explainability needs into a service decision.

Evidence: activity artifacts and acceptance checks

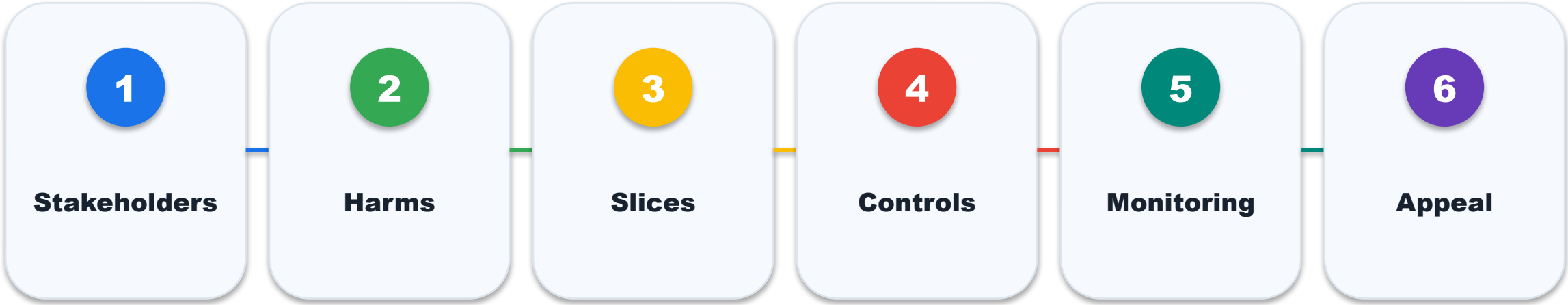
Detailed procedures: Learner Guide + activity README/PDF

- `requirements.csv`
- `architecture_decision.py`
- `expected_decision.json`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Responsible low-code design — mechanism map



Control point Decision evidence must bind stakeholders to appeal through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Responsible low-code design — service decision matrix

OPTION 1

Human-in-loop

Use when

high-impact or uncertain decisions

OPTION 2

Human-on-loop

Use when

automated low-risk with review

OPTION 3

Automated

Use when

reversible, validated low-harm tasks

Exam cue: select from workload evidence—not from product familiarity.

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Responsible low-code design — runnable configuration

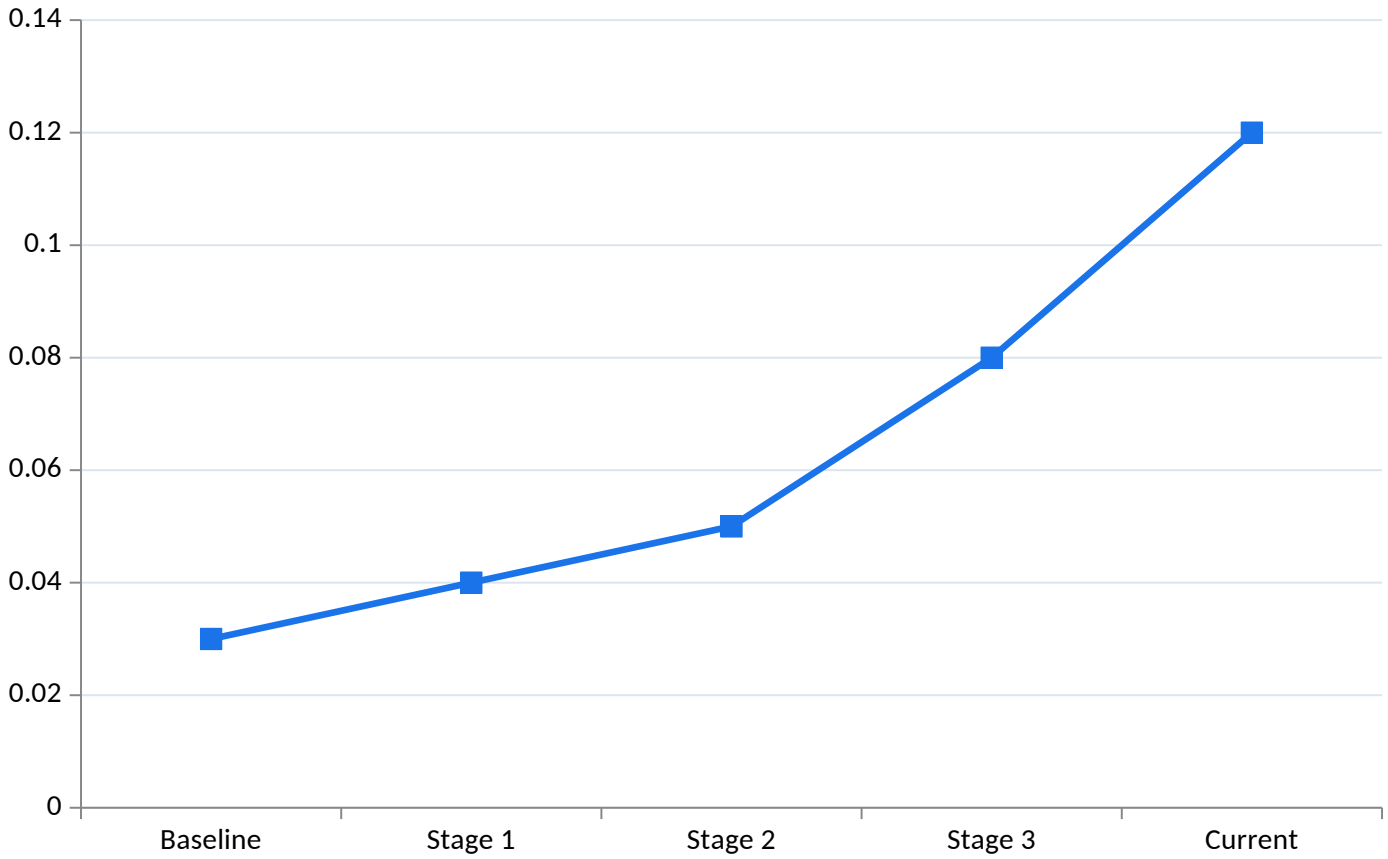
```
slices = ['region','age_band','channel']
max_tpr_gap = 0.05
min_group_n = 100
require_appeal = True
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Responsible low-code design — metric evidence



Equal opportunity gap

Formula

```
max(TPR_group)-
min(TPR_group)
```

Interpret trend, threshold and uncertainty before acting.

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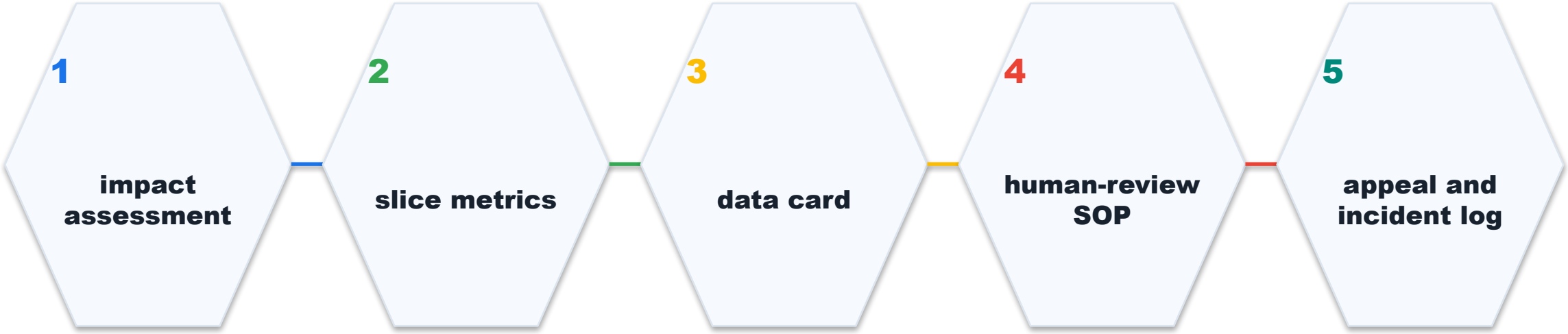
Responsible low-code design — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Hidden disparity	aggregate-only metric	slice evaluation
Automation bias	no override	human authority and appeal
Purpose creep	reused prediction	use limitation and review

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Responsible low-code design — evidence chain



A valid production decision is reproducible from the complete chain.

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Responsible low-code design — exam decision

SCENARIO

A model prioritizes public-service cases where false negatives have unequal community impact.

BEST DECISION

Require human review, group metrics, documented thresholds and a usable appeal process.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

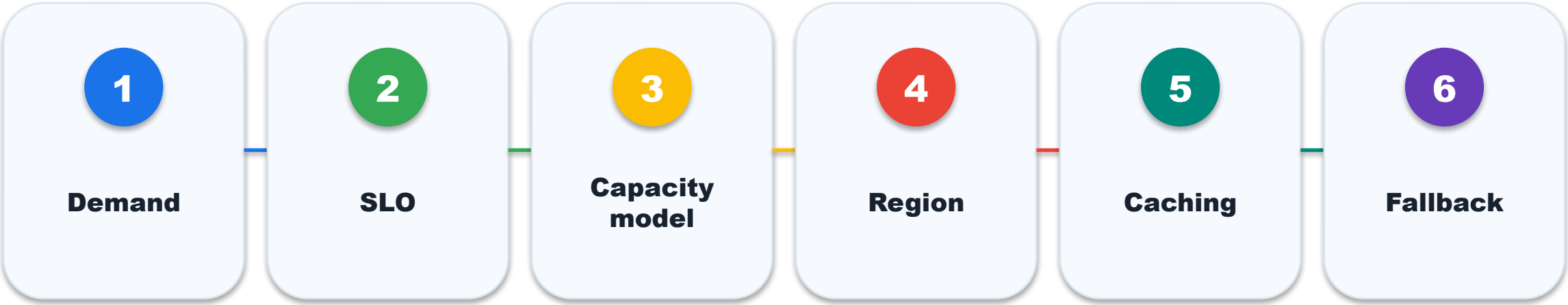
Detailed procedures: Learner Guide + activity README/PDF

- `baseline.csv`
- `current.csv`
- `monitor.py`
- `incident_template.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Cost, latency and availability architecture — mechanism map



Control point Decision evidence must bind demand to fallback through versioned artifacts.

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DOMAIN 1 · DECISION RULE

Cost, latency and availability architecture — service decision matrix

OPTION 1

Batch

Use when

large volume without immediate response

OPTION 2

Online endpoint

Use when

predictable low latency

OPTION 3

Serverless

Use when

bursty stateless inference

Exam cue: select from workload evidence—not from product familiarity.

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Cost, latency and availability architecture — runnable configuration

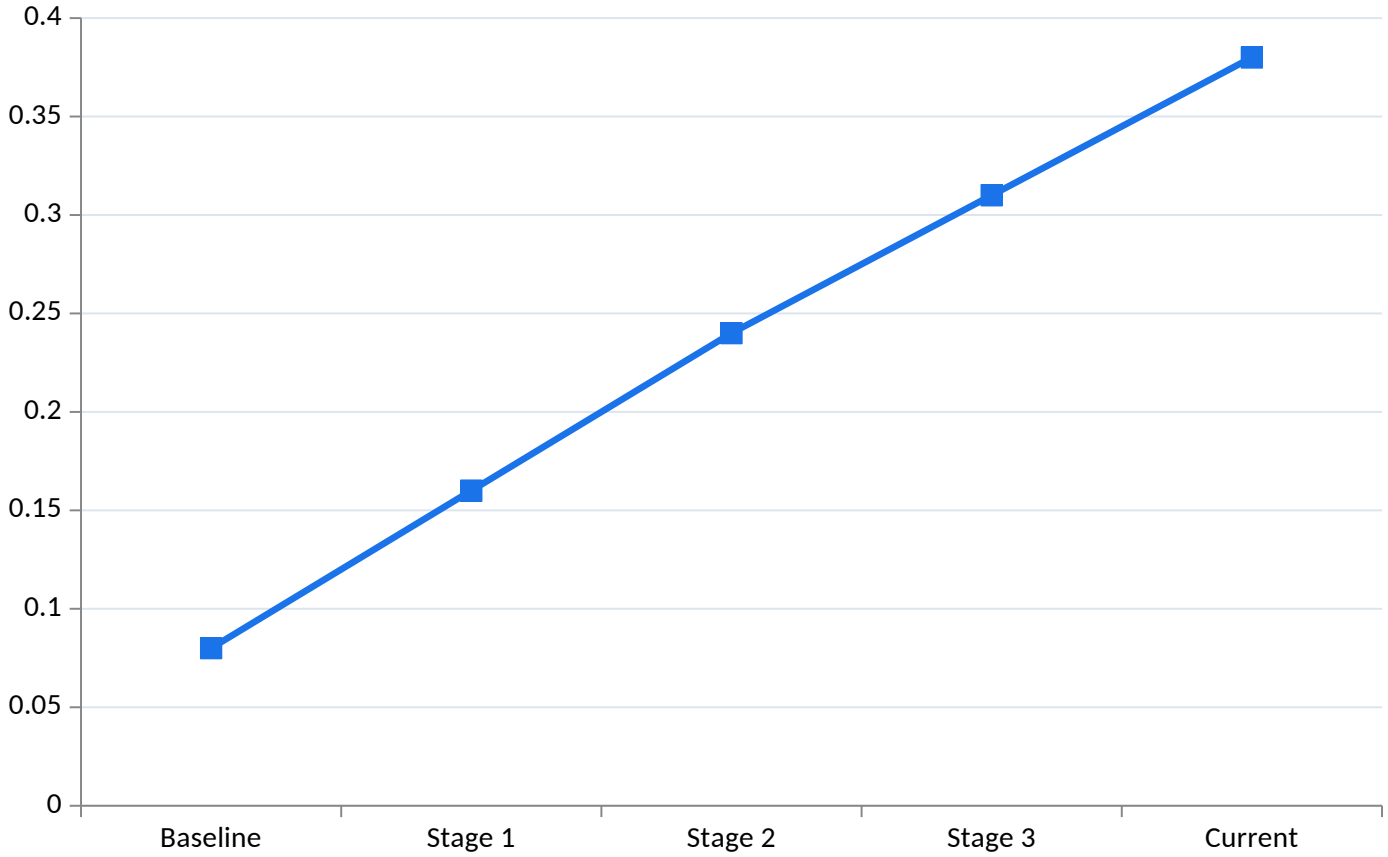
```
target_p95_ms: 300
min_replicas: 2
max_replicas: 20
cache_ttl_s: 900
regional_failover: true
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Cost, latency and availability architecture — metric evidence



Capacity headroom

Formula

```
headroom = 1 - peak_qps/max_qps
```

Interpret trend, threshold and uncertainty before acting.

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Cost, latency and availability architecture — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Tail latency	cold starts or saturation	min replicas and load test
High cost	oversized accelerator	profile and right-size
Regional outage	single region	tested fallback design

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Cost, latency and availability architecture — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 1 · SCENARIO

Cost, latency and availability architecture — exam decision

SCENARIO

A recommendation API sees 10x campaign bursts and must sustain 99.9% monthly availability.

BEST DECISION

Use autoscaled online serving with minimum warm capacity, cached features and tested regional fallback.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 02 evidence — Draft an ML Architecture Decision Record

ACTIVITY 02

Convert business volume, latency, security and explainability needs into a service decision.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `requirements.csv`
- `architecture_decision.py`
- `expected_decision.json`

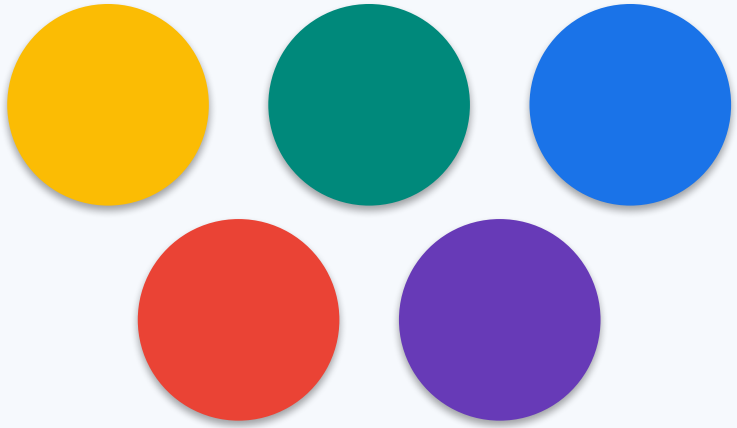
Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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DOMAIN 2 · ~16%

Collaborating to manage data and models

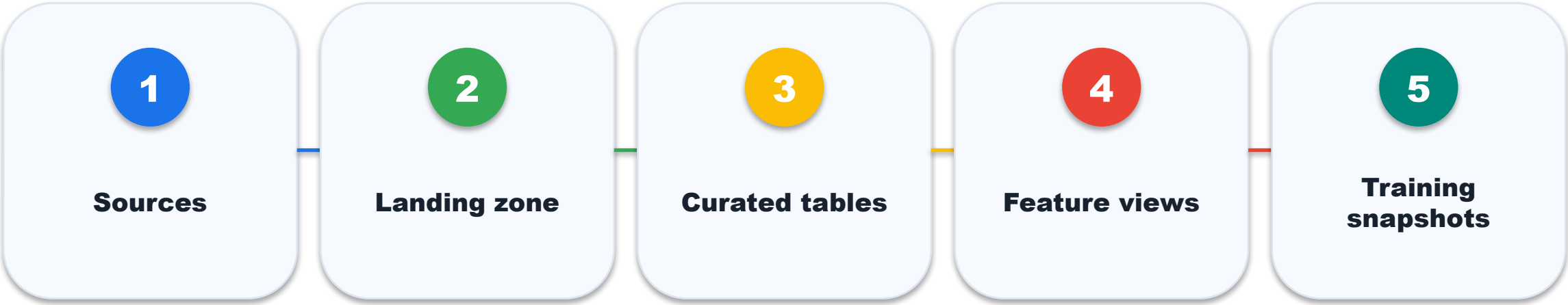
Topics 1, 3 and 5: big data, data quality, feature engineering



8 technical mechanisms · 64 anchored slides

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Data organization by modality — mechanism map



Control point Decision evidence must bind sources to training snapshots through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Data organization by modality — service decision matrix

OPTION 1

BigQuery

Use when

structured analytics and SQL

OPTION 2

Cloud Storage

Use when

large unstructured objects

OPTION 3

BigLake

Use when

governed access across lakes and warehouses

Exam cue: select from workload evidence—not from product familiarity.

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Data organization by modality — runnable configuration

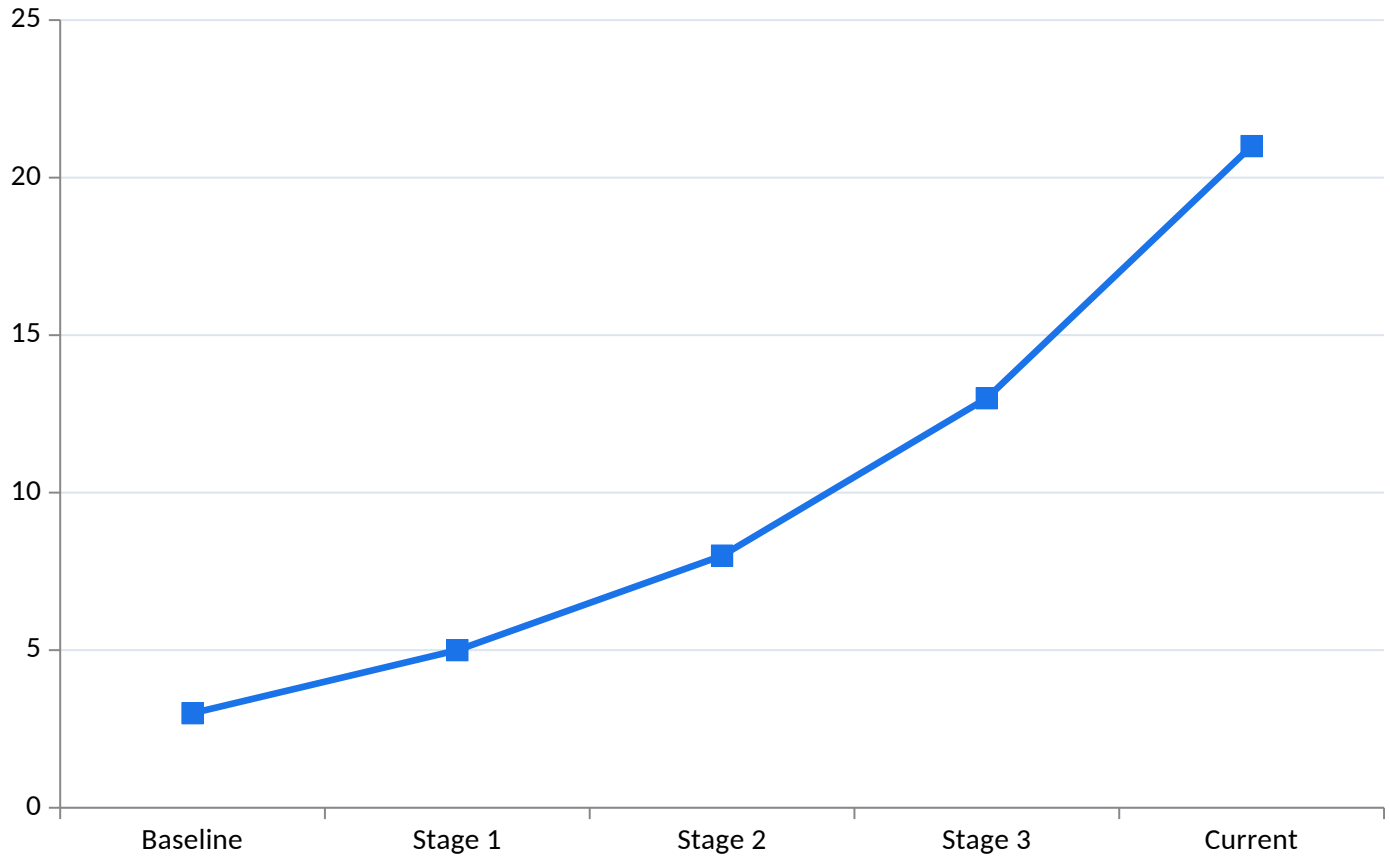
```
gsutil -m cp data/images/*.jpg gs://$BUCKET/raw/images/
bq load --autodetect --source_format=CSV ml.customers
gs://$BUCKET/raw/customers.csv
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Data organization by modality — metric evidence



Freshness lag

Formula

```
lag = ingestion_time - event_time
```

Interpret trend, threshold and uncertainty before acting.

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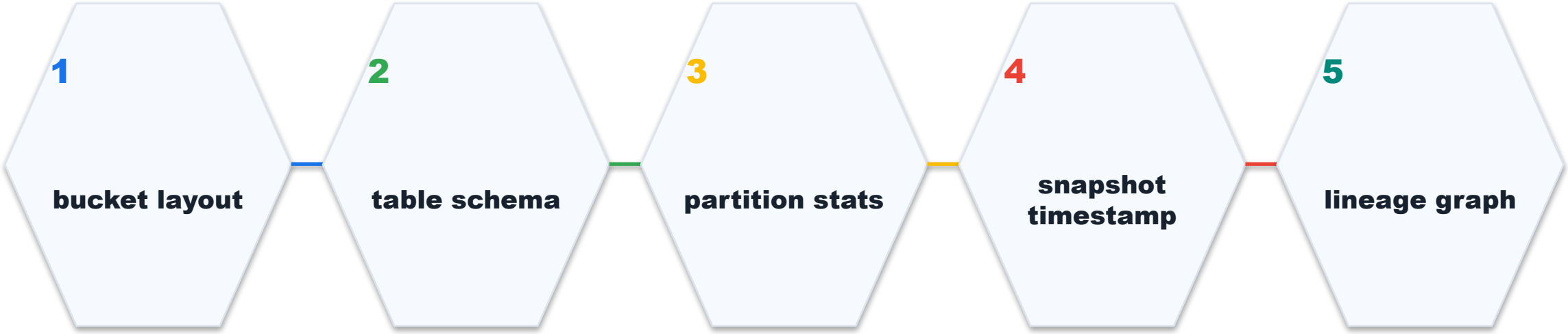
Data organization by modality — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Schema drift	producer changed fields	contract checks
Duplicate entities	weak keys	deduplicate by event id
Training skew	mutable source	snapshot and version

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Data organization by modality — evidence chain



A valid production decision is reproducible from the complete chain.

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Data organization by modality — exam decision

SCENARIO

A multimodal claims model uses PDFs, images and tabular policy history from three teams.

BEST DECISION

Land objects in Cloud Storage, curated entities in BigQuery, and bind a point-in-time training manifest.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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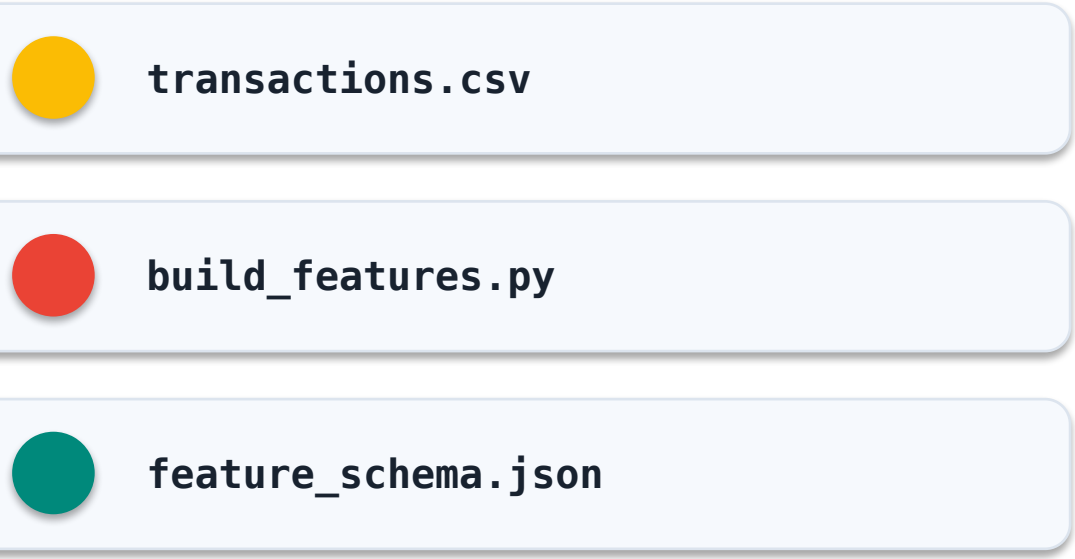
Activity 06 evidence — Build a Data Quality and Feature Pipeline

ACTIVITY 06

Validate schema, repair missing values, encode features and preserve a training manifest.

Evidence: activity artifacts and acceptance checks

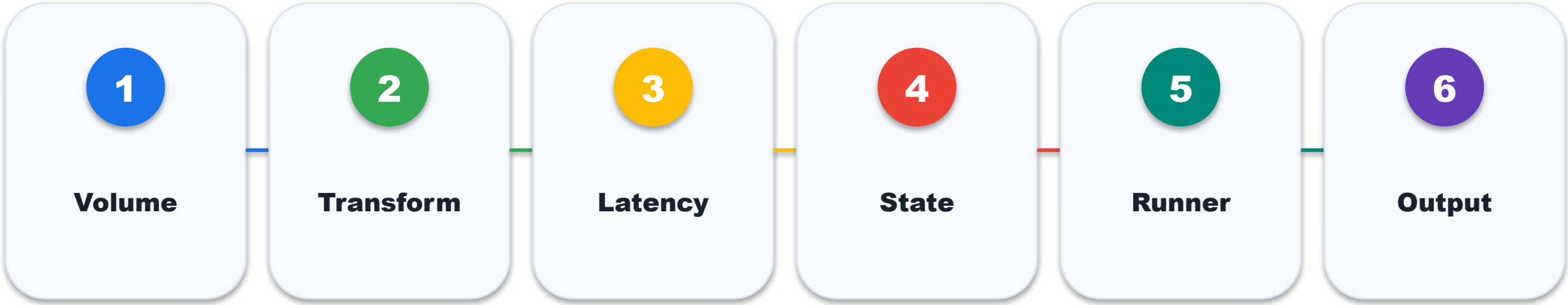
Detailed procedures: Learner Guide + activity README/PDF



Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Preprocessing tool selection — mechanism map



Control point Decision evidence must bind volume to output through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Preprocessing tool selection — service decision matrix

OPTION 1

pandas/sklearn

Use when

fits memory and iteration speed matters

OPTION 2

Dataflow

Use when

managed Beam batch/stream pipelines

OPTION 3

Spark

Use when

existing distributed Spark ecosystem

Exam cue: select from workload evidence—not from product familiarity.

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Preprocessing tool selection — runnable configuration

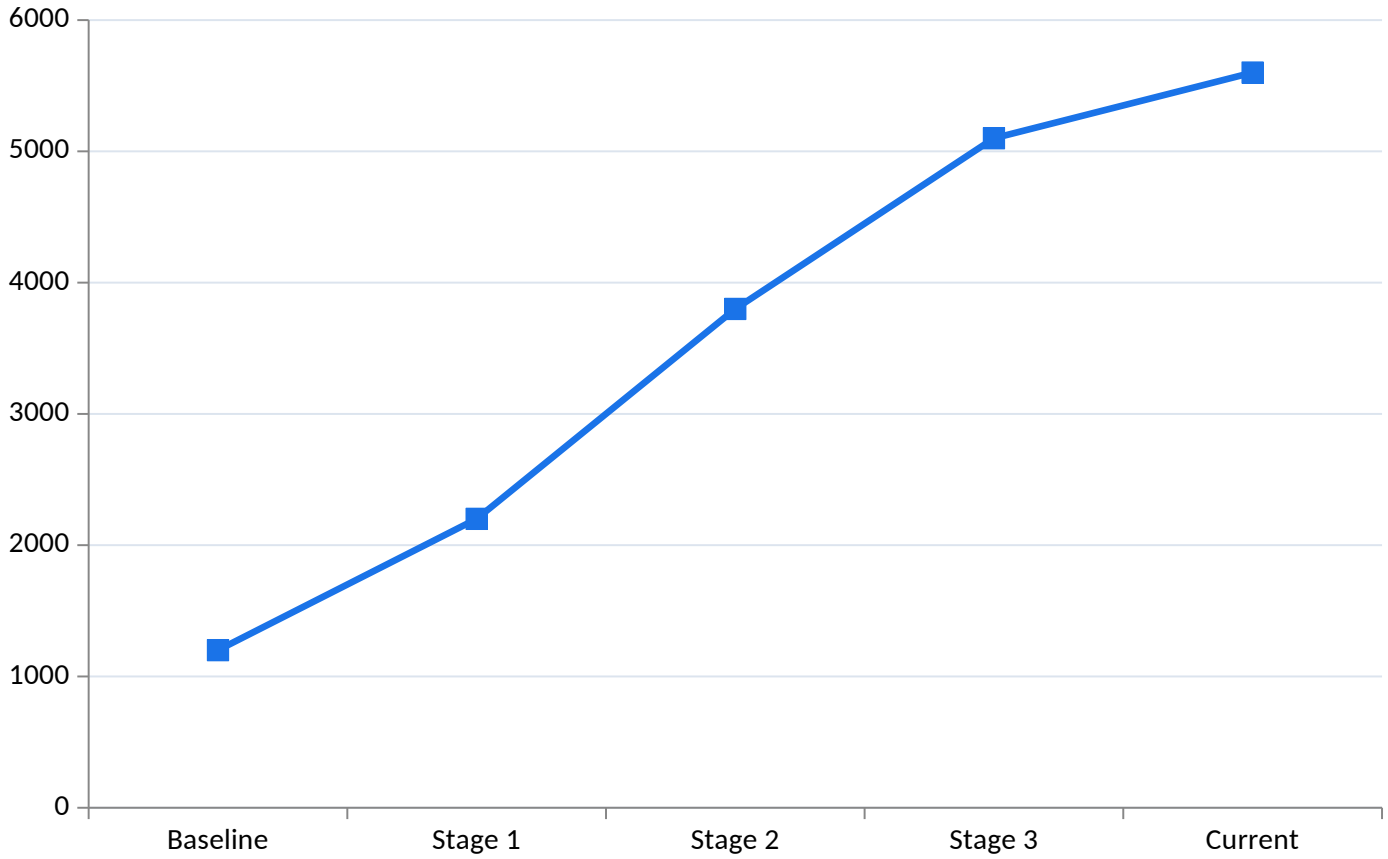
```
with beam.Pipeline(options=opts) as p:
  rows = p | beam.io.ReadFromText(uri)
  clean = rows | beam.Map(validate_and_transform)
  clean | beam.io.WriteToBigQuery(table)
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Preprocessing tool selection — metric evidence



Throughput vs workers

Formula

```
records_per_s =
processed / elapsed_s
```

Interpret trend, threshold and uncertainty before acting.

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Preprocessing tool selection — failure diagnosis

Observable symptom	Likely mechanism	Control or test
OOM	in-memory transform too large	distributed runner
Late data	watermark misconfigured	event-time windows
Hot key	skewed grouping	fan-out or rekey

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Preprocessing tool selection — evidence chain



A valid production decision is reproducible from the complete chain.

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Preprocessing tool selection — exam decision

SCENARIO

A 2 TB daily feed joins clickstream windows and must finish before a 06:00 training job.

BEST DECISION

Use Dataflow batch with autoscaling, dead-letter handling and a measured completion SLO.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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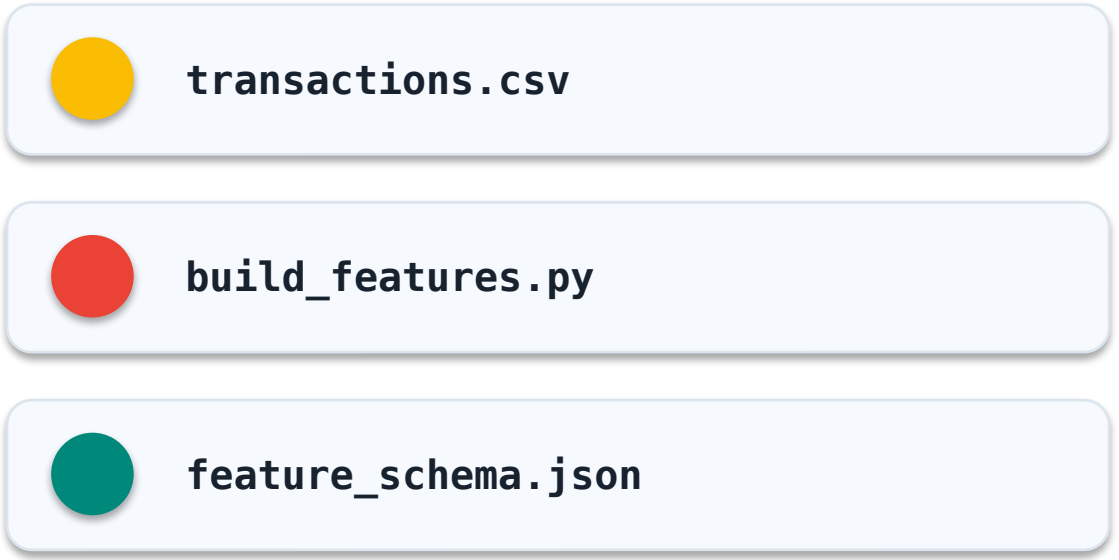
Activity 06 evidence — Build a Data Quality and Feature Pipeline

ACTIVITY 06

Validate schema, repair missing values, encode features and preserve a training manifest.

Evidence: activity artifacts and acceptance checks

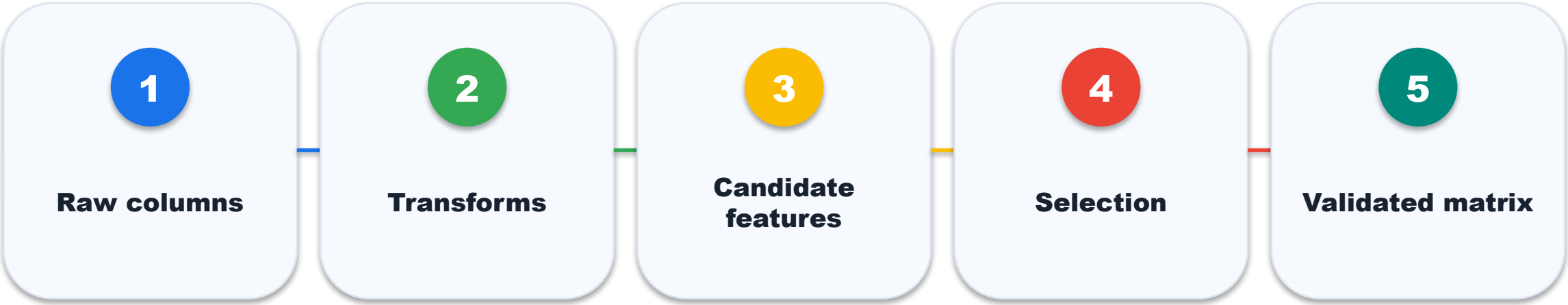
Detailed procedures: Learner Guide + activity README/PDF



Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Feature engineering and selection — mechanism map



Control point Decision evidence must bind raw columns to validated matrix through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Feature engineering and selection — service decision matrix

OPTION 1

Standardization

Use when

linear and distance models

OPTION 2

Vocabulary/ embedding

Use when

high-cardinality text

OPTION 3

Cross feature

Use when

known interaction with bounded cardinality

Exam cue: select from workload evidence—not from product familiarity.

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Feature engineering and selection — runnable configuration

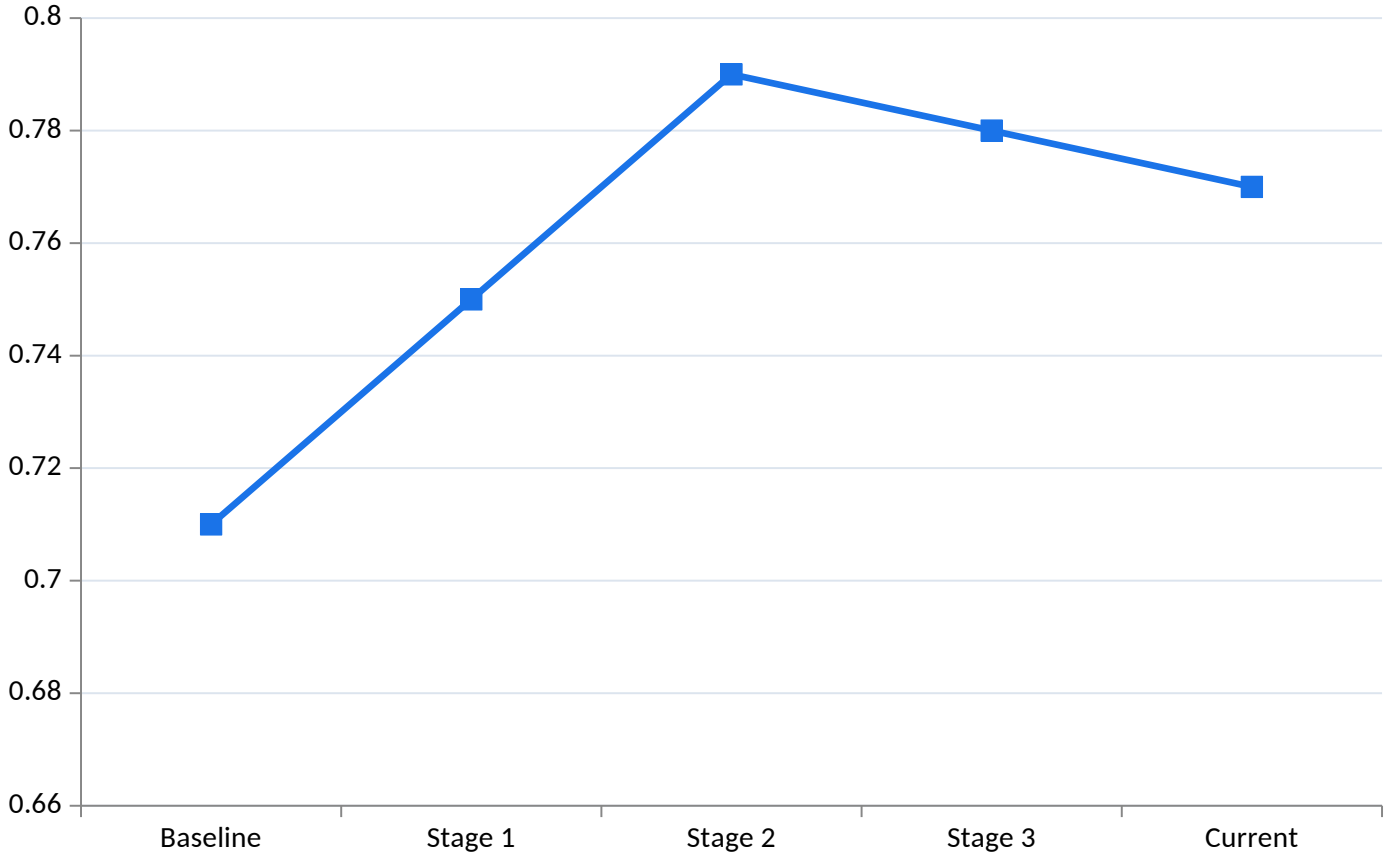
```
preprocess = ColumnTransformer([
 ('num', StandardScaler(), numeric),
 ('cat', OneHotEncoder(handle_unknown='ignore'),
categorical)])
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Feature engineering and selection — metric evidence



Validation AUC by feature set

Formula

```
lift = AUC_candidate - AUC_baseline
```

Interpret trend, threshold and uncertainty before acting.

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Feature engineering and selection — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Leakage	future value encoded	time-aware review
Cardinality blow-up	unbounded categories	hash or vocabulary cap
Train/serve mismatch	duplicate code	shared transform artifact

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Feature engineering and selection — evidence chain



A valid production decision is reproducible from the complete chain.

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Feature engineering and selection — exam decision

SCENARIO

A fraud model sees merchant IDs never observed during training and drifts by region.

BEST DECISION

Use unknown-safe categorical encoding, region slices and versioned preprocessing artifacts.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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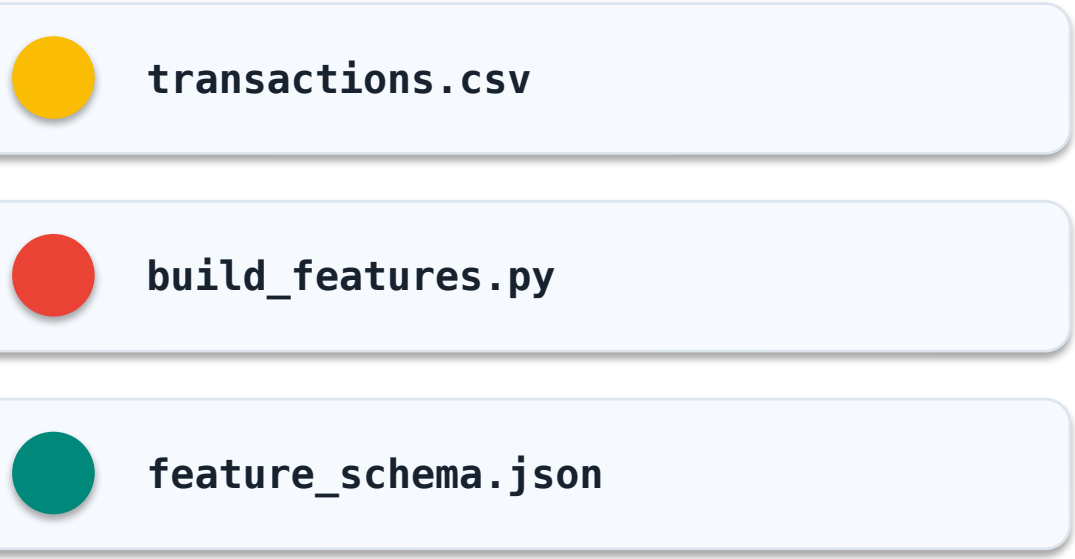
Activity 06 evidence — Build a Data Quality and Feature Pipeline

ACTIVITY 06

Validate schema, repair missing values, encode features and preserve a training manifest.

Evidence: activity artifacts and acceptance checks

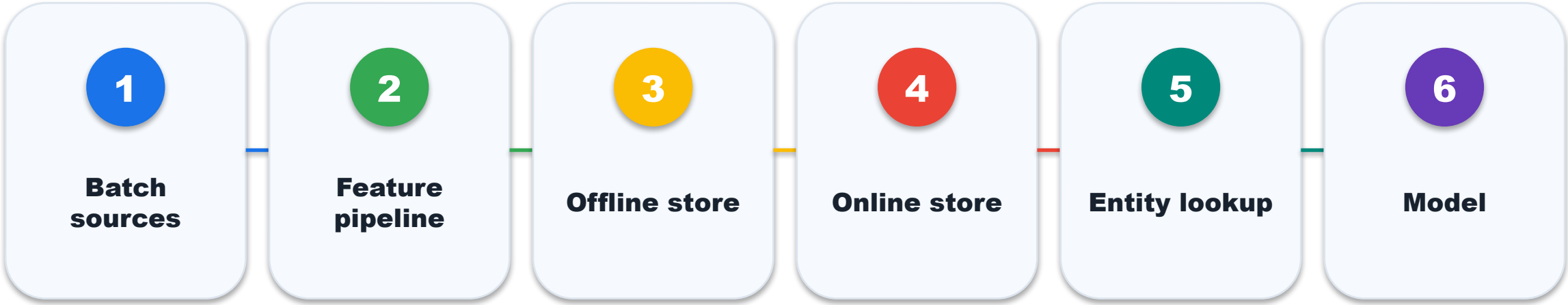
Detailed procedures: Learner Guide + activity README/PDF



Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Feature Store architecture — mechanism map



Control point Decision evidence must bind batch sources to model through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Feature Store architecture — service decision matrix

OPTION 1

Offline features

Use when

training and batch scoring

OPTION 2

Online features

Use when

low-latency current values

OPTION 3

Embedding retrieval

Use when

nearest-neighbor similarity

Exam cue: select from workload evidence—not from product familiarity.

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Feature Store architecture — runnable configuration

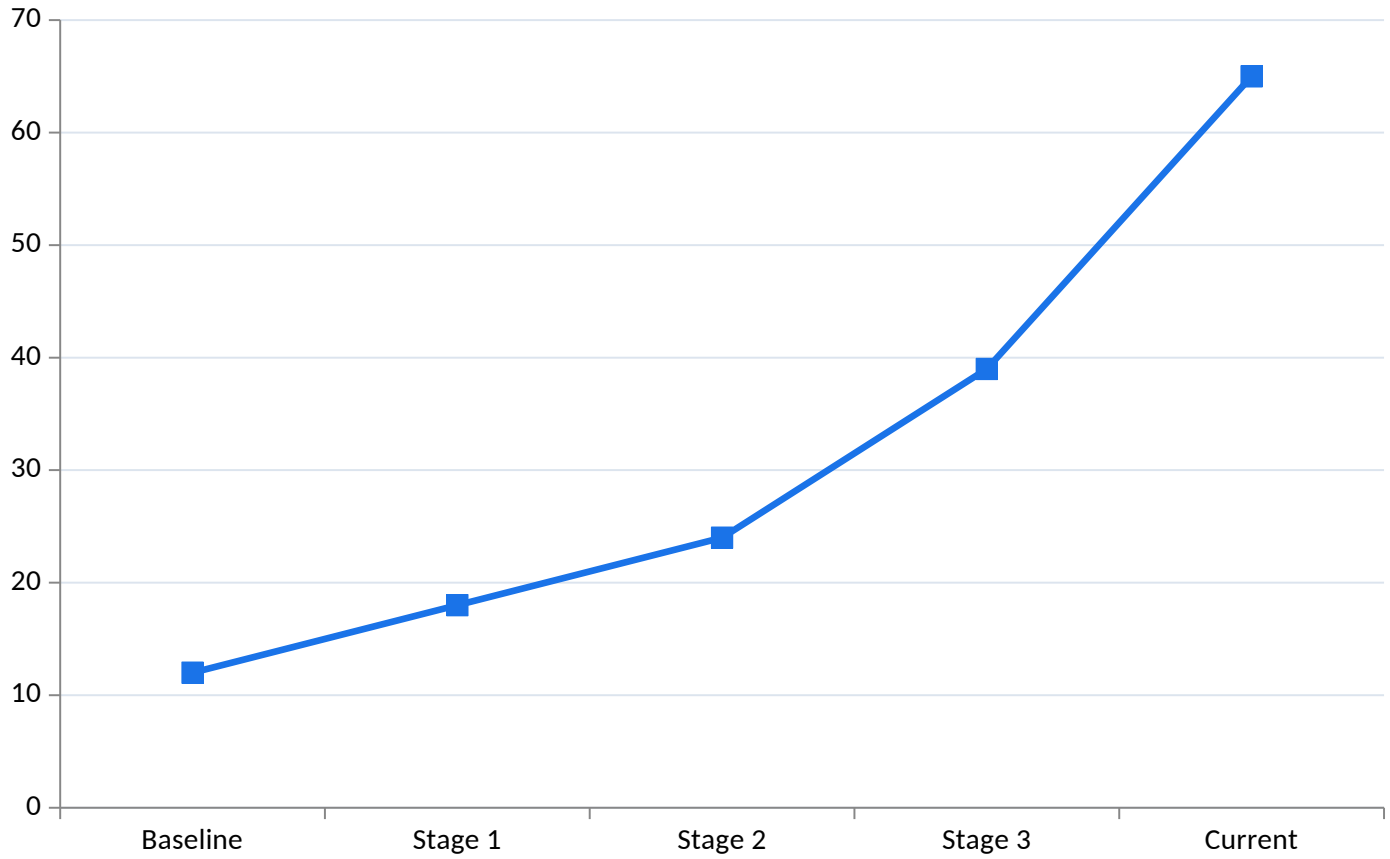
```
entity_id: customer_id
feature_view: customer_realtime
features: [txn_count_1h, spend_7d, risk_score]
online_ttl: 3600
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Feature Store architecture — metric evidence



Online feature freshness

Formula

```
freshness = now - feature_timestamp
```

Interpret trend, threshold and uncertainty before acting.

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Feature Store architecture — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Stale feature	pipeline lag	freshness alert
Wrong entity	key mismatch	contract and lookup test
Point-in-time leak	latest-value join	as-of join

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Feature Store architecture — evidence chain



A valid production decision is reproducible from the complete chain.

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Feature Store architecture — exam decision

SCENARIO

A real-time fraud endpoint needs 50 ms feature lookups and reproducible historical training values.

BEST DECISION

Use online and offline feature paths with one definition and explicit point-in-time correctness tests.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 11 evidence — Verify Feature Store Point-in-Time Parity

ACTIVITY 11

Model offline/online feature definitions and detect stale or leaked features.

Evidence: activity artifacts and acceptance checks

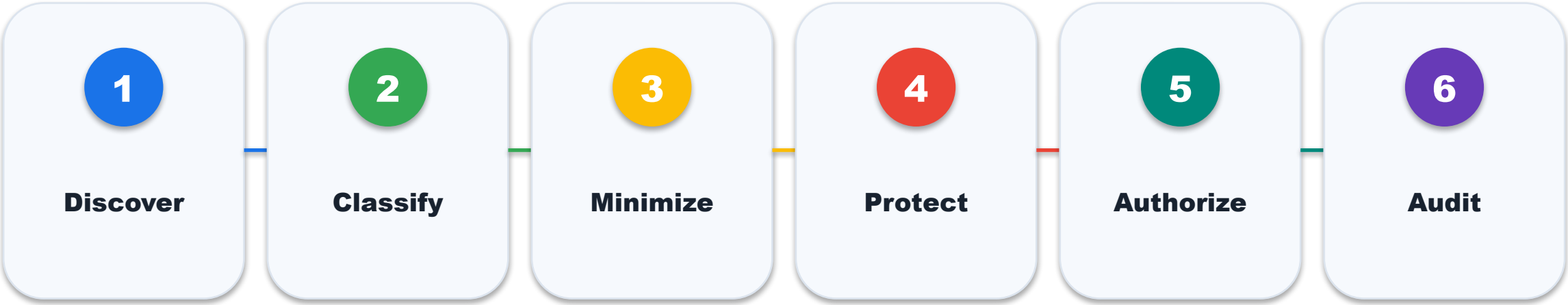
Detailed procedures: Learner Guide + activity README/PDF

- `offline_features.csv`
- `online_features.csv`
- `check_parity.py`
- `feature_view.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Privacy and sensitive-data controls — mechanism map



Control point Decision evidence must bind discover to audit through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Privacy and sensitive-data controls — service decision matrix

OPTION 1

DLP inspection

Use when

discover PII at scale

OPTION 2

Tokenization

Use when

retain joins without clear identifiers

OPTION 3

Aggregation

Use when

reduce re-identification risk

Exam cue: select from workload evidence—not from product familiarity.

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Privacy and sensitive-data controls — runnable configuration

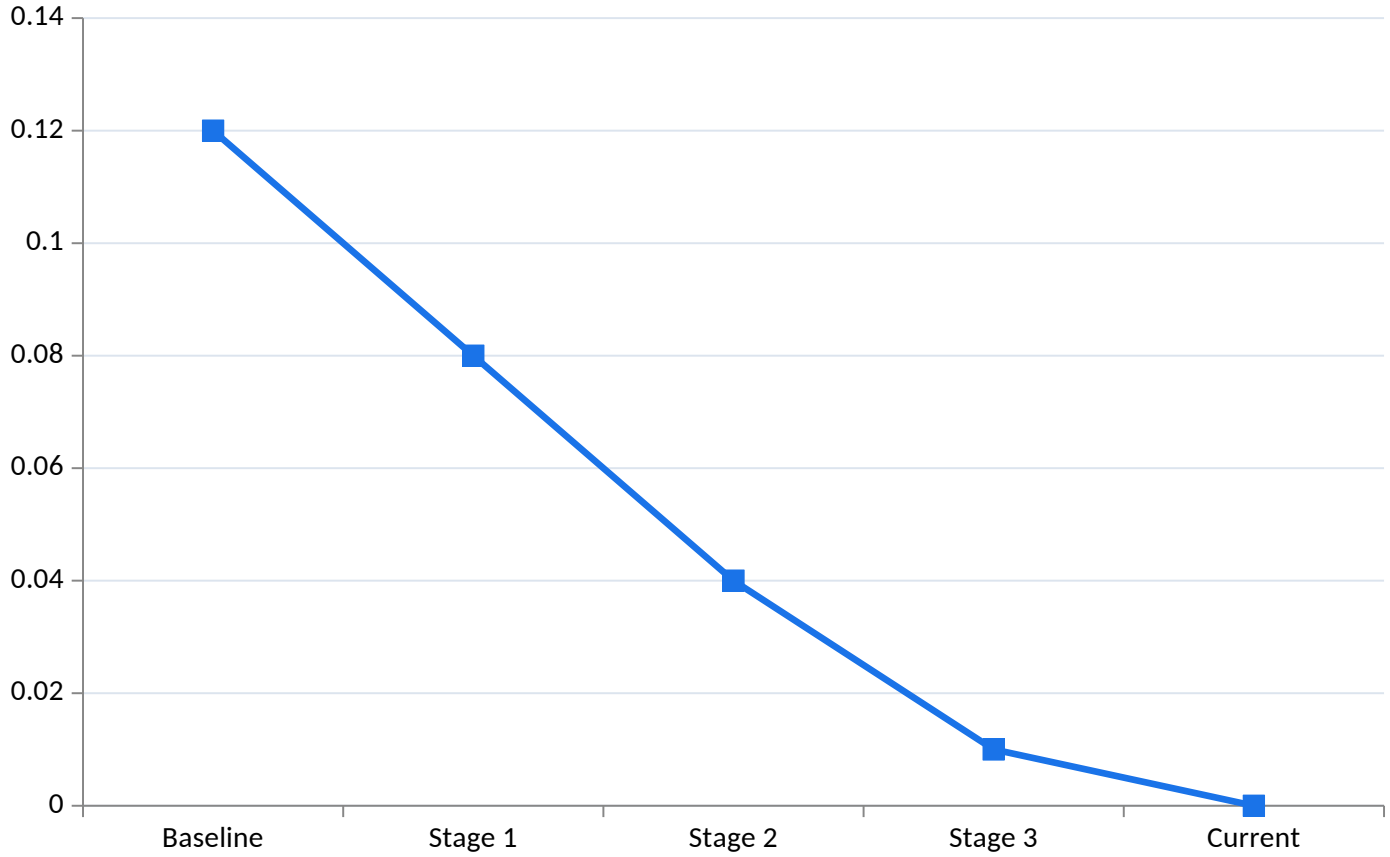
```
gcloud dlp jobs create inspect pii-scan --location=global \
+  --storage-config-file=storage.json --inspect-config-
file=inspect.json
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Privacy and sensitive-data controls — metric evidence



PII exposure rate

Formula

```
exposure =
unprotected_sensitive_fi
elds / sensitive_fields
```

Interpret trend, threshold and uncertainty before acting.

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Privacy and sensitive-data controls — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Open bucket	inherited broad role	least privilege and policy scan
PII in notebook	local export	managed storage and VPC controls
Log leakage	payload logging	redact before logging

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Privacy and sensitive-data controls — evidence chain



A valid production decision is reproducible from the complete chain.

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Privacy and sensitive-data controls — exam decision

SCENARIO

A healthcare team shares a training dataset across projects while patient identifiers must not leave a boundary.

BEST DECISION

De-identify, tokenize approved join keys, restrict perimeters and verify access in audit logs.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 01 evidence — Create a Safe Google Cloud Training Environment

ACTIVITY 01

Create a Google account, Cloud project, billing budget, IAM boundary and Cloud Shell workspace.

Evidence: activity artifacts and acceptance checks

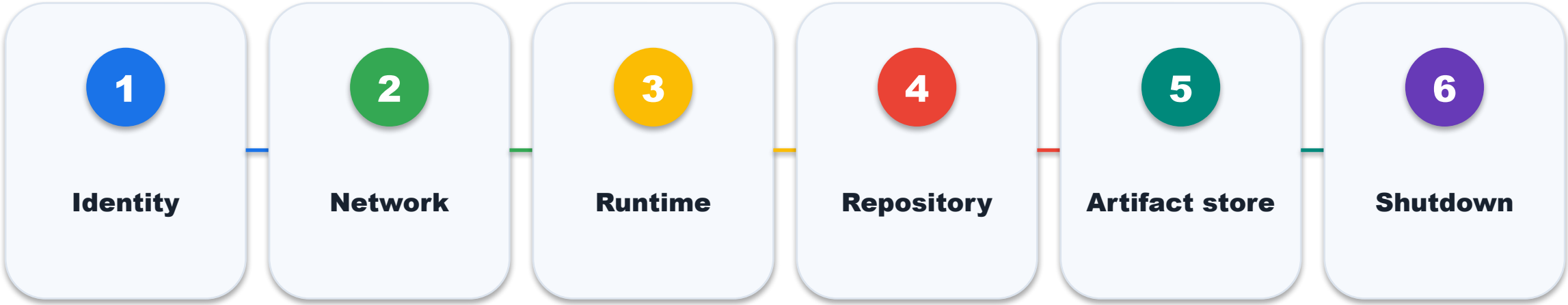
Detailed procedures: Learner Guide + activity README/PDF

- `budget-policy.json`
- `check_environment.sh`
- `iam-role-matrix.csv`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Secure Workbench and Colab Enterprise — mechanism map



Control point Decision evidence must bind identity to shutdown through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Secure Workbench and Colab Enterprise — service decision matrix

OPTION 1

Workbench managed notebook

Use when

integrated Vertex AI and controlled project

OPTION 2

Colab Enterprise

Use when

managed collaborative notebooks

OPTION 3

Local notebook

Use when

only for non-sensitive lightweight work

Exam cue: select from workload evidence—not from product familiarity.

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Secure Workbench and Colab Enterprise — runnable configuration

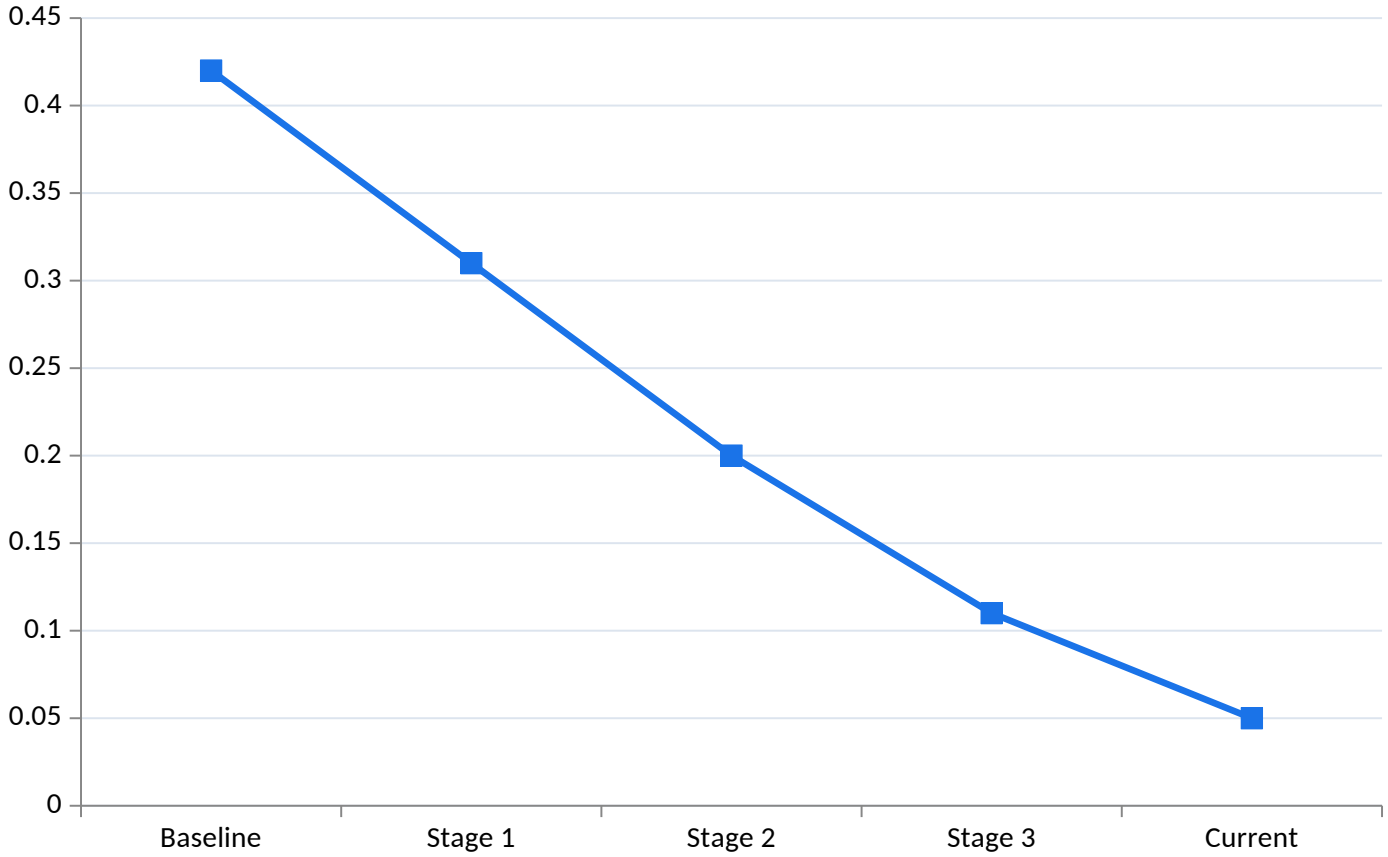
```
disable_public_ip: true
service_account: ml-dev@$PROJECT.iam.gserviceaccount.com
idle_shutdown_minutes: 60
repository: approved-ml-repo
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Secure Workbench and Colab Enterprise — metric evidence



Idle compute waste

Formula

```
waste =
idle_instance_hours /
total_instance_hours
```

Interpret trend, threshold and uncertainty before acting.

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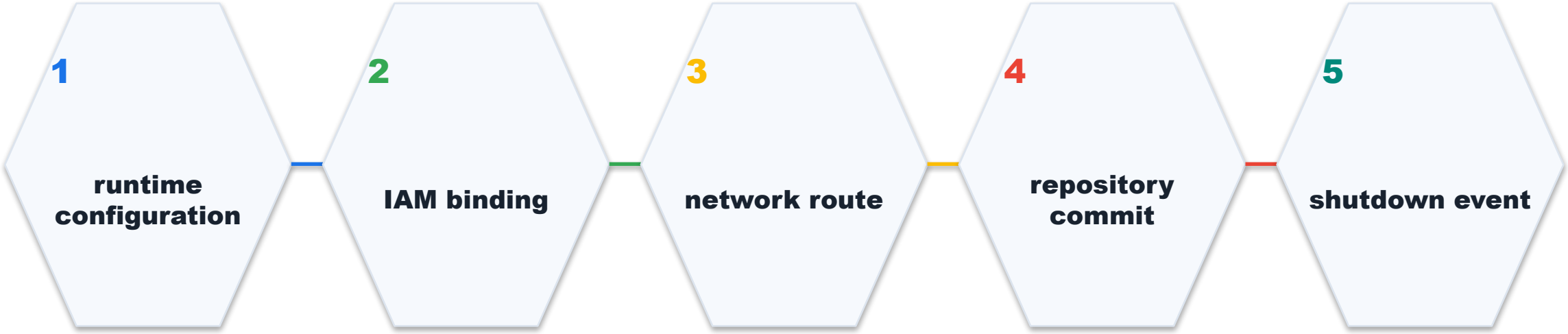
Secure Workbench and Colab Enterprise — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Credential sprawl	user keys in cells	service account and ADC
Data exfiltration	public egress	private IP and perimeter
Cost leak	notebook left on	idle shutdown

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Secure Workbench and Colab Enterprise — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 2 · SCENARIO

Secure Workbench and Colab Enterprise — exam decision

SCENARIO

A regulated analytics team needs collaborative GPUs without public IPs or downloaded service-account keys.

BEST DECISION

Use managed notebooks with private networking, attached least-privilege identity and idle shutdown.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 01 evidence — Create a Safe Google Cloud Training Environment

ACTIVITY 01

Create a Google account, Cloud project, billing budget, IAM boundary and Cloud Shell workspace.

Evidence: activity artifacts and acceptance checks

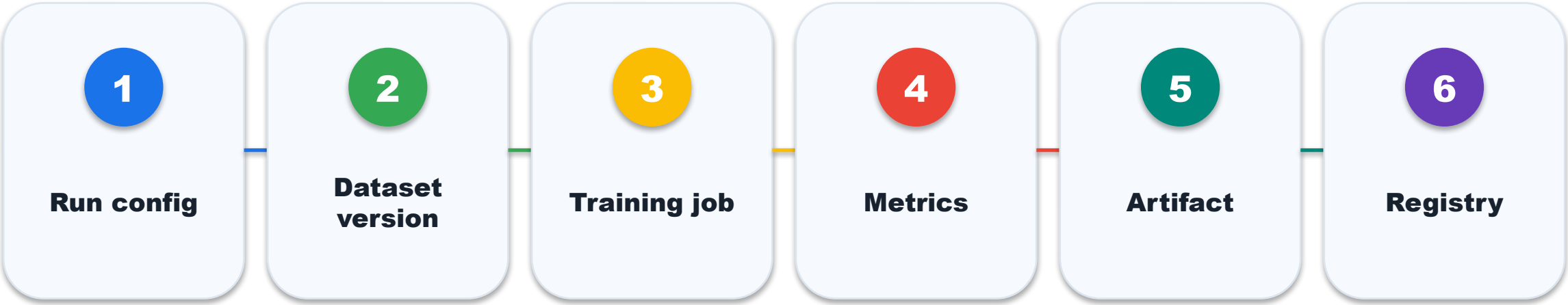
Detailed procedures: Learner Guide + activity README/PDF

- `budget-policy.json`
- `check_environment.sh`
- `iam-role-matrix.csv`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Experiment tracking and lineage — mechanism map



Control point Decision evidence must bind run config to registry through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Experiment tracking and lineage — service decision matrix

OPTION 1

Vertex AI Experiments

Use when

managed run comparison

OPTION 2

ML Metadata

Use when

pipeline lineage

OPTION 3

Git + manifest

Use when

code and configuration provenance

Exam cue: select from workload evidence—not from product familiarity.

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Experiment tracking and lineage — runnable configuration

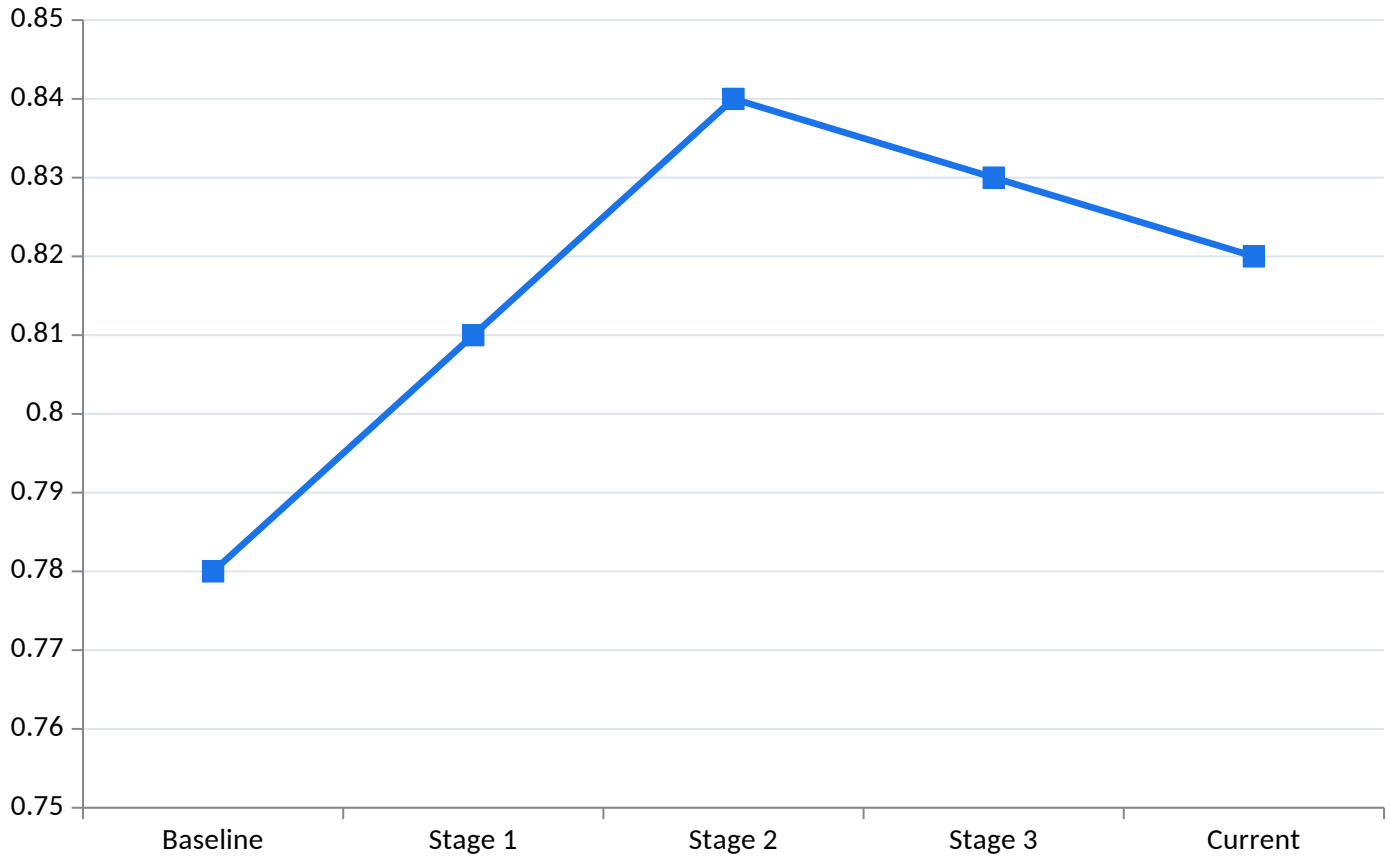
```
run.log_params({'lr':0.001,'batch_size':128,'data_uri':DATA
_URI})
run.log_metrics({'val_auc':0.842,'val_loss':0.391})
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Experiment tracking and lineage — metric evidence



Validation AUC by run

Formula

```
best = argmax(metric subject to fairness and latency gates)
```

Interpret trend, threshold and uncertainty before acting.

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Experiment tracking and lineage — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Unrepeatable run	missing dataset version	immutable URI and manifest
Metric cherry-pick	no primary metric	pre-register gates
Artifact orphan	manual upload	pipeline metadata

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Experiment tracking and lineage — evidence chain



A valid production decision is reproducible from the complete chain.

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Experiment tracking and lineage — exam decision

SCENARIO

Two teams report different best models because they trained on mutable tables and recorded only accuracy.

BEST DECISION

Bind runs to immutable data and code, predefine metrics, and compare through one experiment tracker.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 09 evidence — Analyze Vertex AI Hyperparameter Trials

ACTIVITY 09

Define a Vizier-style search space and select a trial using quality and cost evidence.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `trials.csv`
- `tuning_config.yaml`
- `select_trial.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Predictive and gen AI evaluation — mechanism map



Control point Decision evidence must bind task rubric to release gate through versioned artifacts.

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DOMAIN 2 · DECISION RULE

Predictive and gen AI evaluation — service decision matrix

OPTION 1

Classification metrics

Use when

labeled predictive outcomes

OPTION 2

LLM-as-judge

Use when

scaled rubric scoring with calibration

OPTION 3

Human evaluation

Use when

high-stakes nuance and preference

Exam cue: select from workload evidence—not from product familiarity.

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Predictive and gen AI evaluation — runnable configuration

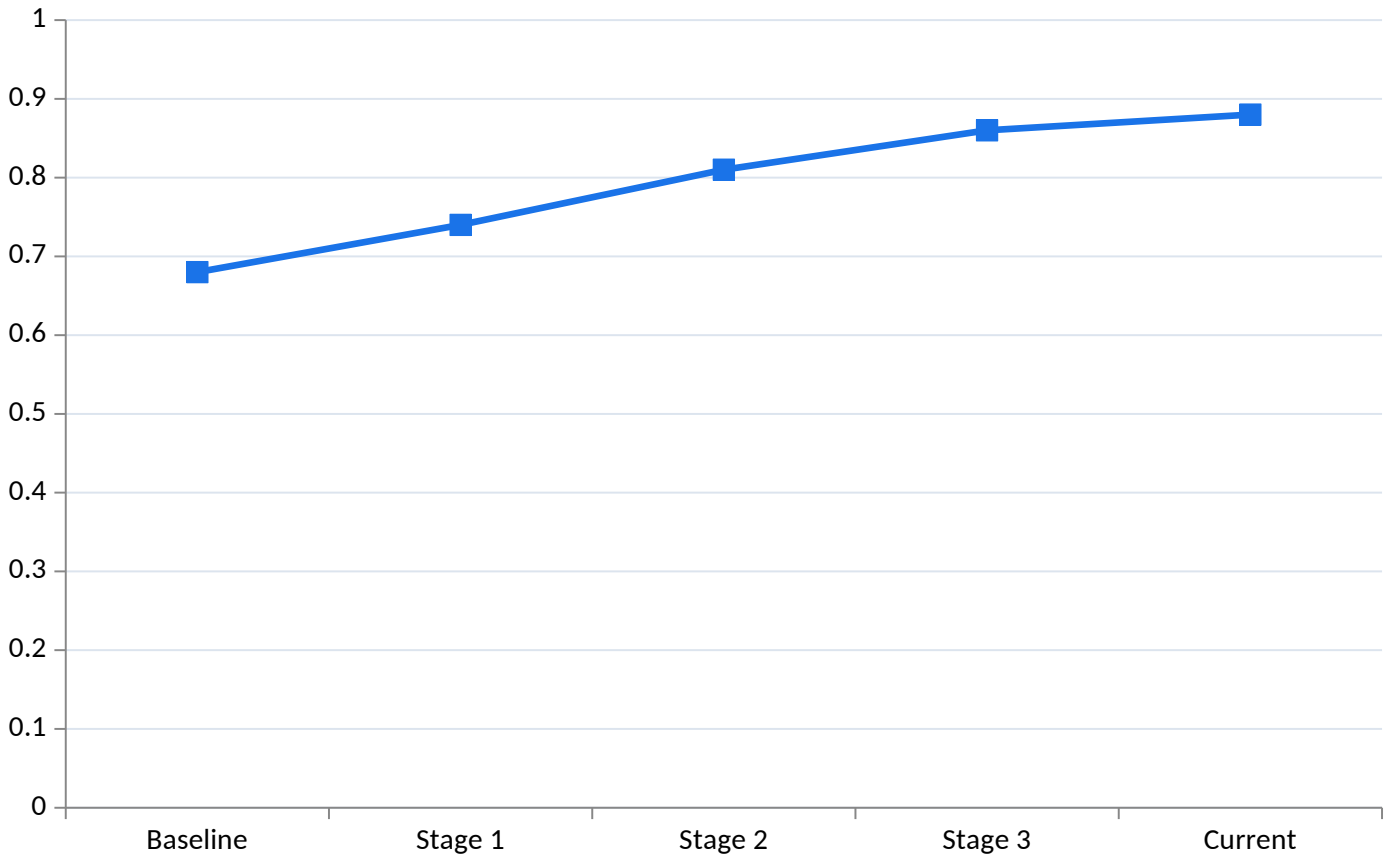
```
rubric =
{'groundedness':0.35,'correctness':0.35,'safety':0.20,'styl
e':0.10}
release_if = 'score>=0.82 and safety_violations==0'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Predictive and gen AI evaluation — metric evidence



Evaluation agreement

Formula

```
agreement =
matching_ratings /
reviewed_items
```

Interpret trend, threshold and uncertainty before acting.

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Predictive and gen AI evaluation — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Judge bias	uncalibrated model judge	human calibration sample
Contamination	eval in tuning set	holdout governance
Aggregate masking	no slices	report by task and risk

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Predictive and gen AI evaluation — evidence chain



A valid production decision is reproducible from the complete chain.

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Predictive and gen AI evaluation — exam decision

SCENARIO

A gen-AI summarizer improves ROUGE but starts omitting critical contraindications in one document type.

BEST DECISION

Use risk-weighted rubric, document-type slices and human adjudication; block release on critical omissions.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `baseline.csv`
- `current.csv`
- `monitor.py`
- `incident_template.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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DOMAIN 3 · ~21%

Scaling prototypes into ML models

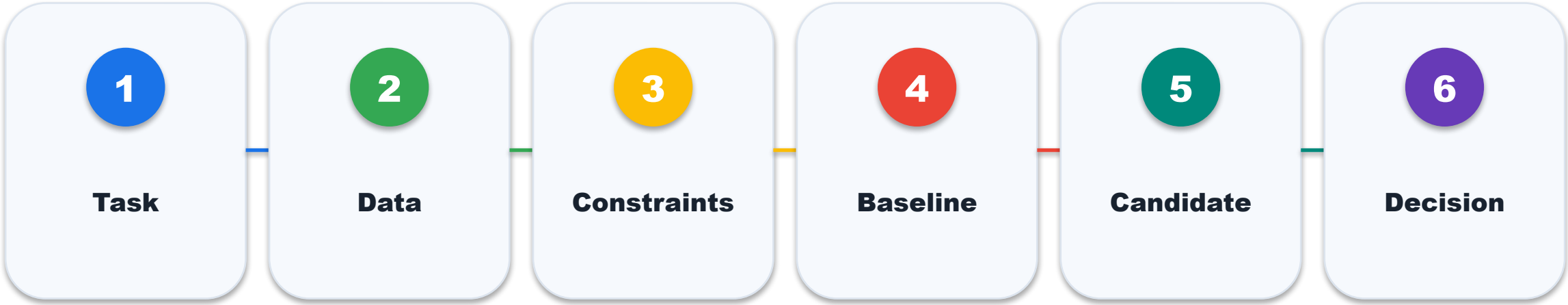
Topics 4–6: TensorFlow, feature engineering, enterprise ML

8 technical mechanisms · 64 anchored slides



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Model type and product decision — mechanism map



Control point Decision evidence must bind task to decision through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Model type and product decision — service decision matrix

OPTION 1

ARIMA_PLUS

Use when

interpretable time-series forecast

OPTION 2

DNN

Use when

large nonlinear supervised task

OPTION 3

LLM

Use when

language generation or reasoning

Exam cue: select from workload evidence—not from product familiarity.

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Model type and product decision — runnable configuration

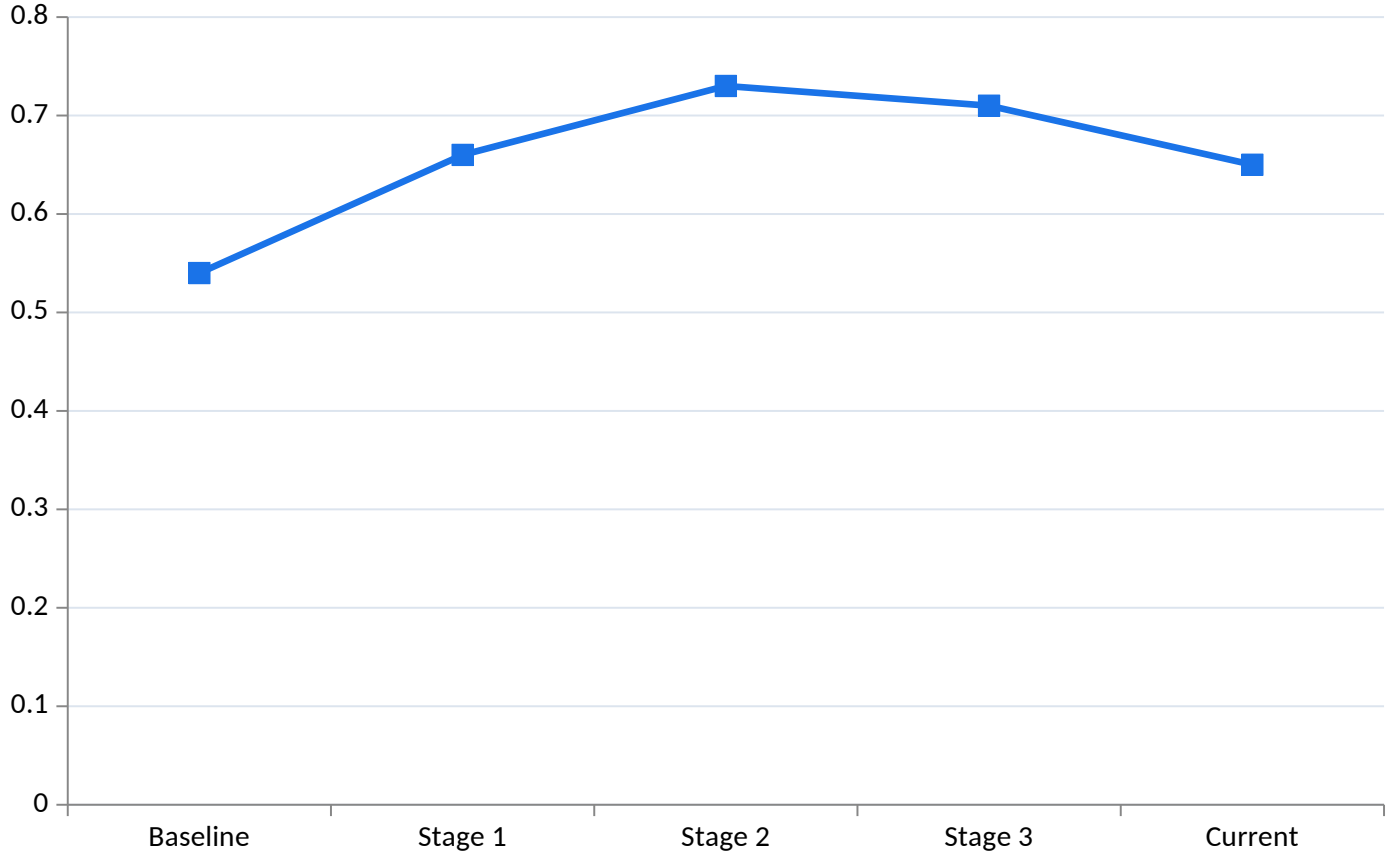
```
decision =
{'task':'forecast','horizon_days':30,'need_intervals':True,
            'product':'BigQuery ML ARIMA_PLUS'}
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Model type and product decision — metric evidence



Decision utility

Formula

```
utility = quality -
cost_penalty -
latency_penalty -
risk_penalty
```

Interpret trend, threshold and uncertainty before acting.

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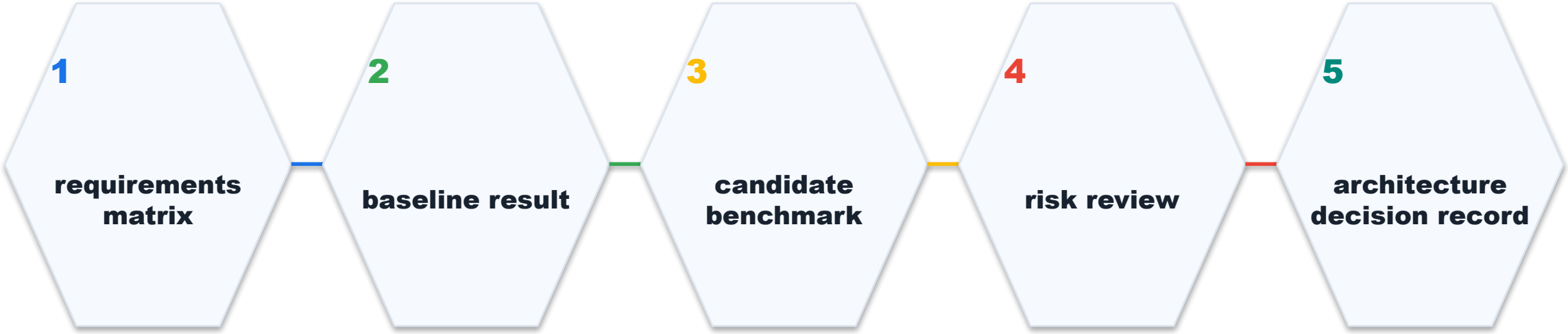
Model type and product decision — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Over-complex model	baseline omitted	establish simple baseline
Wrong metric	business loss ignored	map metric to decision
Unclear ownership	product mismatch	operating-model review

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Model type and product decision — evidence chain



A valid production decision is reproducible from the complete chain.

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Model type and product decision — exam decision

SCENARIO

A planner needs 30-day demand intervals from structured history and must explain seasonality.

BEST DECISION

Choose ARIMA_PLUS in BigQuery ML before a DNN; validate interval coverage and holiday effects.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 02 evidence — Draft an ML Architecture Decision Record

ACTIVITY 02

Convert business volume, latency, security and explainability needs into a service decision.

Evidence: activity artifacts and acceptance checks

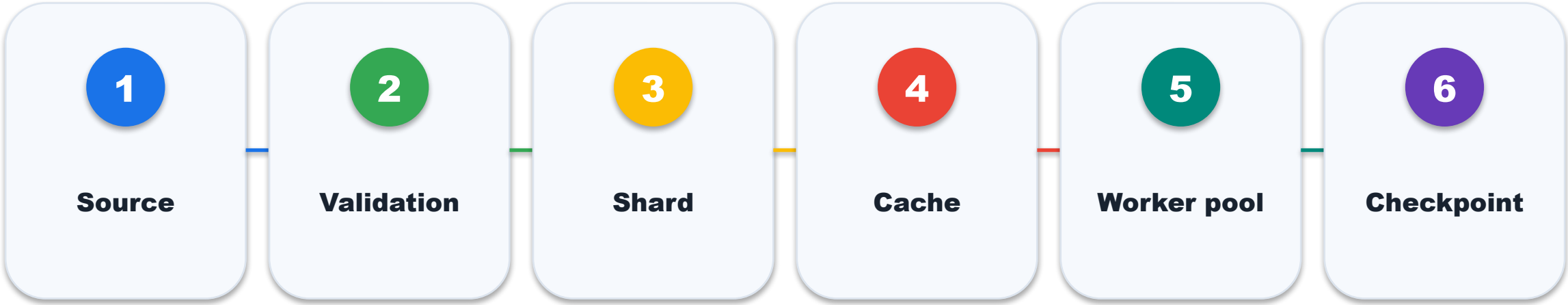
Detailed procedures: Learner Guide + activity README/PDF

- `requirements.csv`
- `architecture_decision.py`
- `expected_decision.json`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Training data ingestion — mechanism map



Control point Decision evidence must bind source to checkpoint through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Training data ingestion — service decision matrix

OPTION 1

Cloud Storage shards

Use when

images, text and TFRecord

OPTION 2

BigQuery export

Use when

warehouse-managed tabular data

OPTION 3

Streaming ingest

Use when

continuous examples with later snapshot

Exam cue: select from workload evidence—not from product familiarity.

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Training data ingestion — runnable configuration

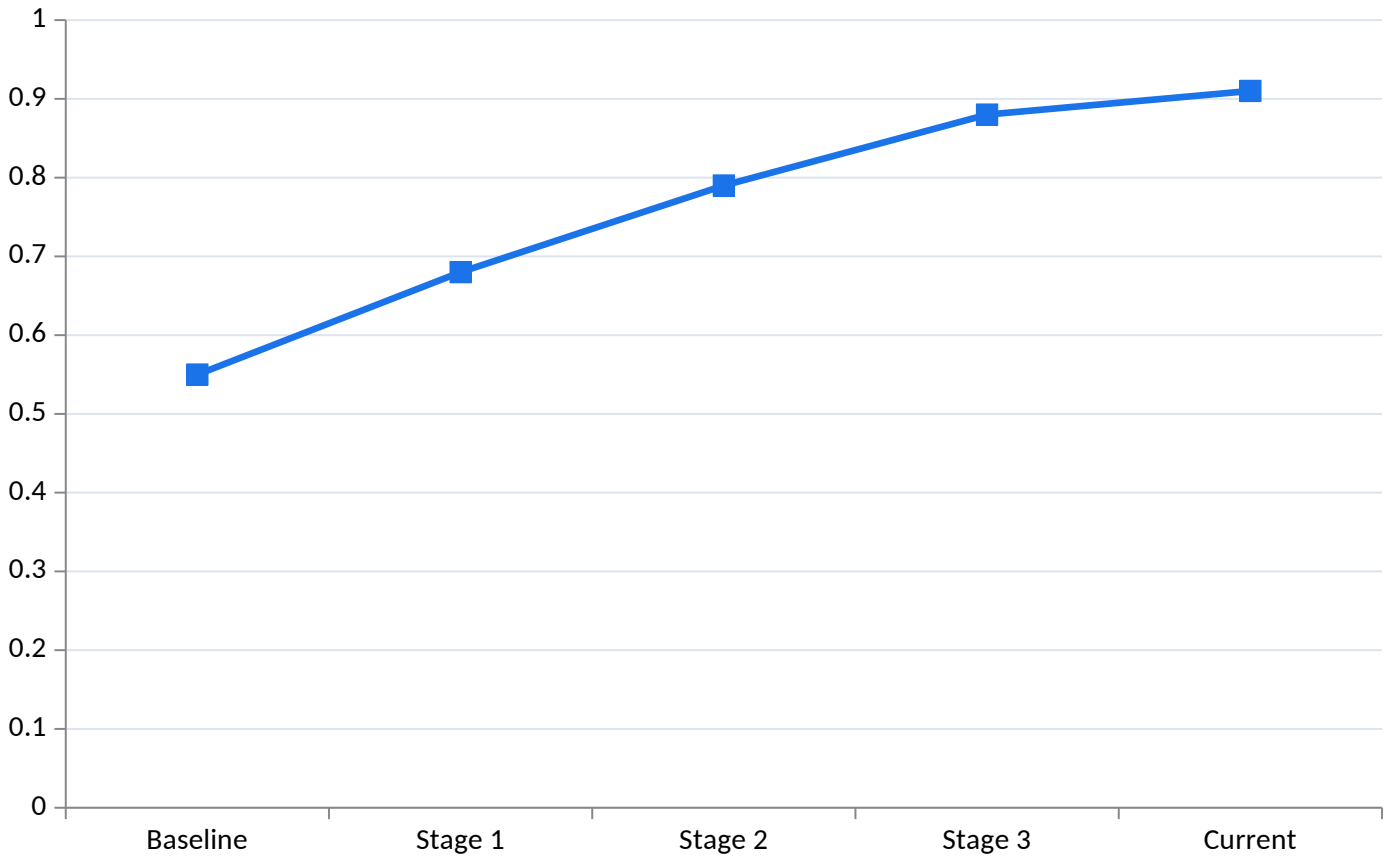
```
gcloud storage cp gs://$BUCKET/manifests/train-*.jsonl
./manifests/
python validate_manifest.py --schema schema.json --input
manifests/
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Training data ingestion — metric evidence



Input pipeline utilization

Formula

```
utilization =
data_wait_free_step_time
/ total_step_time
```

Interpret trend, threshold and uncertainty before acting.

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Training data ingestion — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Workers starve	tiny unshuffled files	shard and parallel-read
Corrupt examples	no validation	fail-fast schema checks
Nonrepeatable split	random at runtime	persist split manifest

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Training data ingestion — evidence chain



A valid production decision is reproducible from the complete chain.

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Training data ingestion — exam decision

SCENARIO

A vision job uses 12 million images and GPUs idle while workers open millions of tiny files.

BEST DECISION

Create balanced shards, parallel reads and prefetch; measure accelerator utilization and input wait.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 10 evidence — Plan Distributed PyTorch GPU Training

ACTIVITY 10

Configure torchrun-style workers, checkpoints and scaling-efficiency diagnostics.

Evidence: activity artifacts and acceptance checks

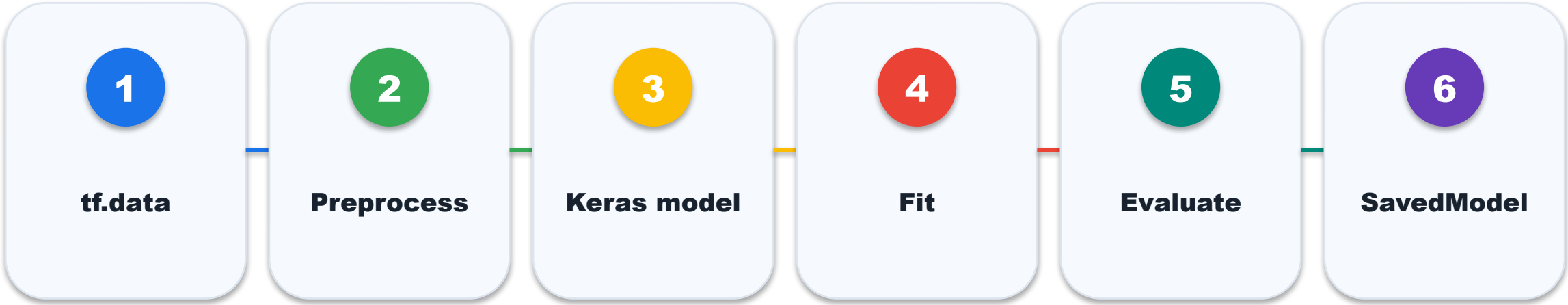
Detailed procedures: Learner Guide + activity README/PDF

- `train_distributed.py`
- `Dockerfile`
- `worker_pool.yaml`
- `scaling_results.csv`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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TensorFlow and Keras training — mechanism map



Control point Decision evidence must bind tf.data to savedmodel through versioned artifacts.

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DOMAIN 3 · DECISION RULE

TensorFlow and Keras training — service decision matrix

OPTION 1

Sequential API

Use when

single-input linear stack

OPTION 2

Functional API

Use when

multi-input or branching graph

OPTION 3

Subclassing

Use when

custom behavior requiring code control

Exam cue: select from workload evidence—not from product familiarity.

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TensorFlow and Keras training — runnable configuration

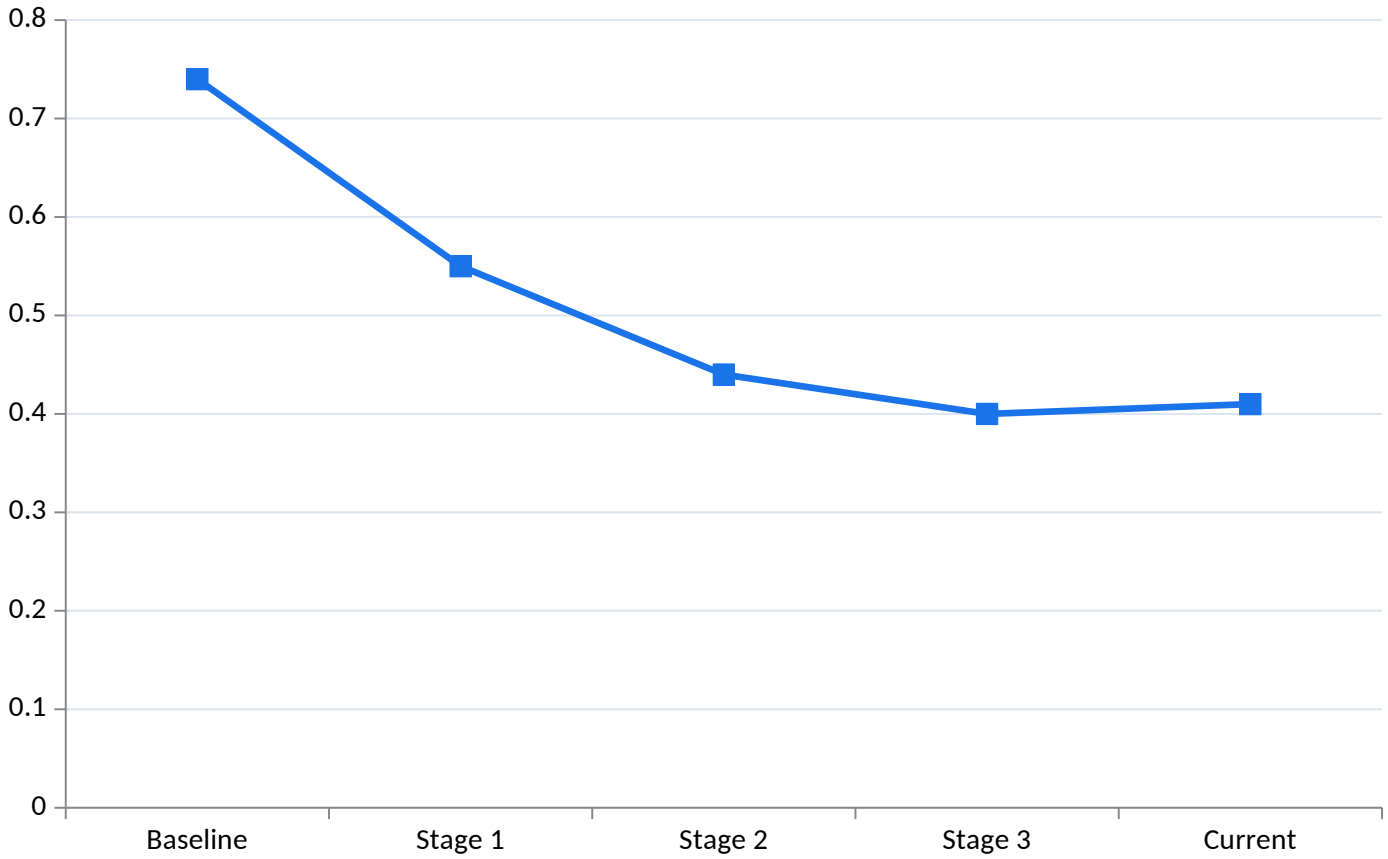
```
ds =
tf.data.Dataset.from_tensor_slices((X,y)).shuffle(10000).ba
tch(128).prefetch(tf.data.AUTOTUNE)
model.fit(ds, epochs=20, callbacks=[early_stop])
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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TensorFlow and Keras training — metric evidence



Validation loss

Formula

```
stop when val_loss fails
to improve for patience
epochs
```

Interpret trend, threshold and uncertainty before acting.

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TensorFlow and Keras training — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Overfit	training continues after val plateau	early stopping
Slow input	no prefetch	AUTOTUNE pipeline
Serving failure	custom op missing	export and load test

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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TensorFlow and Keras training — evidence chain



A valid production decision is reproducible from the complete chain.

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TensorFlow and Keras training — exam decision

SCENARIO

A tabular neural network improves training loss but validation loss rises after epoch 12.

BEST DECISION

Use early stopping and regularization, then verify the exported signature with held-out examples.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 07 evidence — Train a TensorFlow/Keras Model

ACTIVITY 07

Use tf.data, a Keras network, early stopping and SavedModel verification.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `train_tensorflow.py`
- `requirements.txt`
- `expected_history.csv`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Vertex AI custom training — mechanism map



Control point Decision evidence must bind package/container to model registry through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Vertex AI custom training — service decision matrix

OPTION 1

Prebuilt container

Use when

supported framework and conventional entry point

OPTION 2

Custom container

Use when

system packages or custom runtime

OPTION 3

Custom training package

Use when

Python package on managed image

Exam cue: select from workload evidence—not from product familiarity.

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Vertex AI custom training — runnable configuration

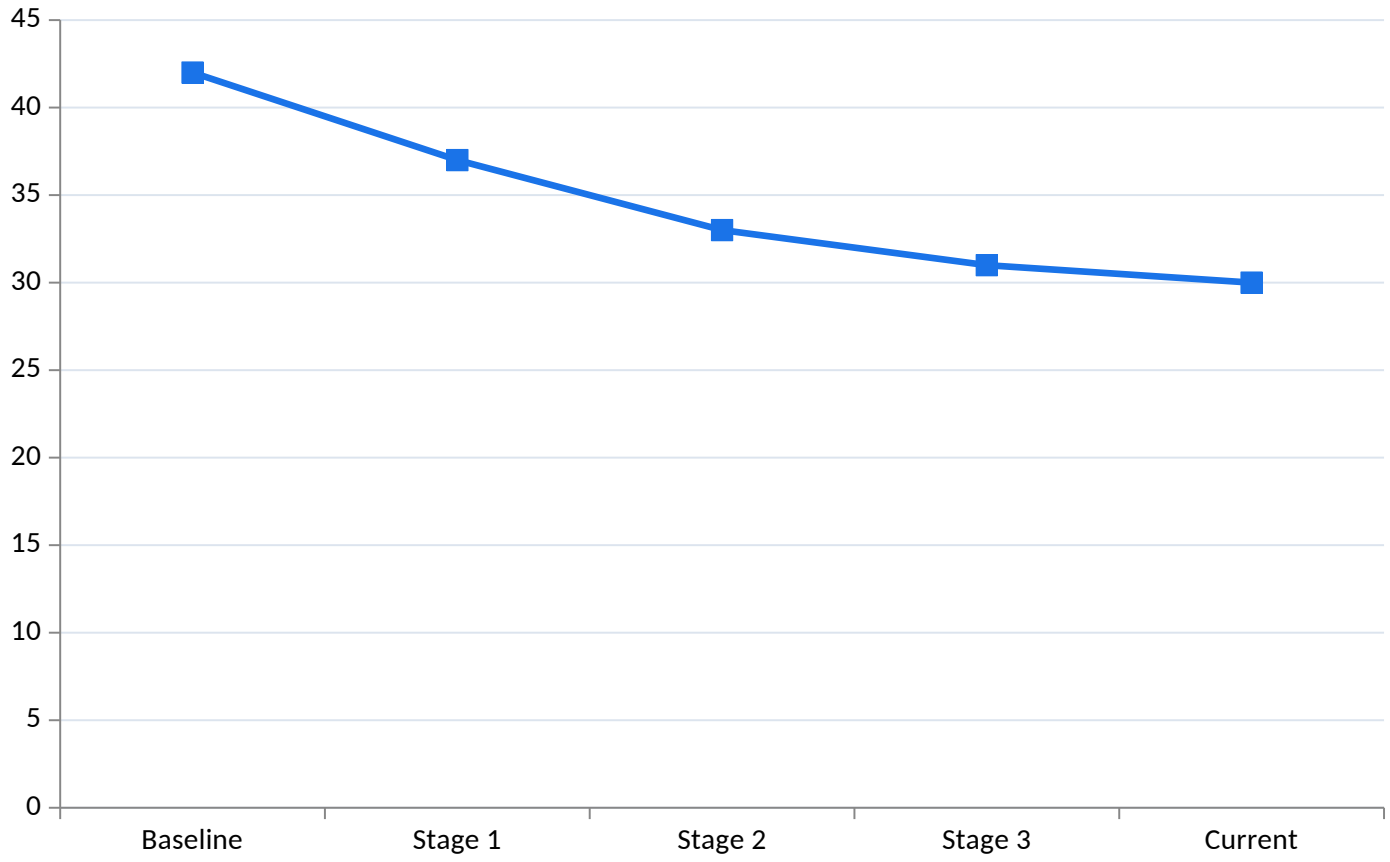
```
gcloud ai custom-jobs create --region=$REGION --display-
name=churn-train \
+ --config=config/custom_job.yaml
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Vertex AI custom training — metric evidence



Training cost per successful run

Formula

```
cost_success =
total_job_cost /
successful_jobs
```

Interpret trend, threshold and uncertainty before acting.

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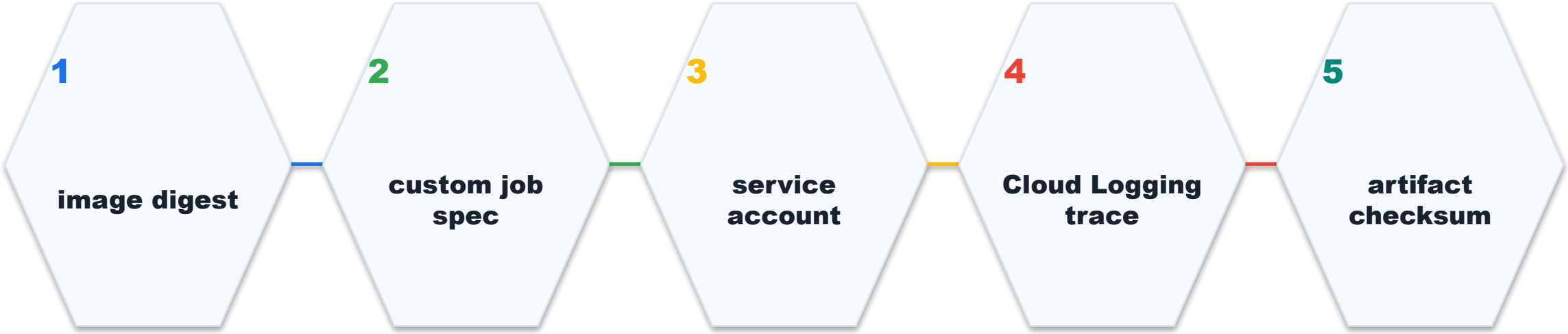
Vertex AI custom training — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Module not found	package omitted dependency	lock and build image
Permission denied	training SA lacks access	least-privilege roles
No artifact	wrong AIP_MODEL_DIR	write to managed output URI

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Vertex AI custom training — evidence chain



A valid production decision is reproducible from the complete chain.

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Vertex AI custom training — exam decision

SCENARIO

A sklearn pipeline requires a native library unavailable in Vertex prebuilt images.

BEST DECISION

Use a custom container pinned by digest and test it locally before CustomJob submission.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 08 evidence — Package a Vertex AI Custom Training Job

ACTIVITY 08

Build a reproducible training package and container with managed output paths.

Evidence: activity artifacts and acceptance checks

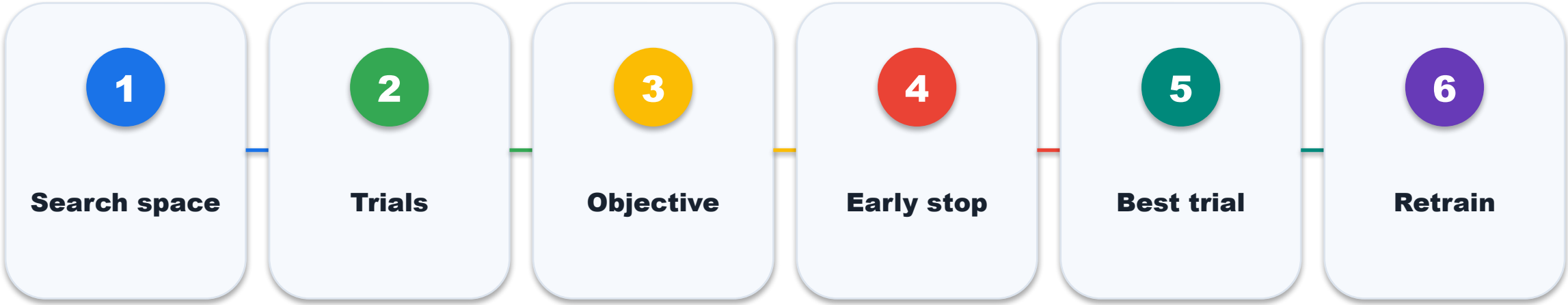
Detailed procedures: Learner Guide + activity README/PDF

- `train.py`
- `Dockerfile`
- `custom_job.yaml`
- `requirements.txt`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Hyperparameter tuning — mechanism map



Control point Decision evidence must bind search space to retrain through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Hyperparameter tuning — service decision matrix

OPTION 1

Grid

Use when

small discrete exhaustive space

OPTION 2

Random

Use when

broad mixed space

OPTION 3

Bayesian/Vizier

Use when

expensive trials with guided search

Exam cue: select from workload evidence—not from product familiarity.

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Hyperparameter tuning — runnable configuration

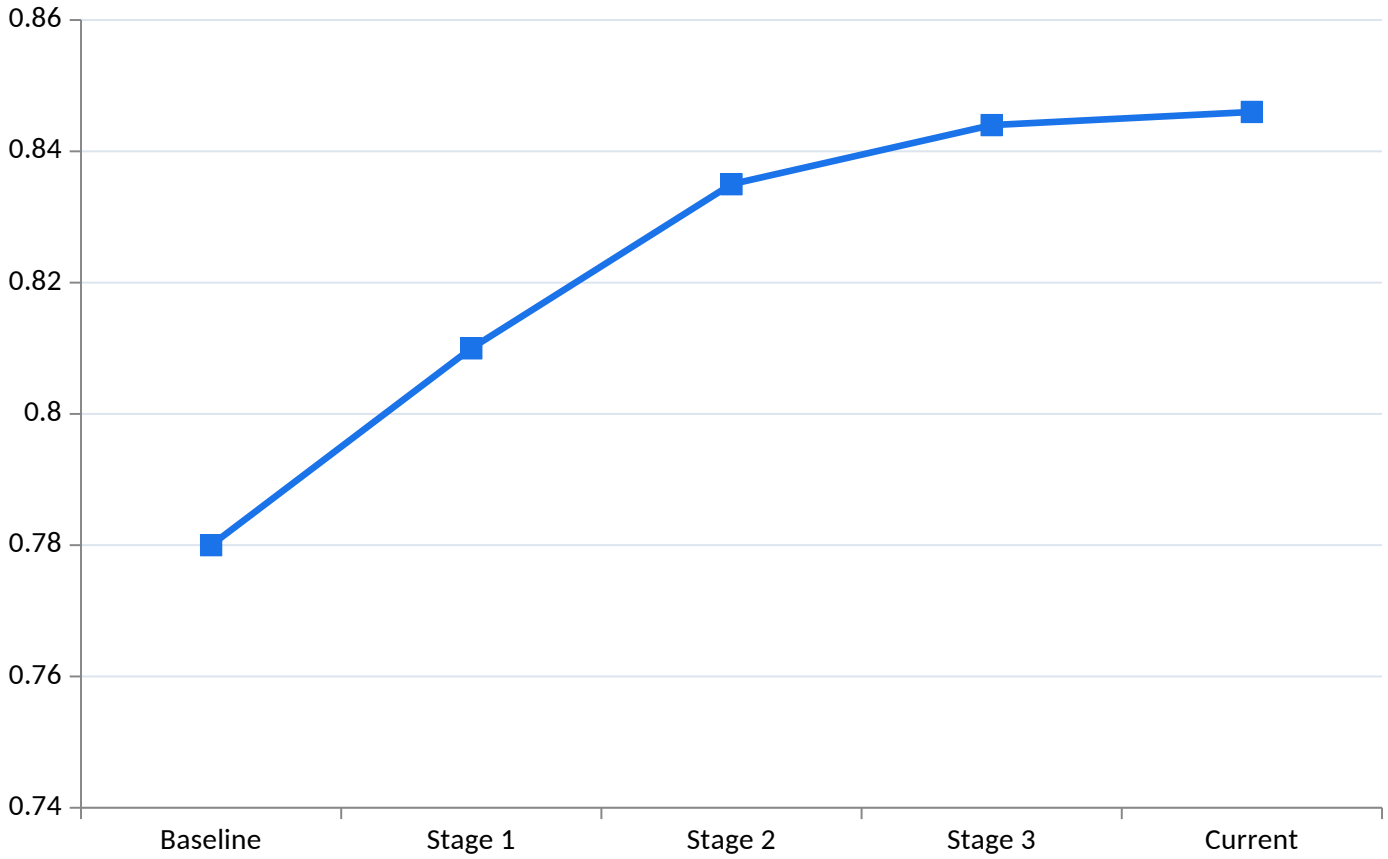
```
metric_spec: {val_auc: maximize}
parameters:
- {parameterId: learning_rate, doubleValueSpec: {minValue:
0.0001, maxValue: 0.1, scaleType: UNIT_LOG_SCALE}}
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Hyperparameter tuning — metric evidence



Best objective by trial

Formula

```
best_n =
max(metric_1...metric_n)
```

Interpret trend, threshold and uncertainty before acting.

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Hyperparameter tuning — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Wrong objective	metric name mismatch	emit exact metric
Budget waste	poor trials run fully	early stopping
Overfit to validation	too many searches	final untouched test

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Hyperparameter tuning — evidence chain



A valid production decision is reproducible from the complete chain.

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Hyperparameter tuning — exam decision

SCENARIO

A DNN training run costs $80; the team needs learning rate, depth and dropout optimization.

BEST DECISION

Use Vizier/Bayesian search with log-scaled learning rate, early stopping and an untouched final test.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 09 evidence — Analyze Vertex AI Hyperparameter Trials

ACTIVITY 09

Define a Vizier-style search space and select a trial using quality and cost evidence.

Evidence: activity artifacts and acceptance checks

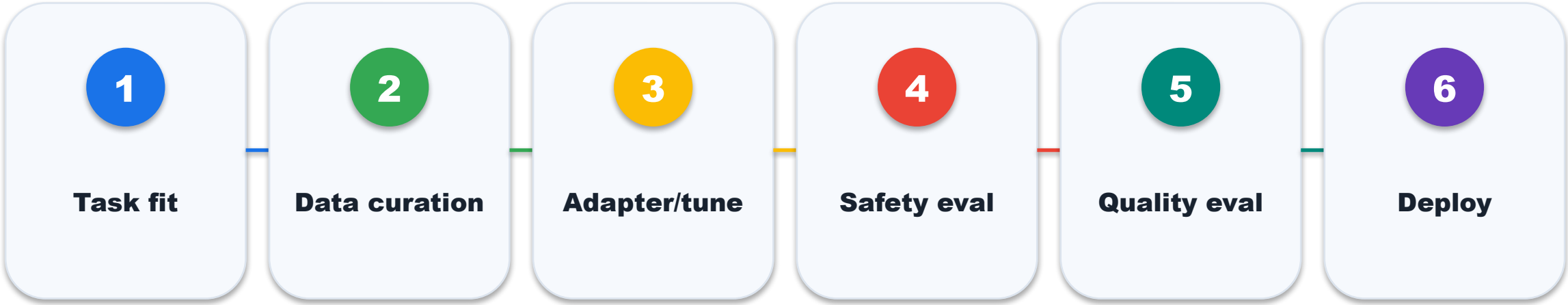
Detailed procedures: Learner Guide + activity README/PDF

- `trials.csv`
- `tuning_config.yaml`
- `select_trial.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Foundation model fine-tuning — mechanism map



Control point Decision evidence must bind task fit to deploy through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Foundation model fine-tuning — service decision matrix

OPTION 1

Prompt/RAG

Use when

knowledge or instruction gap

OPTION 2

Adapter tuning

Use when

task style with modest data

OPTION 3

Full fine-tune

Use when

rarely justified; high control and cost

Exam cue: select from workload evidence—not from product familiarity.

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Foundation model fine-tuning — runnable configuration

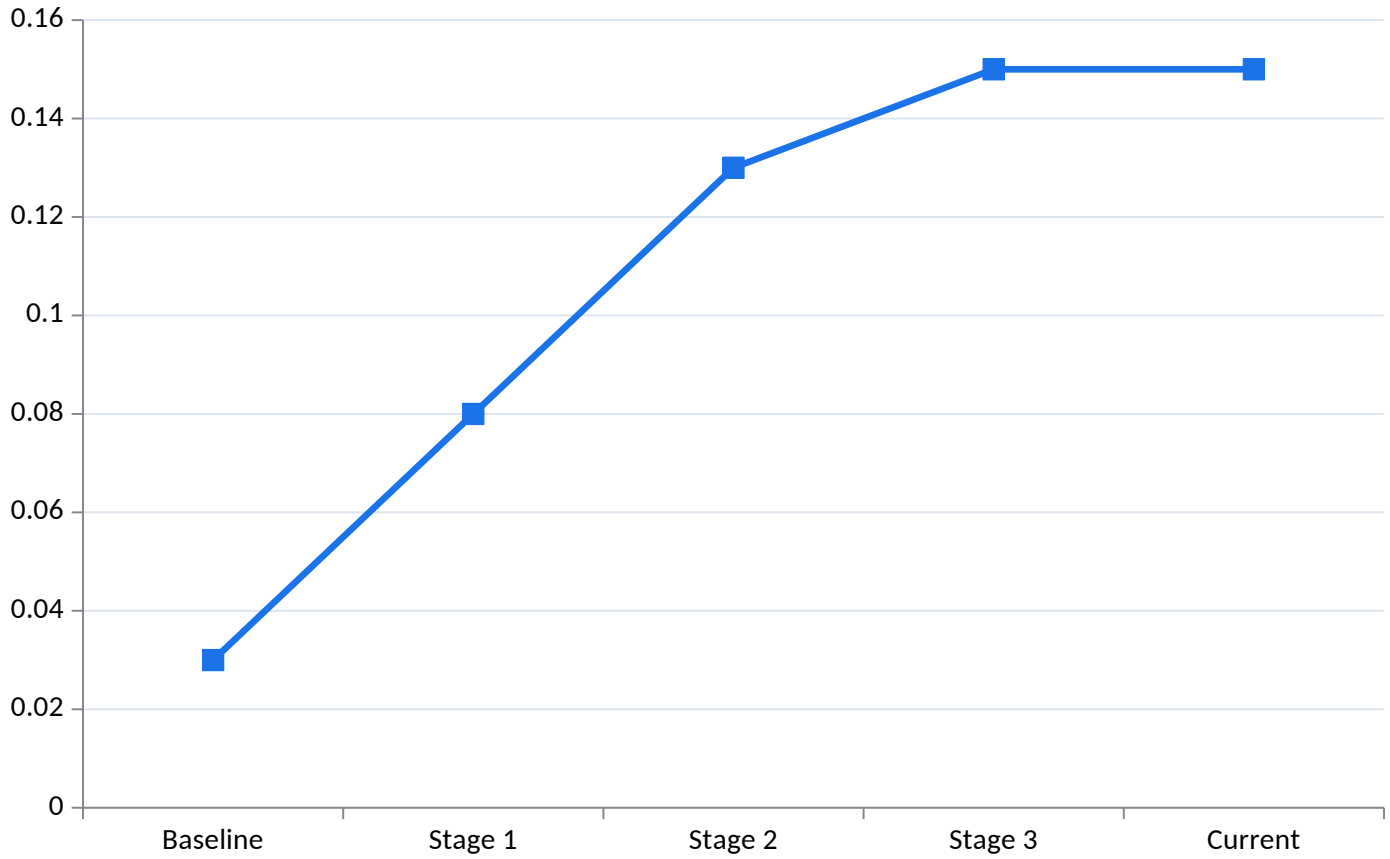
```
train.jsonl: {'input_text':'...','output_text':'...'}
validation.jsonl: {'input_text':'...','output_text':'...'}
max_duplicate_rate: 0.01
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Foundation model fine-tuning — metric evidence



Quality vs training examples

Formula

```
gain = tuned_score - base_score
```

Interpret trend, threshold and uncertainty before acting.

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Foundation model fine-tuning — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Catastrophic behavior	poor examples	curate and safety-gate
No factual freshness	tuning used for knowledge	use retrieval
Sensitive memorization	PII in examples	DLP and dedup

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Foundation model fine-tuning — evidence chain



A valid production decision is reproducible from the complete chain.

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Foundation model fine-tuning — exam decision

SCENARIO

A service needs consistent structured incident summaries but incident facts change every minute.

BEST DECISION

Tune structure/style only; retrieve live facts and block memorization-prone sensitive examples.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

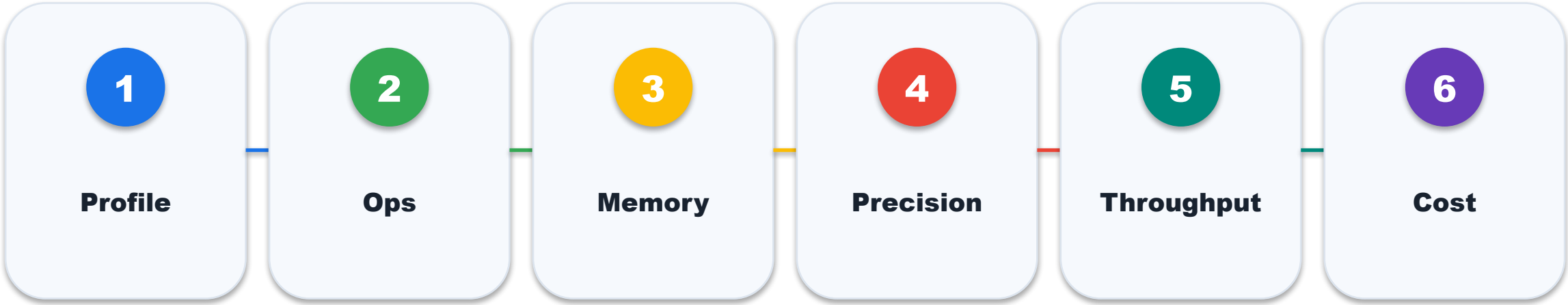
Detailed procedures: Learner Guide + activity README/PDF

- `baseline.csv`
- `current.csv`
- `monitor.py`
- `incident_template.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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CPU, GPU and TPU selection — mechanism map



Control point Decision evidence must bind profile to cost through versioned artifacts.

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DOMAIN 3 · DECISION RULE

CPU, GPU and TPU selection — service decision matrix

OPTION 1

CPU

Use when

small classical models and irregular control flow

OPTION 2

GPU

Use when

broad deep-learning ecosystem

OPTION 3

TPU

Use when

large TensorFlow/JAX workloads optimized for matrix ops

Exam cue: select from workload evidence—not from product familiarity.

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CPU, GPU and TPU selection — runnable configuration

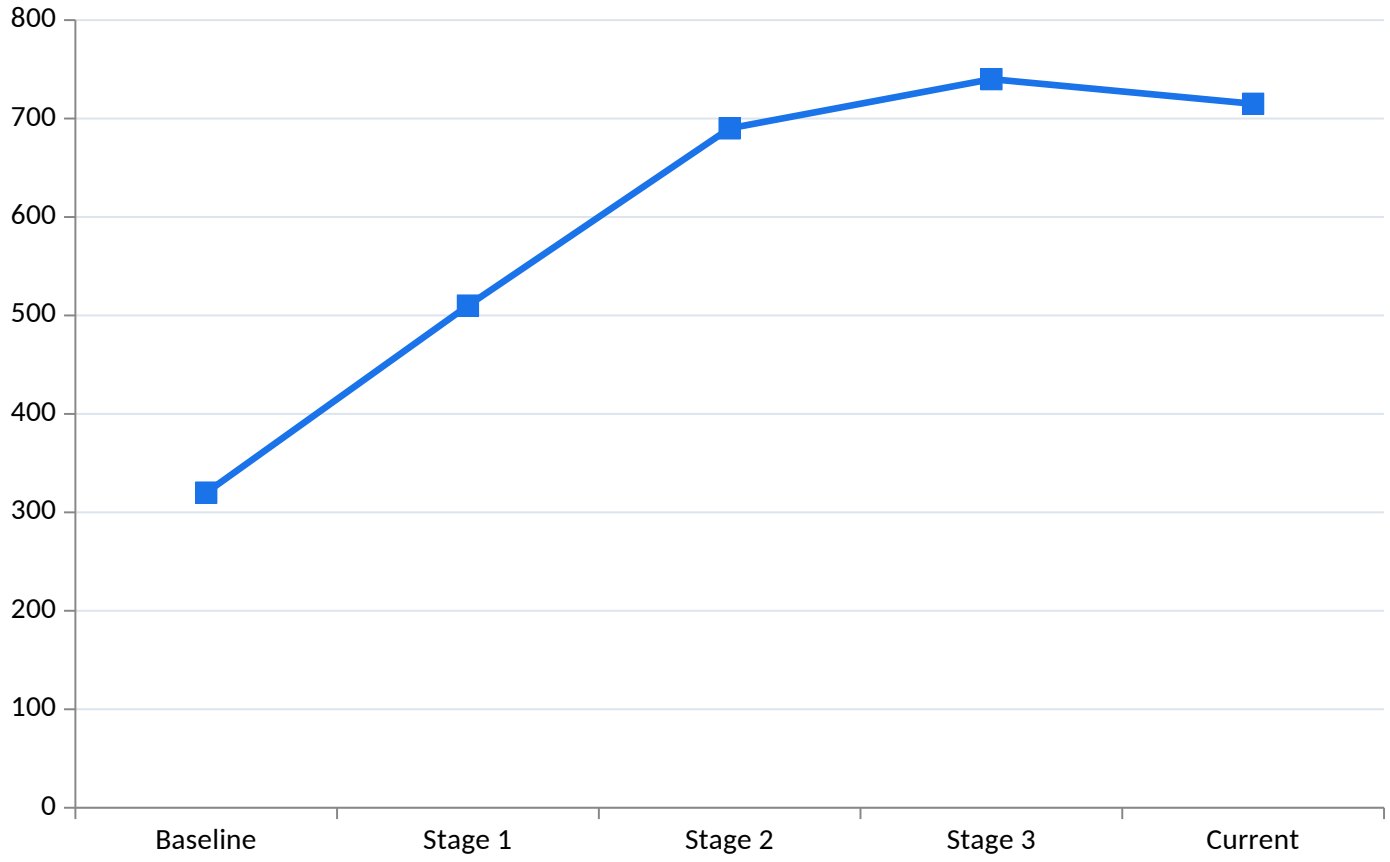
```
acceleratorType: NVIDIA_TESLA_T4
acceleratorCount: 2
machineSpec: {machineType: n1-standard-16}
replicaCount: 1
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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CPU, GPU and TPU selection — metric evidence



Examples per dollar

Formula

```
efficiency =
examples_per_second /
hourly_cost
```

Interpret trend, threshold and uncertainty before acting.

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CPU, GPU and TPU selection — failure diagnosis

Observable symptom	Likely mechanism	Control or test
OOM	batch/model exceeds memory	gradient accumulation or larger device
Low utilization	input bottleneck	profile pipeline
Quota error	accelerator unavailable	reserve quota or alternate region

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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CPU, GPU and TPU selection — evidence chain



A valid production decision is reproducible from the complete chain.

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CPU, GPU and TPU selection — exam decision

SCENARIO

A transformer fits on one GPU but throughput misses the nightly window by 3x.

BEST DECISION

Profile first; compare larger GPU and data-parallel options by examples per dollar and deadline.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 10 evidence — Plan Distributed PyTorch GPU Training

ACTIVITY 10

Configure torchrun-style workers, checkpoints and scaling-efficiency diagnostics.

Evidence: activity artifacts and acceptance checks

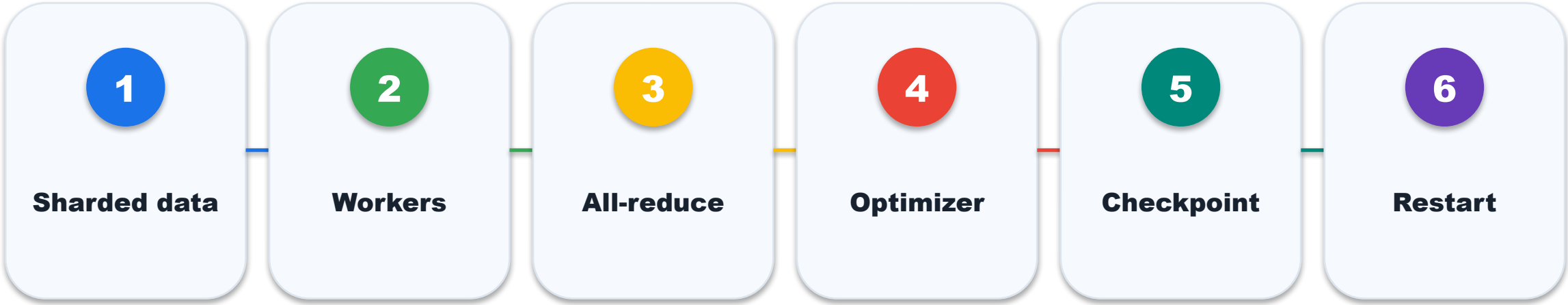
Detailed procedures: Learner Guide + activity README/PDF

- `train_distributed.py`
- `Dockerfile`
- `worker_pool.yaml`
- `scaling_results.csv`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Distributed training — mechanism map



Control point Decision evidence must bind sharded data to restart through versioned artifacts.

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DOMAIN 3 · DECISION RULE

Distributed training — service decision matrix

OPTION 1

Data parallel

Use when

replicate model; split batches

OPTION 2

Model parallel

Use when

split model too large for one device

OPTION 3

Parameter server

Use when

asynchronous large-scale pattern

Exam cue: select from workload evidence—not from product familiarity.

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Distributed training — runnable configuration

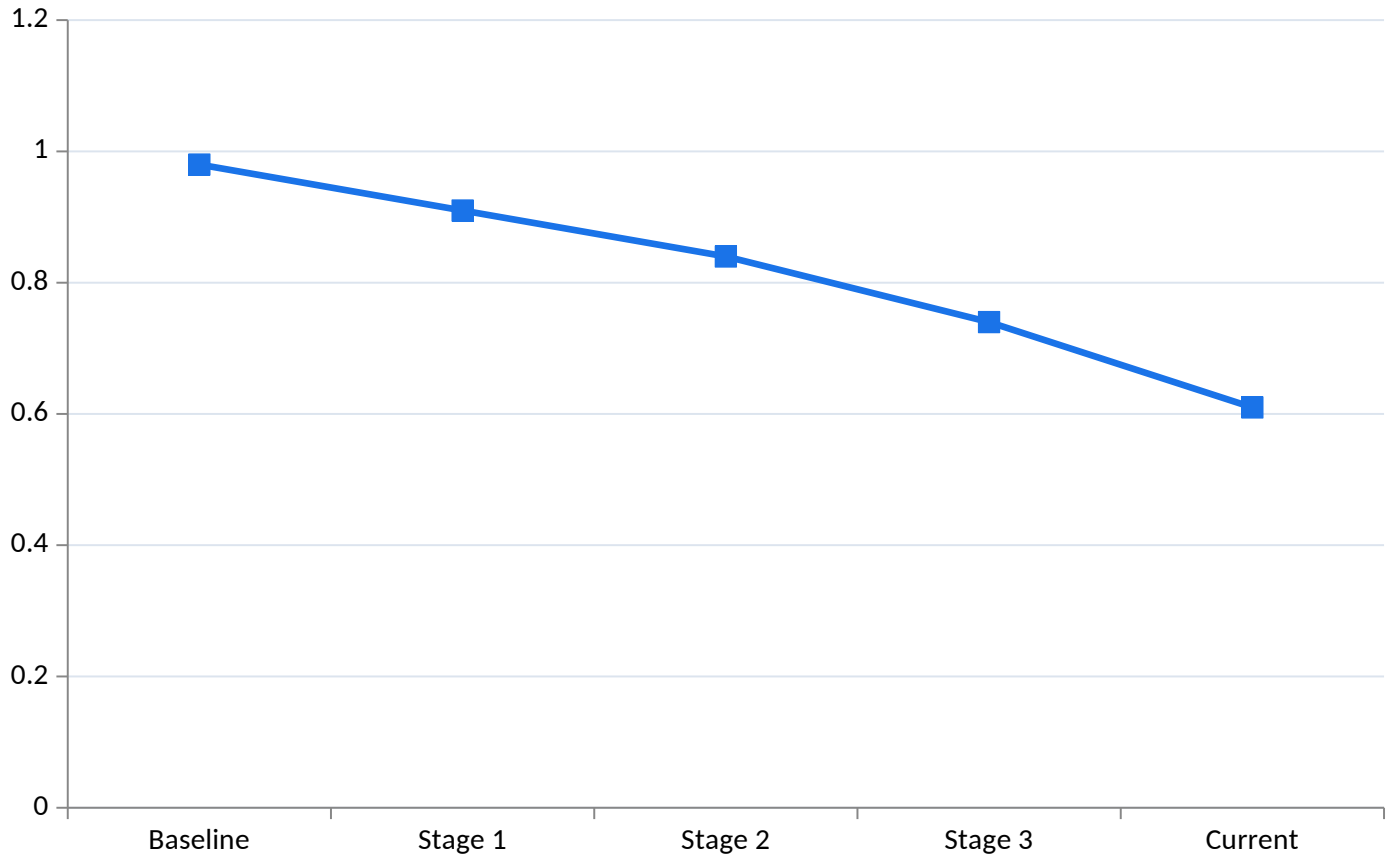
```
python -m torch.distributed.run --nproc_per_node=4 train.py
\
+  --backend nccl --checkpoint_uri gs://$BUCKET/checkpoints
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Distributed training — metric evidence



Scaling efficiency

Formula

```
efficiency = speedup / worker_count
```

Interpret trend, threshold and uncertainty before acting.

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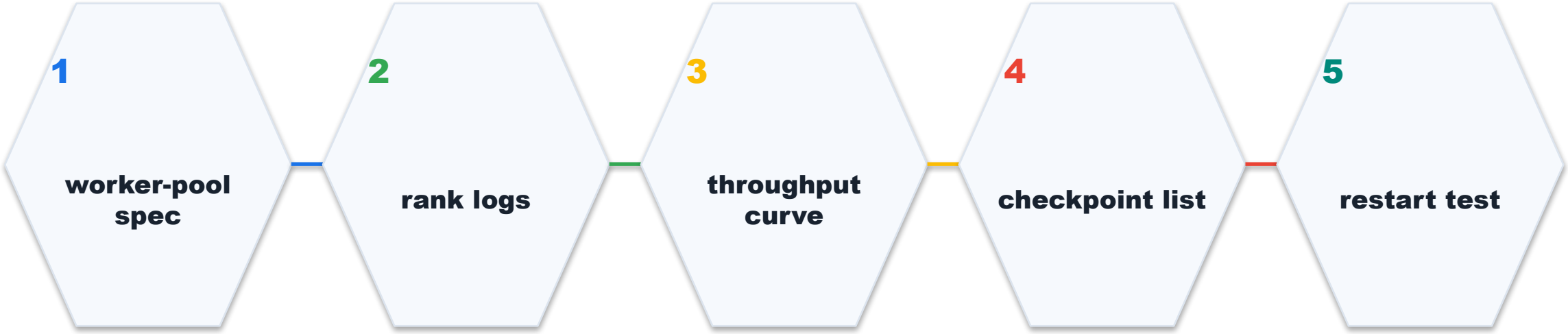
Distributed training — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Hang	rank/network mismatch	validate rendezvous and ports
Divergence	effective batch changed	scale learning rate
Lost work	no frequent checkpoint	checkpoint to durable GCS

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Distributed training — evidence chain



A valid production decision is reproducible from the complete chain.

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Distributed training — exam decision

SCENARIO

An ImageNet job must scale from 1 to 8 GPUs and survive preemptible worker loss.

BEST DECISION

Use torchrun data parallelism, durable checkpoints and measure scaling efficiency before adding workers.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 10 evidence — Plan Distributed PyTorch GPU Training

ACTIVITY 10

Configure torchrun-style workers, checkpoints and scaling-efficiency diagnostics.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

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- `Dockerfile`
- `worker_pool.yaml`
- `scaling_results.csv`

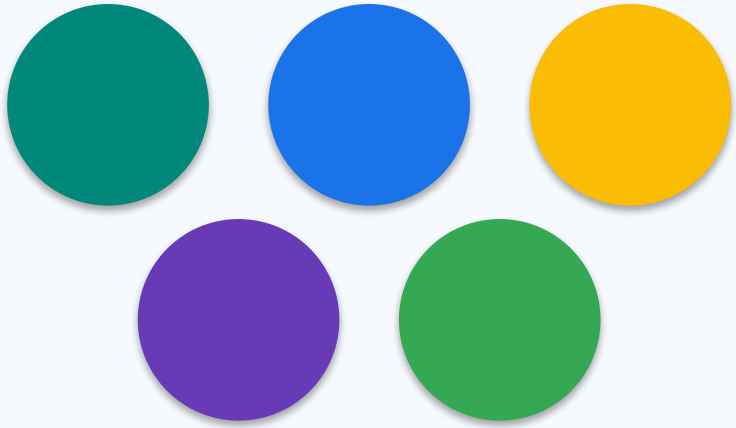
Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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DOMAIN 4 · ~20%

Serving and scaling models

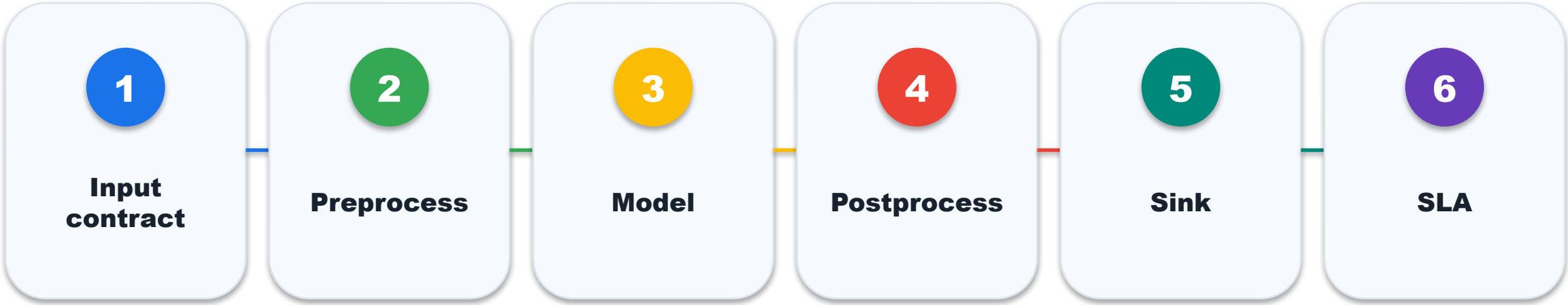
Topics 6–7: batch/online prediction, production systems



8 technical mechanisms · 64 anchored slides

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Batch versus online inference — mechanism map



Control point Decision evidence must bind input contract to sla through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Batch versus online inference — service decision matrix

OPTION 1

Batch prediction

Use when

asynchronous high-volume jobs

OPTION 2

Online endpoint

Use when

interactive low-latency requests

OPTION 3

Streaming

Use when

continuous events with bounded delay

Exam cue: select from workload evidence—not from product familiarity.

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Batch versus online inference — runnable configuration

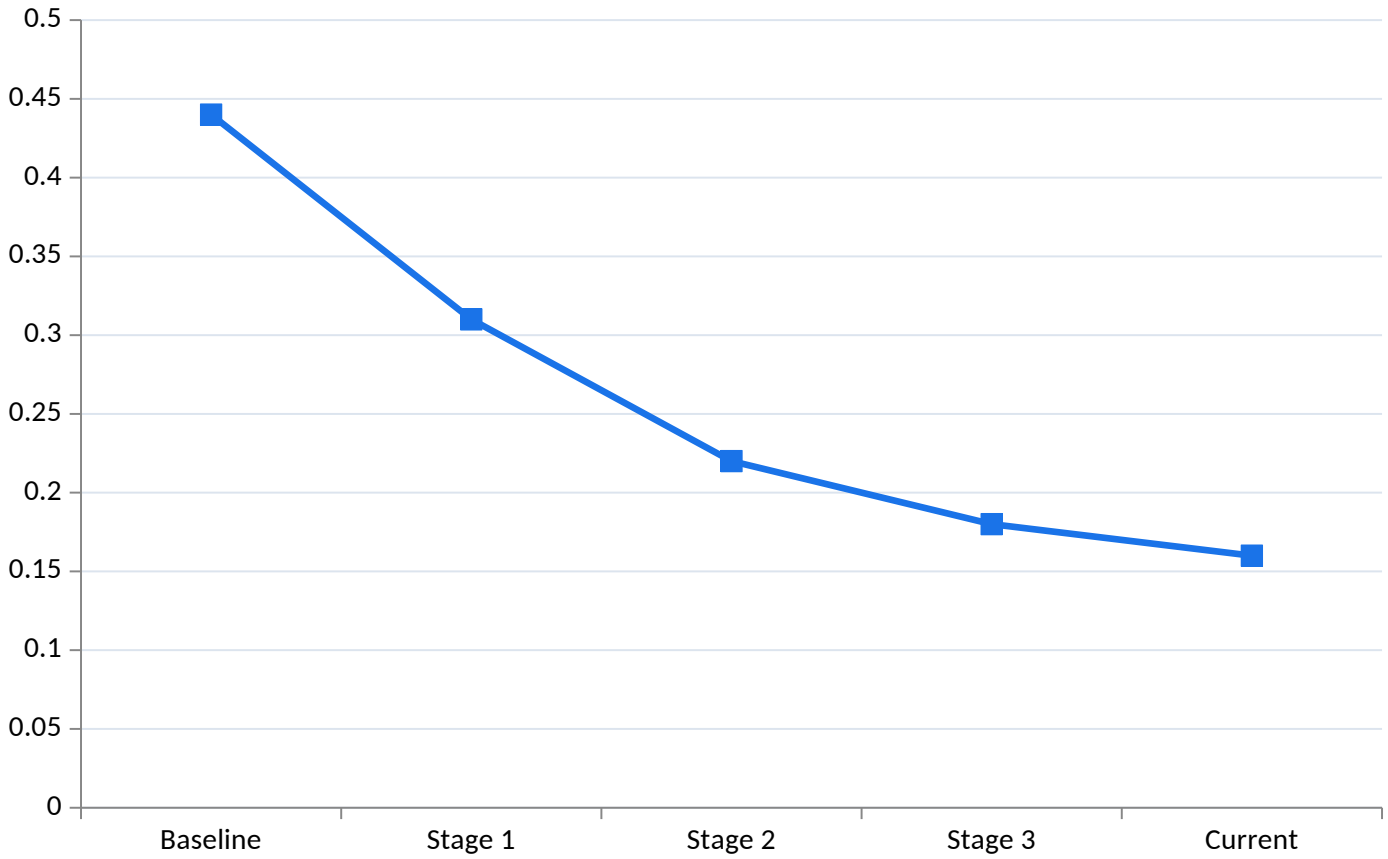
```
gcloud ai batch-prediction-jobs create --region=$REGION --
model=$MODEL_ID \
+
--jsonl-request=gcsSource.uris=gs://$BUCKET/requests.jsonl
\
+
--jsonl-output=gcsDestination.outputUriPrefix=gs://$BUCKET/
predictions/
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Batch versus online inference — metric evidence



Cost per 1,000 predictions

Formula

```
cost_1k =
total_serving_cost /
predictions * 1000
```

Interpret trend, threshold and uncertainty before acting.

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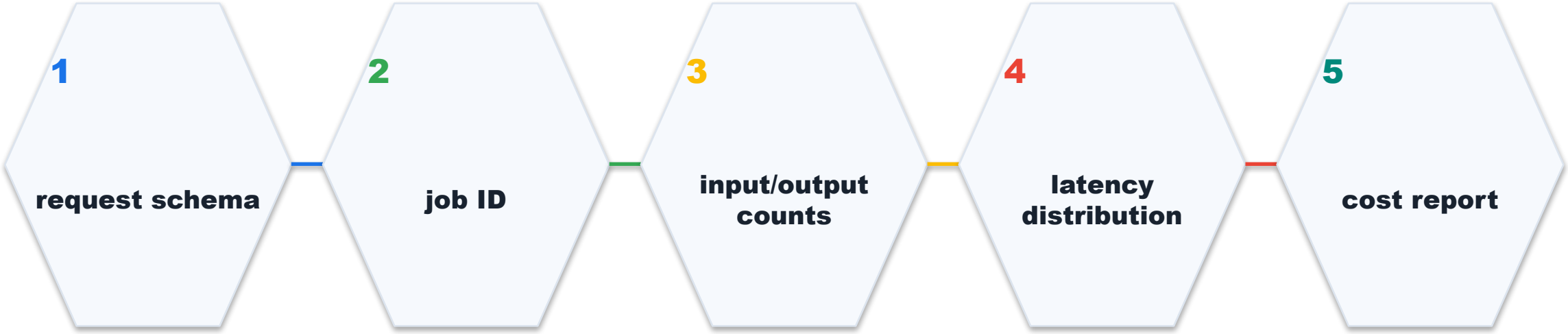
Batch versus online inference — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Timeout	online used for huge payload	batch job
Stale output	batch schedule late	freshness SLO
Duplicate score	retries not idempotent	request IDs

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Batch versus online inference — evidence chain



A valid production decision is reproducible from the complete chain.

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Batch versus online inference — exam decision

SCENARIO

Marketing needs 50 million nightly scores; the UI needs only the latest stored score.

BEST DECISION

Use batch prediction to a governed table; serve the stored score rather than 50 million online calls.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 12 evidence — Register and Canary a Model Version

ACTIVITY 12

Create registry metadata, traffic split, guardrails and rollback evidence.

Evidence: activity artifacts and acceptance checks

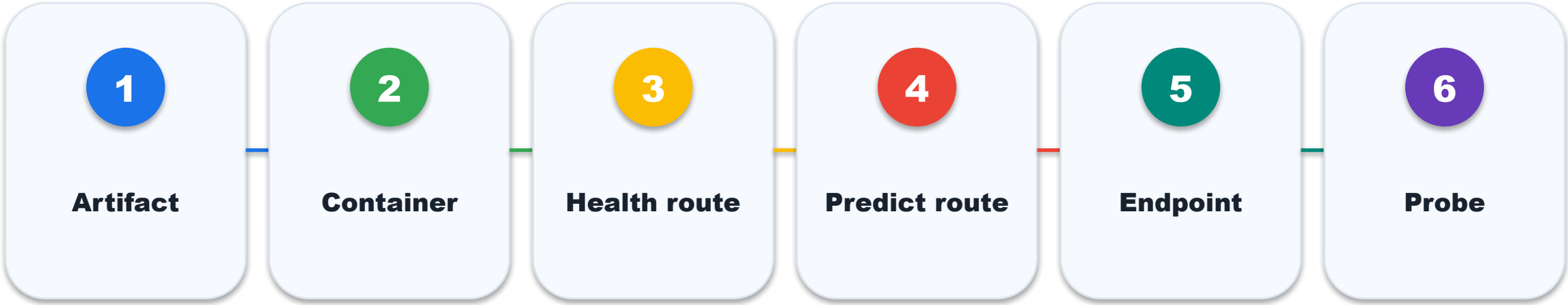
Detailed procedures: Learner Guide + activity README/PDF

- `model_manifest.json`
- `deployment.yaml`
- `canary_metrics.csv`
- `evaluate_canary.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Prebuilt and custom serving containers — mechanism map



Control point Decision evidence must bind artifact to probe through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Prebuilt and custom serving containers — service decision matrix

OPTION 1

Prebuilt

Use when

supported framework and signature

OPTION 2

Custom

Use when

custom dependencies or server

OPTION 3

Cloud Run container

Use when

portable stateless HTTP serving

Exam cue: select from workload evidence—not from product familiarity.

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Prebuilt and custom serving containers — runnable configuration

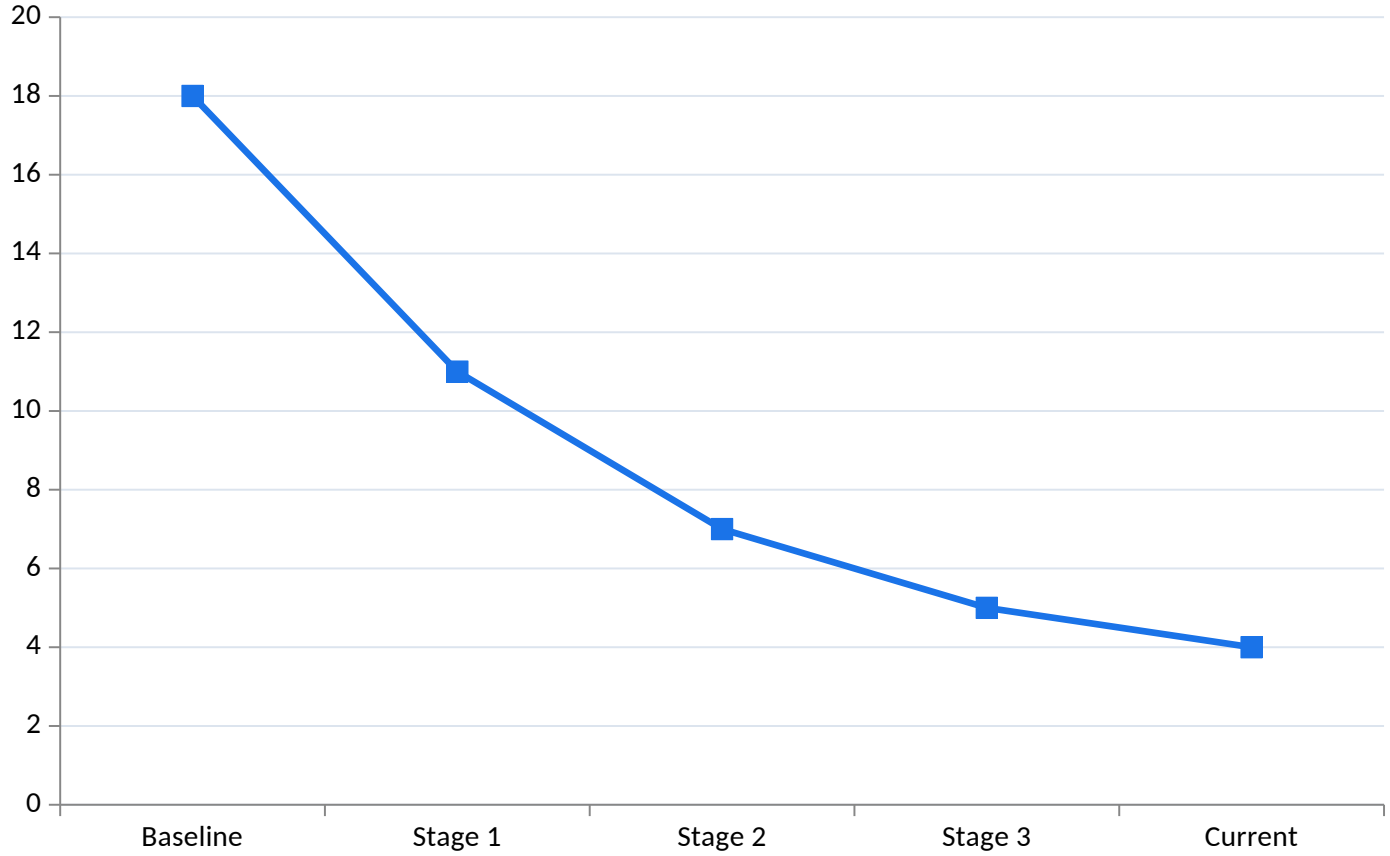
```
FROM python:3.11-slim
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py model.joblib ./
CMD ["gunicorn","-b",":8080","app:app"]
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Prebuilt and custom serving containers — metric evidence



Container startup time

Formula

```
startup =
ready_timestamp -
instance_start
```

Interpret trend, threshold and uncertainty before acting.

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Prebuilt and custom serving containers — failure diagnosis

Observable symptom	Likely mechanism	Control or test
CrashLoop	bad command or missing file	local container smoke test
Unhealthy	wrong probe path/port	implement health route
ABI error	runtime mismatch	pin base image and wheels

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Prebuilt and custom serving containers — evidence chain



A valid production decision is reproducible from the complete chain.

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Prebuilt and custom serving containers — exam decision

SCENARIO

An XGBoost model requires custom preprocessing and a native package not in prebuilt serving images.

BEST DECISION

Build a minimal custom image, pin the digest, generate an SBOM and probe health/predict routes.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 13 evidence — Serve a Model on Cloud Run

ACTIVITY 13

Package a Flask prediction API, test its contract and analyze load-test results.

Evidence: activity artifacts and acceptance checks

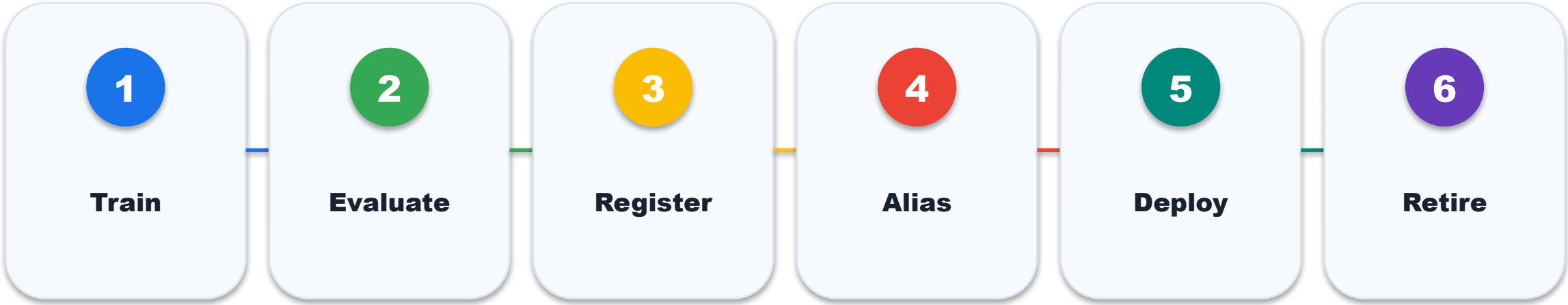
Detailed procedures: Learner Guide + activity README/PDF

- `app.py`
- `Dockerfile`
- `requirements.txt`
- `test_api.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Model Registry and versioning — mechanism map



Control point Decision evidence must bind train to retire through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Model Registry and versioning — service decision matrix

OPTION 1

Version

Use when

immutable model artifact generation

OPTION 2

Alias

Use when

movable label such as champion

OPTION 3

Lineage

Use when

data/code/run relation

Exam cue: select from workload evidence—not from product familiarity.

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Model Registry and versioning — runnable configuration

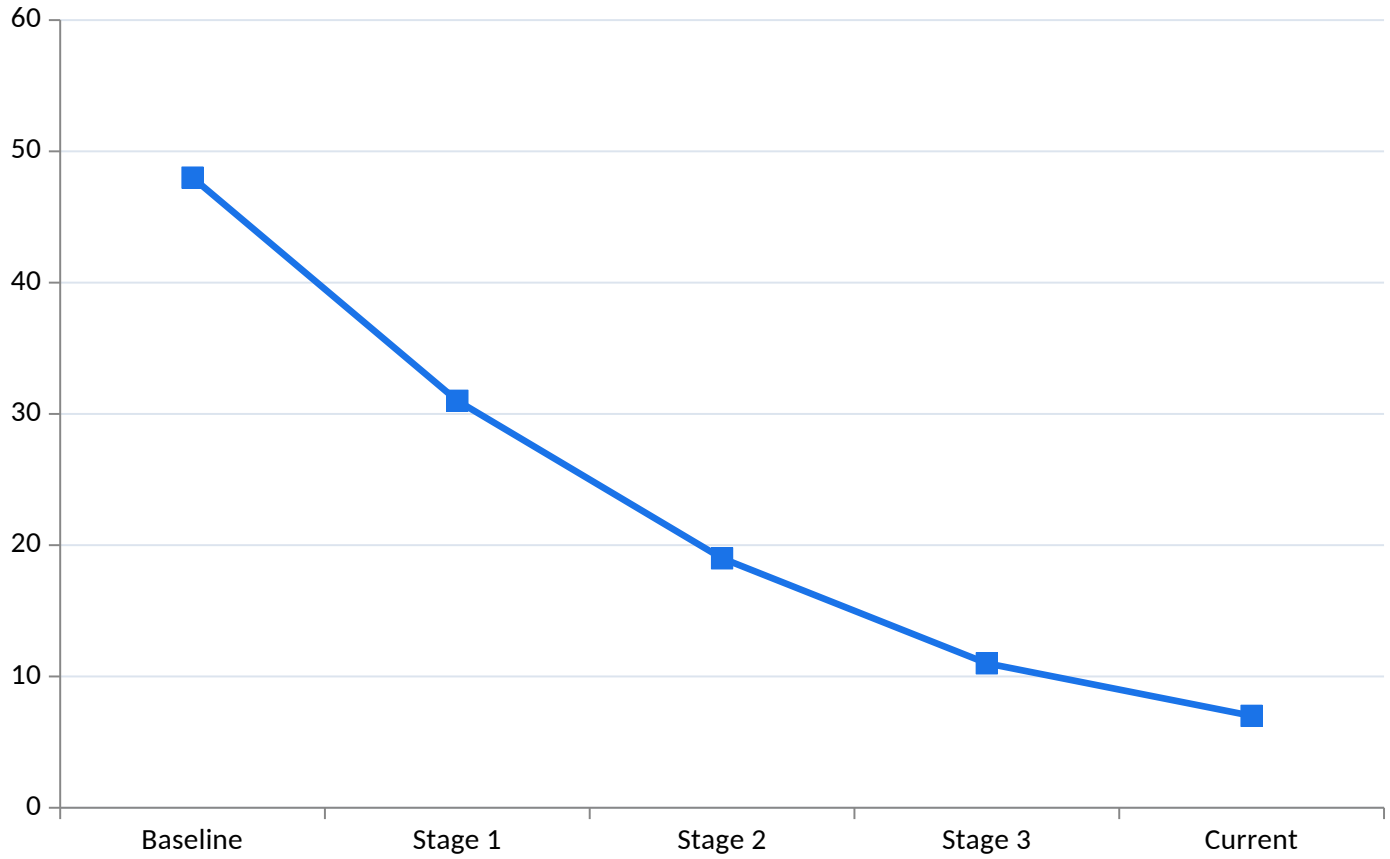
```
model_labels =
{'git_sha':GIT_SHA,'data_version':DATA_VER,'owner':'risk-
ml'}
model_alias = 'candidate'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Model Registry and versioning — metric evidence



Rollback time

Formula

```
MTTR = restore_time - incident_start
```

Interpret trend, threshold and uncertainty before acting.

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Model Registry and versioning — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Unknown model	manual file copied	registry-only deployment
Wrong alias	unreviewed promotion	approval gate
No rollback	old version deleted	retain validated prior version

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Model Registry and versioning — evidence chain



A valid production decision is reproducible from the complete chain.

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Model Registry and versioning — exam decision

SCENARIO

A model release must be traceable to code and data, and rollback must finish within 15 minutes.

BEST DECISION

Register immutable versions with lineage labels; promote aliases only after gates and drill rollback.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 12 evidence — Register and Canary a Model Version

ACTIVITY 12

Create registry metadata, traffic split, guardrails and rollback evidence.

Evidence: activity artifacts and acceptance checks

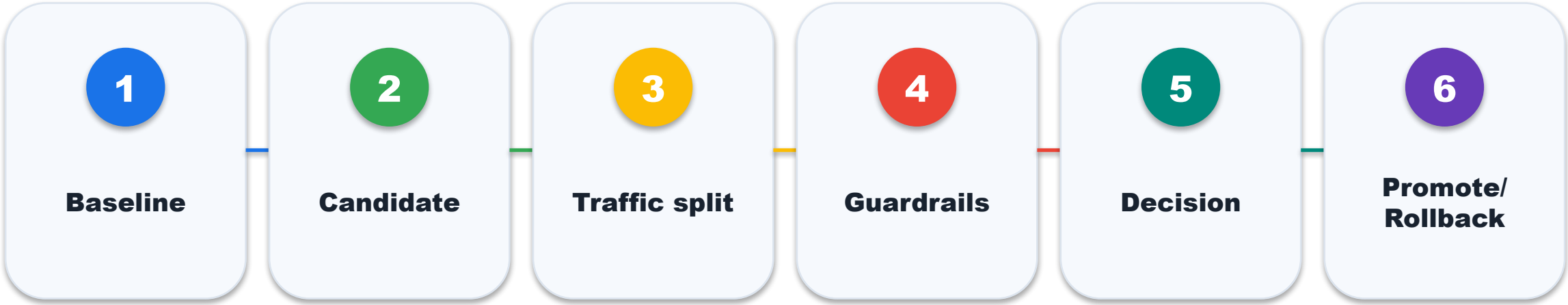
Detailed procedures: Learner Guide + activity README/PDF

- `model_manifest.json`
- `deployment.yaml`
- `canary_metrics.csv`
- `evaluate_canary.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Canary and A/B rollout — mechanism map



Control point Decision evidence must bind baseline to promote/rollback through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Canary and A/B rollout — service decision matrix

OPTION 1

Canary

Use when

risk-limited technical validation

OPTION 2

A/B

Use when

compare user/business outcomes

OPTION 3

Shadow

Use when

observe candidate without affecting users

Exam cue: select from workload evidence—not from product familiarity.

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Canary and A/B rollout — runnable configuration

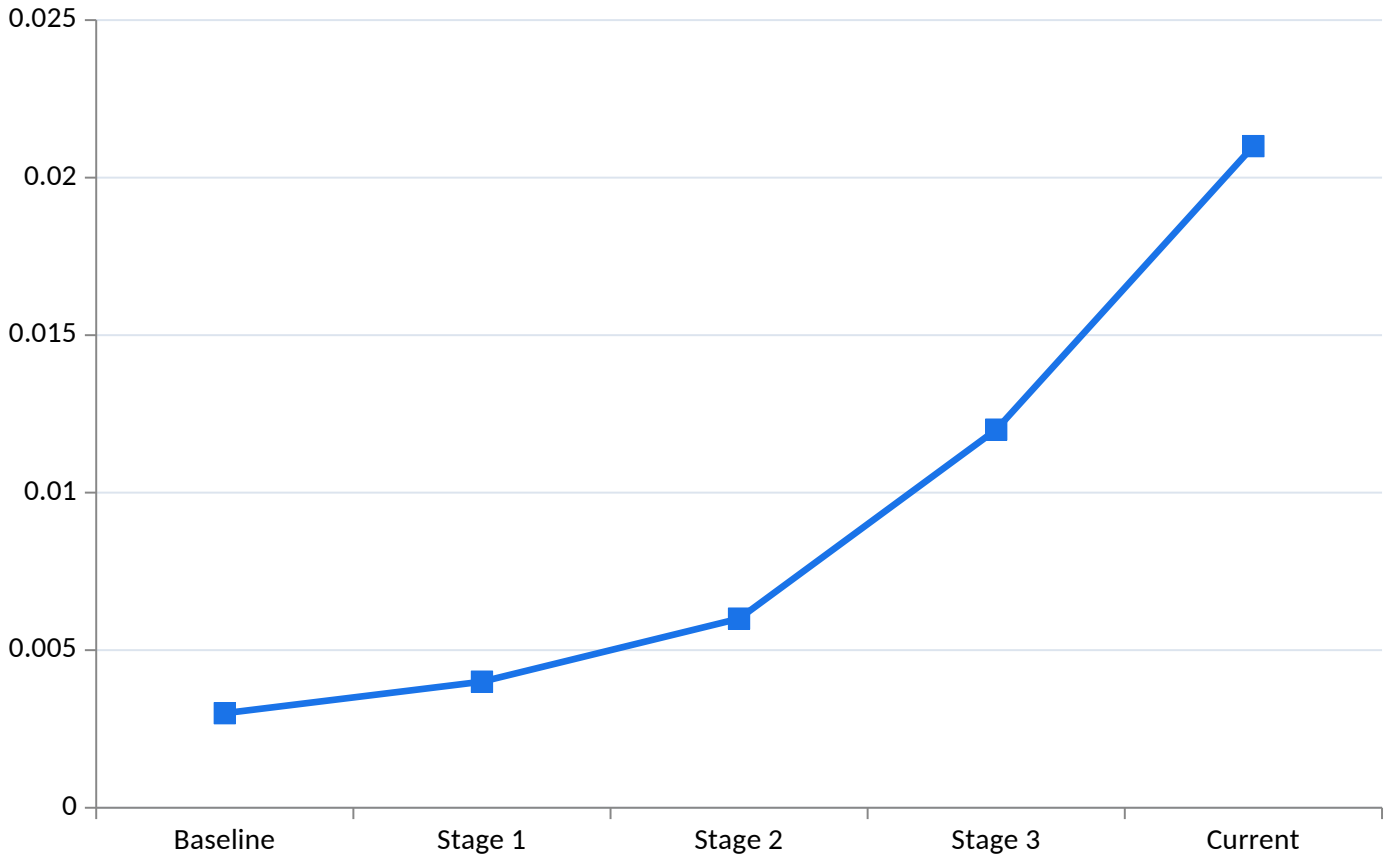
```
trafficSplit:
  deployedModels:
    champion: 90
    candidate: 10
rollback_if: 'p95_ms > 300 or error_rate > 0.01'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Canary and A/B rollout — metric evidence



Canary error rate

Formula

```
error_rate =
failed_requests /
total_requests
```

Interpret trend, threshold and uncertainty before acting.

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Canary and A/B rollout — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Biased sample	non-random routing	stable random assignment
Slow rollback	manual process	automated guardrail
Metric dilution	aggregate traffic	per-version metrics

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Canary and A/B rollout — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 4 · SCENARIO

Canary and A/B rollout — exam decision

SCENARIO

A new ranking model improves offline NDCG but has unknown tail latency at peak traffic.

BEST DECISION

Start with shadow then 10% canary; gate on version-specific latency, errors and business metric.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 12 evidence — Register and Canary a Model Version

ACTIVITY 12

Create registry metadata, traffic split, guardrails and rollback evidence.

Evidence: activity artifacts and acceptance checks

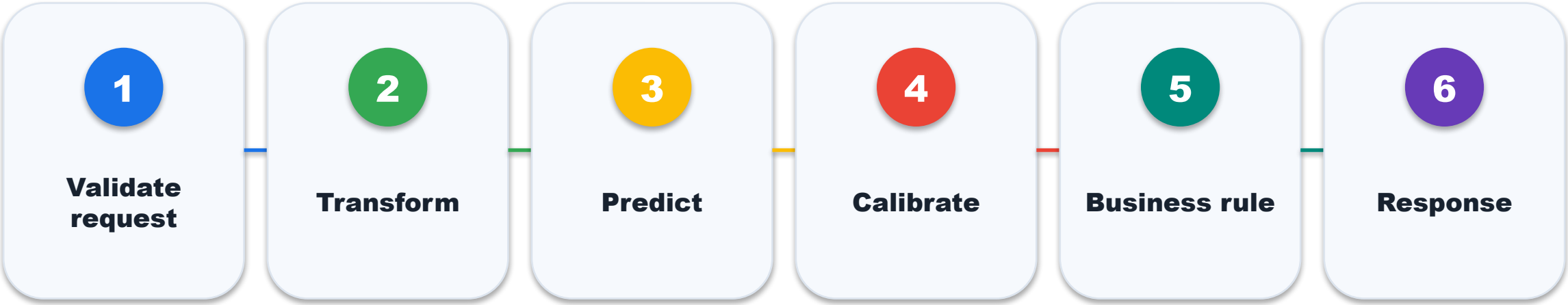
Detailed procedures: Learner Guide + activity README/PDF

- `model_manifest.json`
- `deployment.yaml`
- `canary_metrics.csv`
- `evaluate_canary.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Inference preprocessing and postprocessing — mechanism map



Control point Decision evidence must bind validate request to response through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Inference preprocessing and postprocessing — service decision matrix

OPTION 1

In-container transform

Use when

tight coupling and low latency

OPTION 2

Pipeline transform

Use when

shared batch/online code

OPTION 3

Feature Store

Use when

central online feature retrieval

Exam cue: select from workload evidence—not from product familiarity.

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Inference preprocessing and postprocessing — runnable configuration

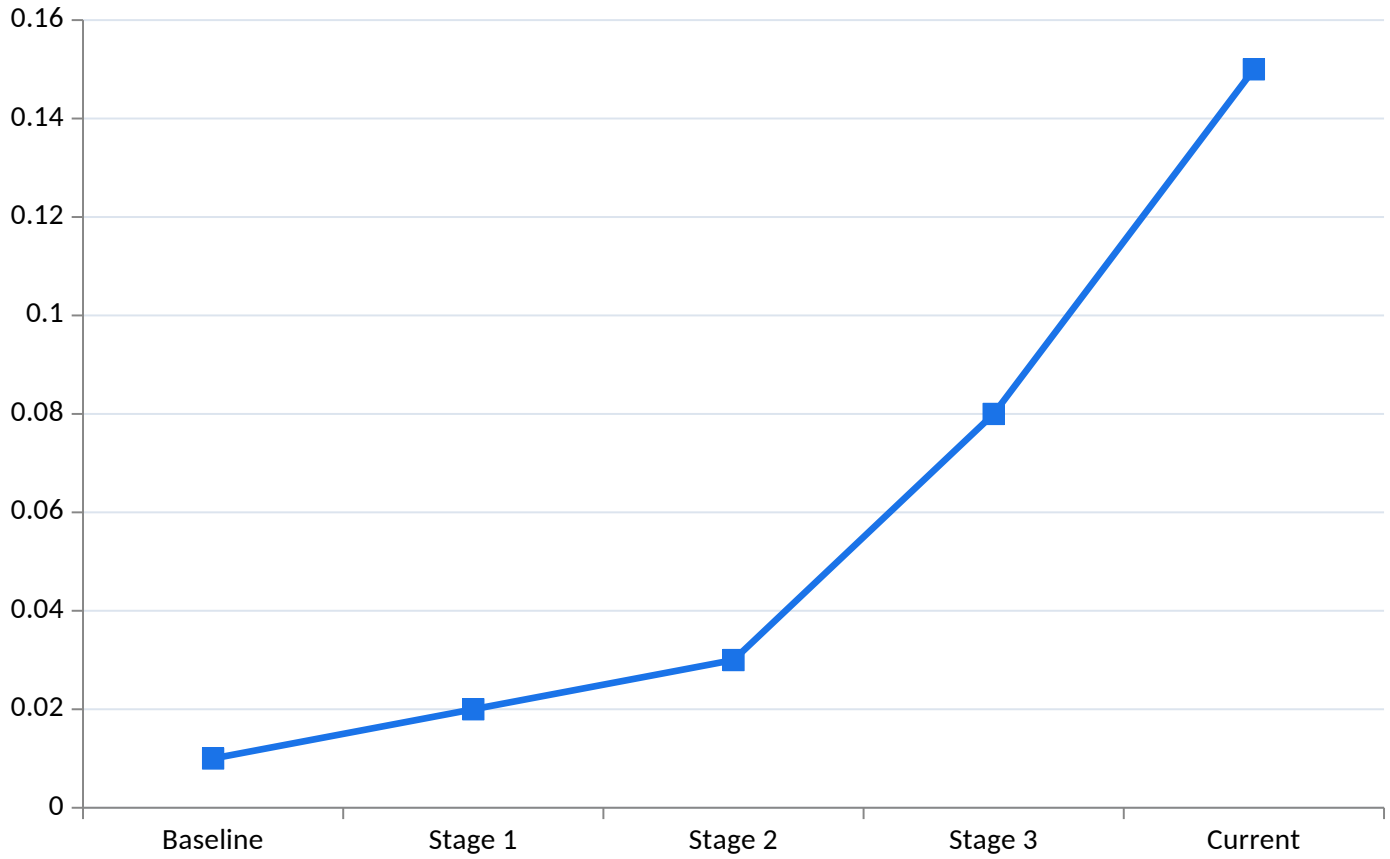
```
assert request['schema_version']=='3'
x = transformer.transform([request['features']])
p = calibrator.predict_proba(x)[0,1]
return {'risk':float(p),'model_version':VERSION}
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Inference preprocessing and postprocessing — metric evidence



Training-serving skew

Formula

```
skew =
distance(train_feature,
serving_feature)
```

Interpret trend, threshold and uncertainty before acting.

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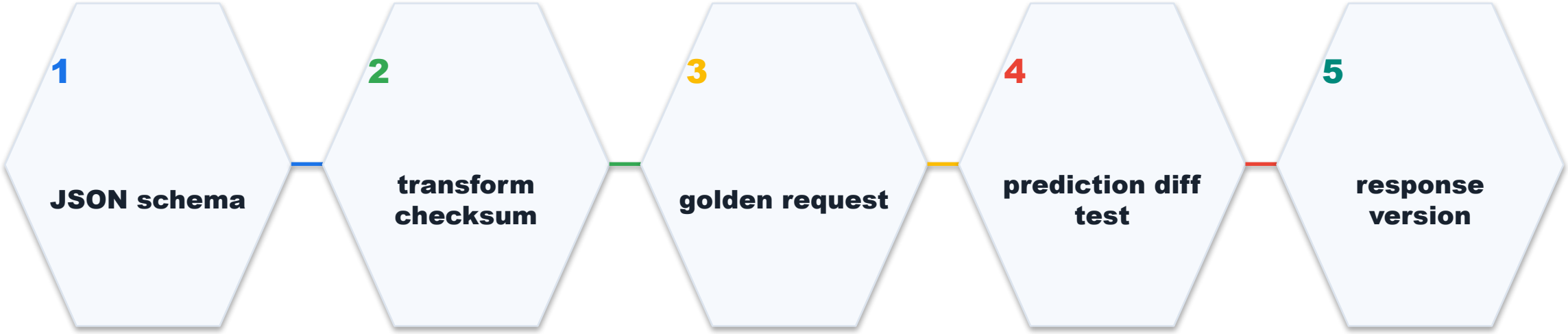
Inference preprocessing and postprocessing — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Shape error	schema changed	versioned contract
Probability drift	calibrator omitted	bundle postprocessor
Silent default	missing feature filled	reject or explicit missingness

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Inference preprocessing and postprocessing — evidence chain



A valid production decision is reproducible from the complete chain.

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Inference preprocessing and postprocessing — exam decision

SCENARIO

The same raw input scores 0.22 offline and 0.71 online after a preprocessing refactor.

BEST DECISION

Compare transformation artifacts and golden requests; deploy preprocessing with the model version.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 13 evidence — Serve a Model on Cloud Run

ACTIVITY 13

Package a Flask prediction API, test its contract and analyze load-test results.

Evidence: activity artifacts and acceptance checks

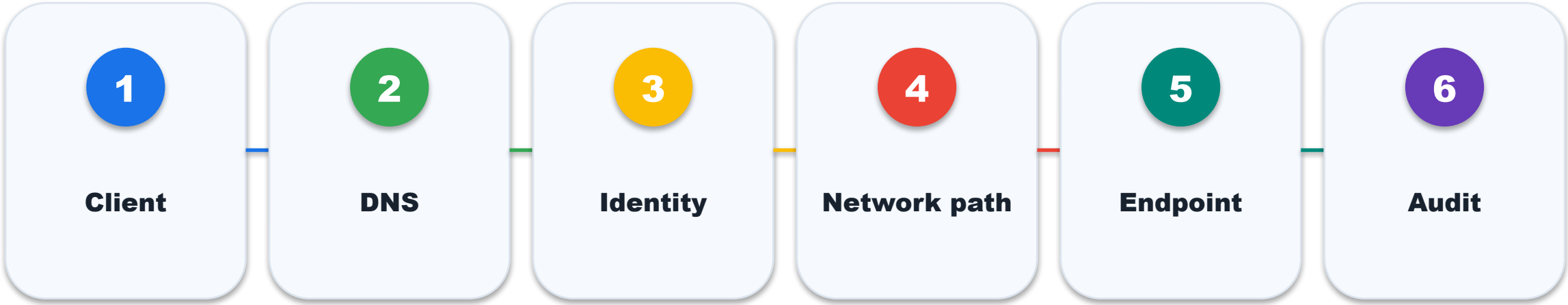
Detailed procedures: Learner Guide + activity README/PDF

- `app.py`
- `Dockerfile`
- `requirements.txt`
- `test_api.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Public and private endpoints — mechanism map



Control point Decision evidence must bind client to audit through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Public and private endpoints — service decision matrix

OPTION 1

Public endpoint

Use when

internet path with IAM/auth controls

OPTION 2

Private Service Connect

Use when

private VPC consumption

OPTION 3

Cloud Run ingress

Use when

service-to-service serverless

Exam cue: select from workload evidence—not from product familiarity.

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Public and private endpoints — runnable configuration

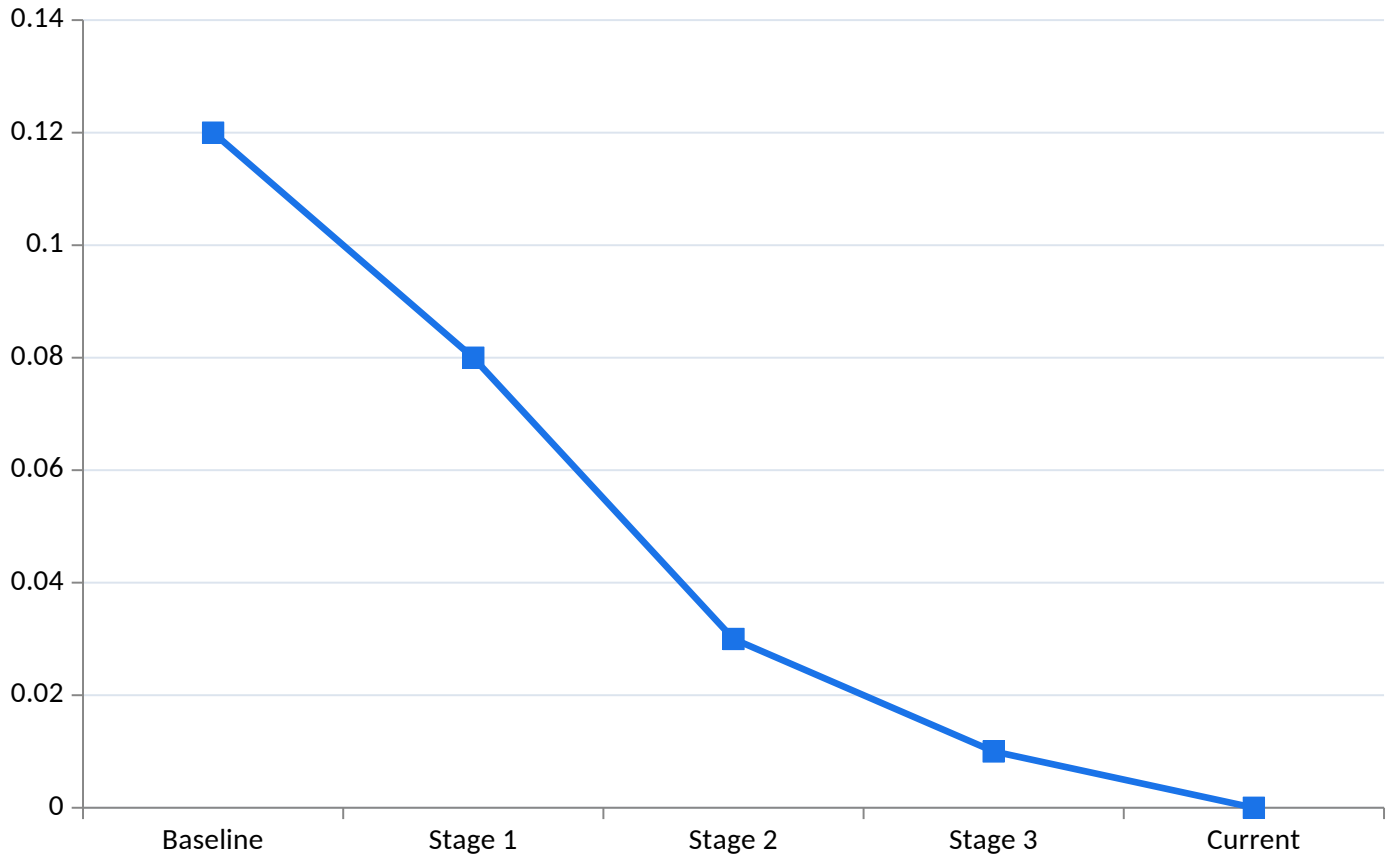
```
ingress: internal
authentication: iam
psc_network: projects/$PROJECT/global/networks/ml-vpc
log_requests: metadata_only
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Public and private endpoints — metric evidence



Unauthorized request rate

Formula

```
denied / total requests
```

Interpret trend, threshold and uncertainty before acting.

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Public and private endpoints — failure diagnosis

Observable symptom	Likely mechanism	Control or test
403	caller identity lacks invoke	grant narrow invoker
No route	PSC/DNS misconfigured	validate endpoint and DNS
Data leak	payload logs	metadata-only logging

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Public and private endpoints — evidence chain



A valid production decision is reproducible from the complete chain.

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Public and private endpoints — exam decision

SCENARIO

A bank model endpoint must be reachable from two VPCs and never expose public ingress.

BEST DECISION

Use private connectivity, service identities and end-to-end connectivity plus audit tests.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 12 evidence — Register and Canary a Model Version

ACTIVITY 12

Create registry metadata, traffic split, guardrails and rollback evidence.

Evidence: activity artifacts and acceptance checks

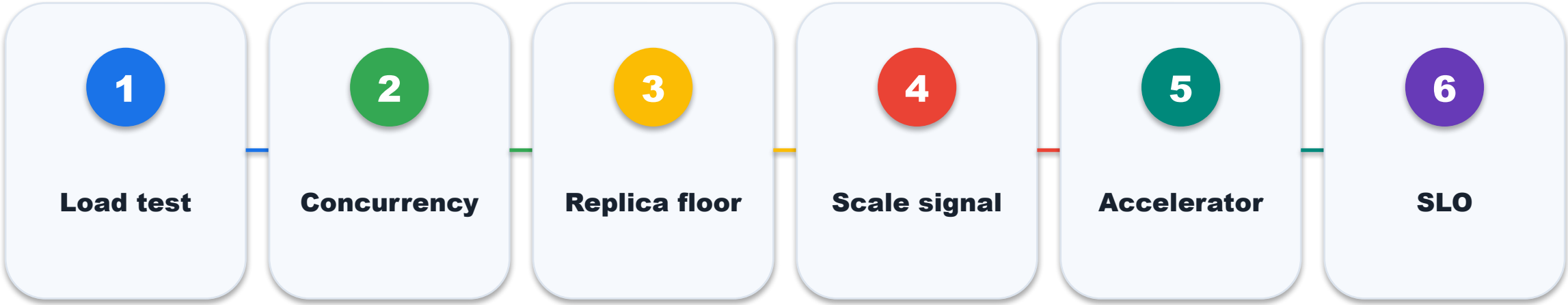
Detailed procedures: Learner Guide + activity README/PDF

- `model_manifest.json`
- `deployment.yaml`
- `canary_metrics.csv`
- `evaluate_canary.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Autoscaling and serving hardware — mechanism map



Control point Decision evidence must bind load test to slo through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Autoscaling and serving hardware — service decision matrix

OPTION 1

CPU autoscale

Use when

lightweight model and variable demand

OPTION 2

GPU endpoint

Use when

high-throughput deep learning

OPTION 3

Edge

Use when

connectivity or privacy requires local inference

Exam cue: select from workload evidence—not from product familiarity.

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Autoscaling and serving hardware — runnable configuration

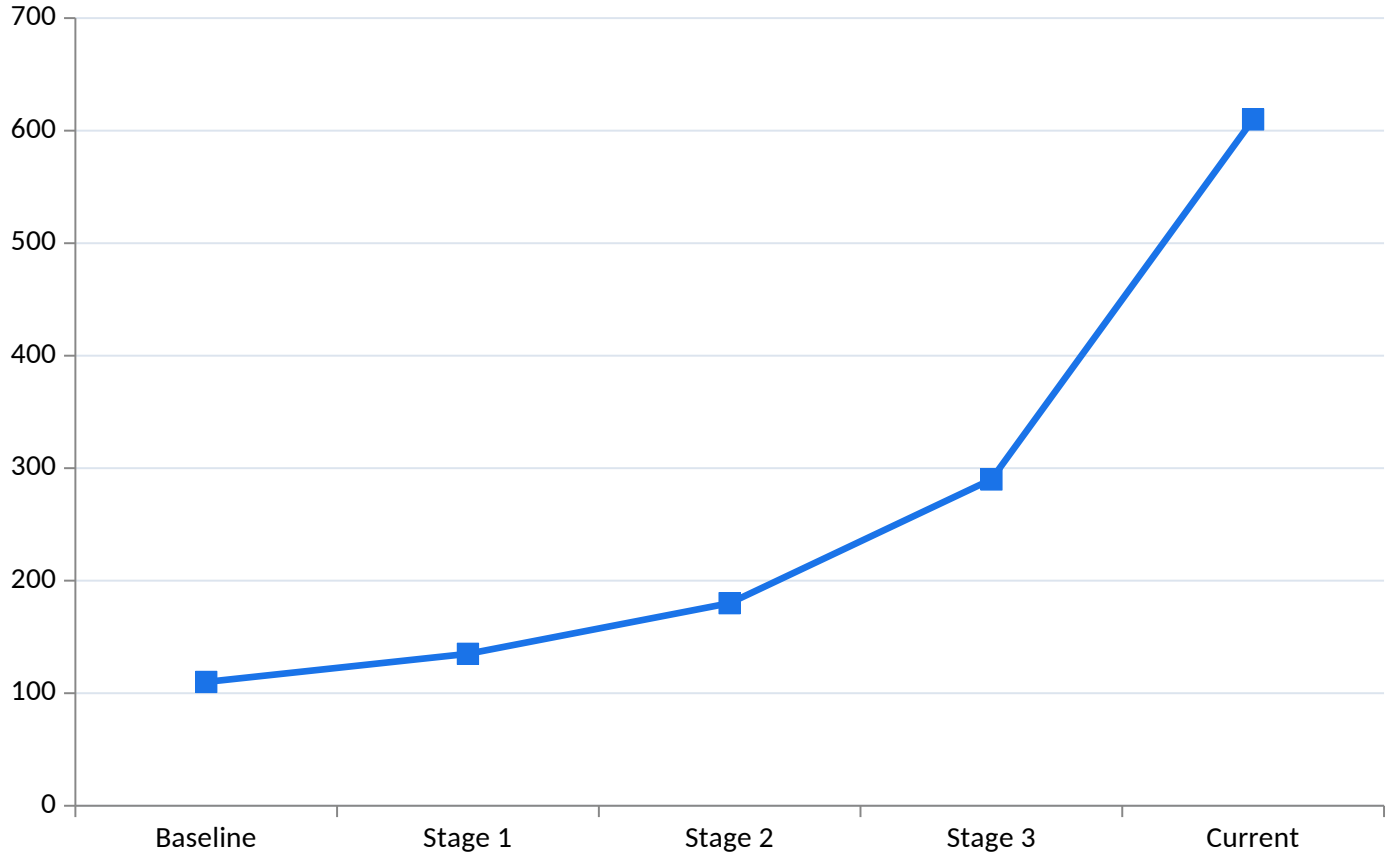
```
minReplicaCount: 2
maxReplicaCount: 12
targetCpuUtilization: 60
acceleratorType: NVIDIA_TESLA_T4
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Autoscaling and serving hardware — metric evidence



p95 latency by QPS

Formula

```
p95 =
percentile(latency_ms,95
)
```

Interpret trend, threshold and uncertainty before acting.

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Autoscaling and serving hardware — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Cold-start spike	scale from zero	warm floor
Queue growth	concurrency too high	capacity test
GPU idle	small batches	dynamic batching or CPU

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Autoscaling and serving hardware — evidence chain



A valid production decision is reproducible from the complete chain.

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Autoscaling and serving hardware — exam decision

SCENARIO

A vision API must handle 5–80 QPS with p95 below 300 ms and control GPU cost.

BEST DECISION

Benchmark CPU/GPU, set a warm floor and autoscale from measured saturation—not guesses.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 13 evidence — Serve a Model on Cloud Run

ACTIVITY 13

Package a Flask prediction API, test its contract and analyze load-test results.

Evidence: activity artifacts and acceptance checks

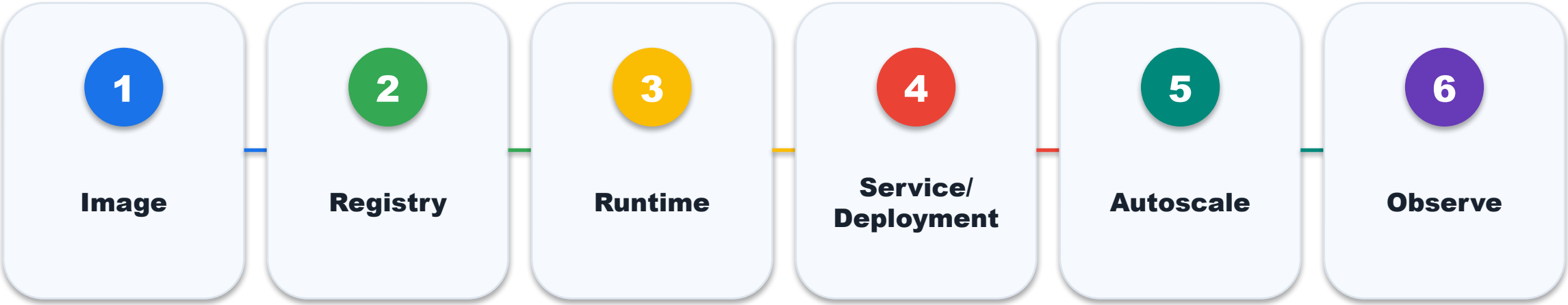
Detailed procedures: Learner Guide + activity README/PDF

- `app.py`
- `Dockerfile`
- `requirements.txt`
- `test_api.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Cloud Run and GKE serving — mechanism map



Control point Decision evidence must bind image to observe through versioned artifacts.

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DOMAIN 4 · DECISION RULE

Cloud Run and GKE serving — service decision matrix

OPTION 1

Cloud Run

Use when

stateless HTTP, rapid ops, bursty load

OPTION 2

GKE

Use when

custom networking, sidecars, accelerators or orchestration

OPTION 3

Vertex endpoint

Use when

managed ML deployment and monitoring

Exam cue: select from workload evidence—not from product familiarity.

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Cloud Run and GKE serving — runnable configuration

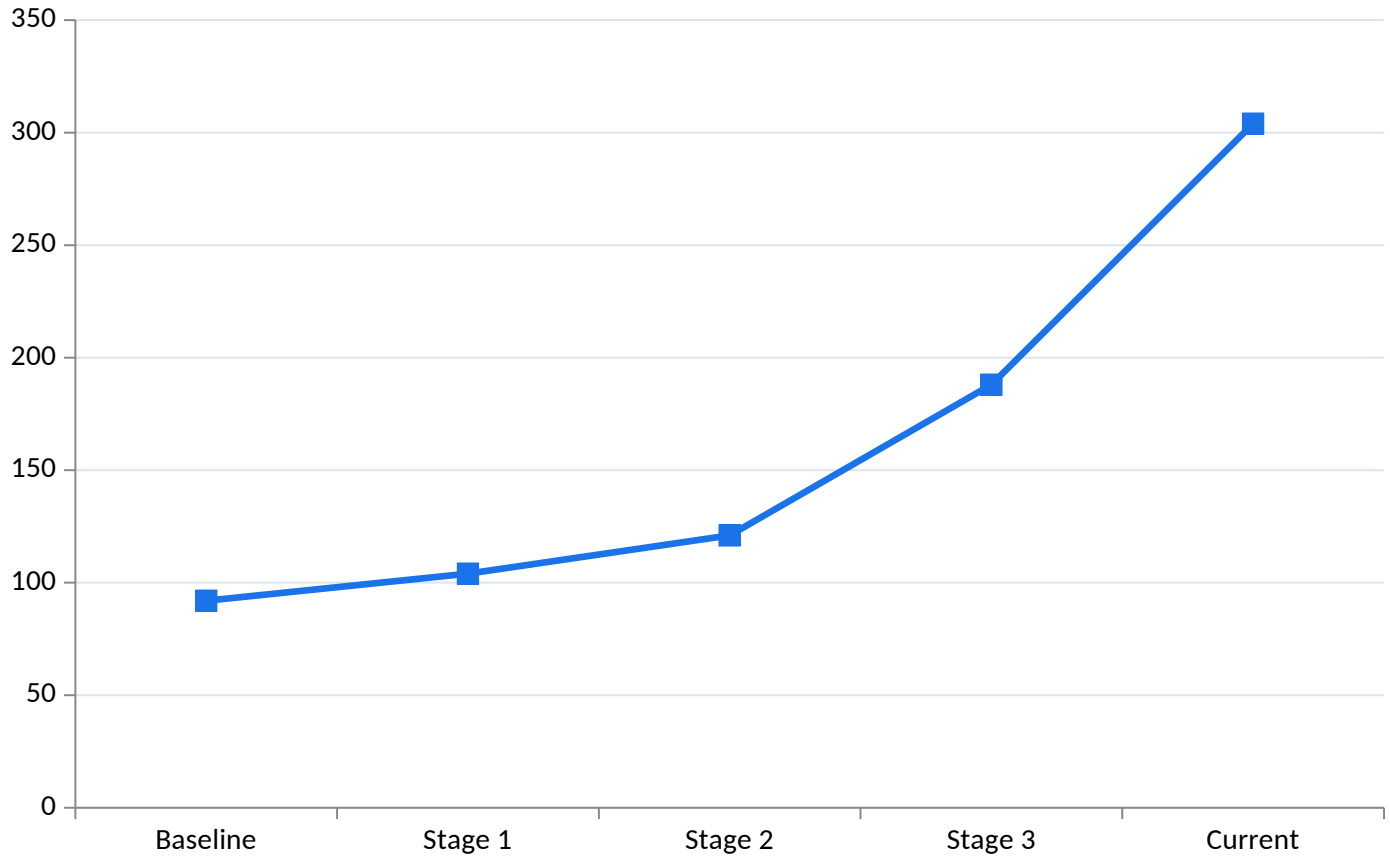
```
gcloud run deploy churn-api --image=$IMAGE --region=$REGION
\
+ --no-allow-unauthenticated --min=1 --max=20 --cpu=2 --
memory=2Gi
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Cloud Run and GKE serving — metric evidence



Request latency

Formula

```
latency = response_time
- request_time
```

Interpret trend, threshold and uncertainty before acting.

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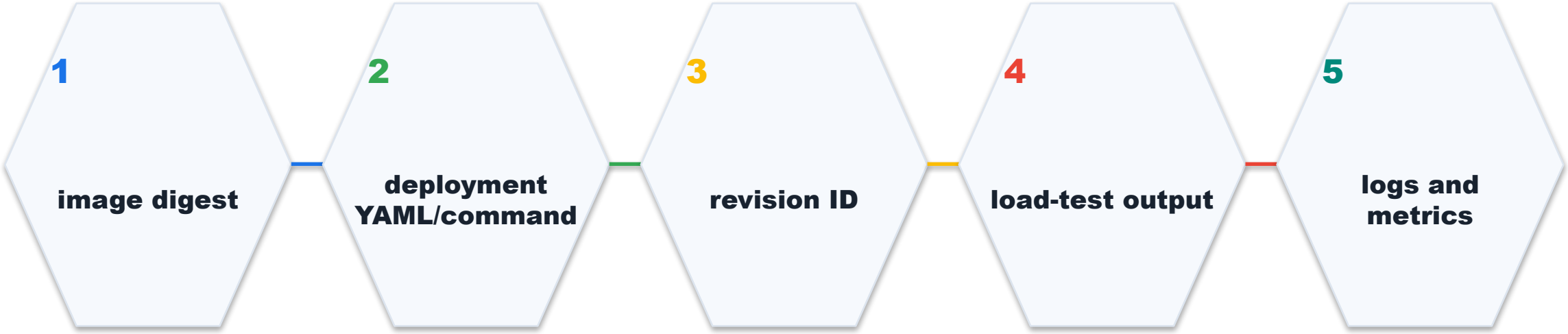
Cloud Run and GKE serving — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Revision not ready	port or startup failure	read revision logs
OOM kill	memory limit	profile and right-size
5xx burst	downstream saturation	timeouts and circuit breaker

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Cloud Run and GKE serving — evidence chain



A valid production decision is reproducible from the complete chain.

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Cloud Run and GKE serving — exam decision

SCENARIO

A scikit-learn API is stateless and bursty; a future GPU version may need custom scheduling.

BEST DECISION

Deploy CPU baseline to Cloud Run; use GKE only when its control requirements become real.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 13 evidence — Serve a Model on Cloud Run

ACTIVITY 13

Package a Flask prediction API, test its contract and analyze load-test results.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `app.py`
- `Dockerfile`
- `requirements.txt`
- `test_api.py`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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DOMAIN 5 · ~18%

Automating and orchestrating ML pipelines

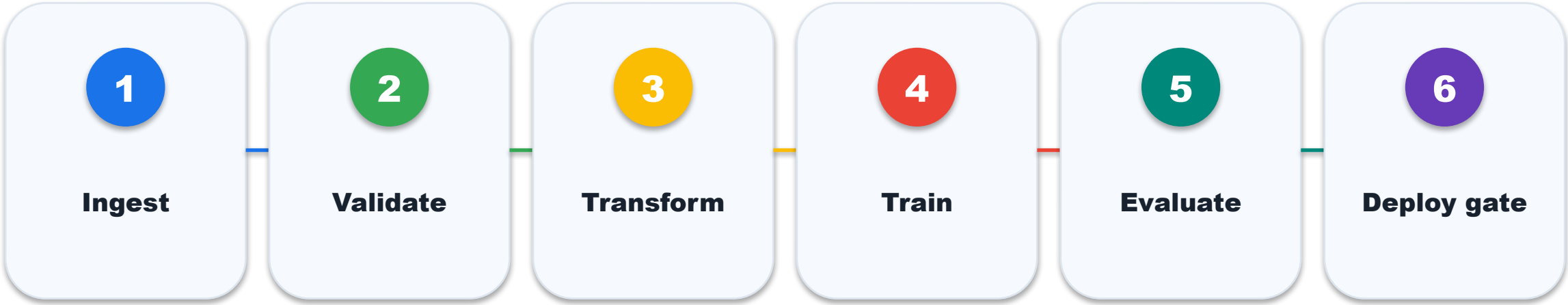
Topic 8: MLOps, CI/CD and repeatable workflows



8 technical mechanisms · 64 anchored slides

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Pipeline component contracts — mechanism map



Control point Decision evidence must bind ingest to deploy gate through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Pipeline component contracts — service decision matrix

OPTION 1

Artifact input

Use when

versioned file/model output

OPTION 2

Parameter

Use when

small configuration value

OPTION 3

Metric

Use when

numeric gate and comparison

Exam cue: select from workload evidence—not from product familiarity.

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Pipeline component contracts — runnable configuration

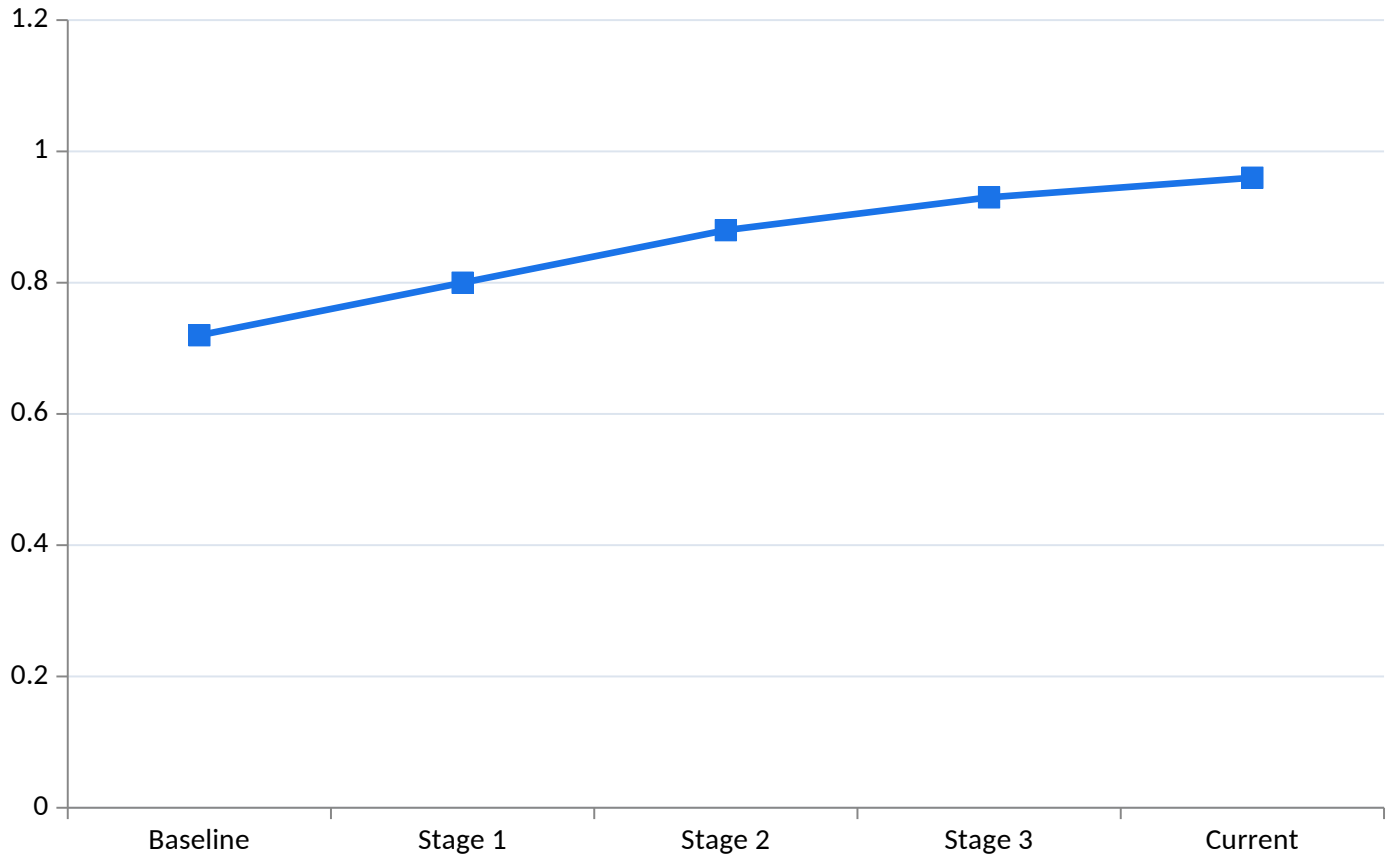
```
@component
def evaluate(model: Input[Model], metrics:
Output[Metrics]):
    score = run_eval(model.path)
    metrics.log_metric('auc', score)
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Pipeline component contracts — metric evidence



Pipeline pass rate

Formula

```
passes / scheduled_runs
```

Interpret trend, threshold and uncertainty before acting.

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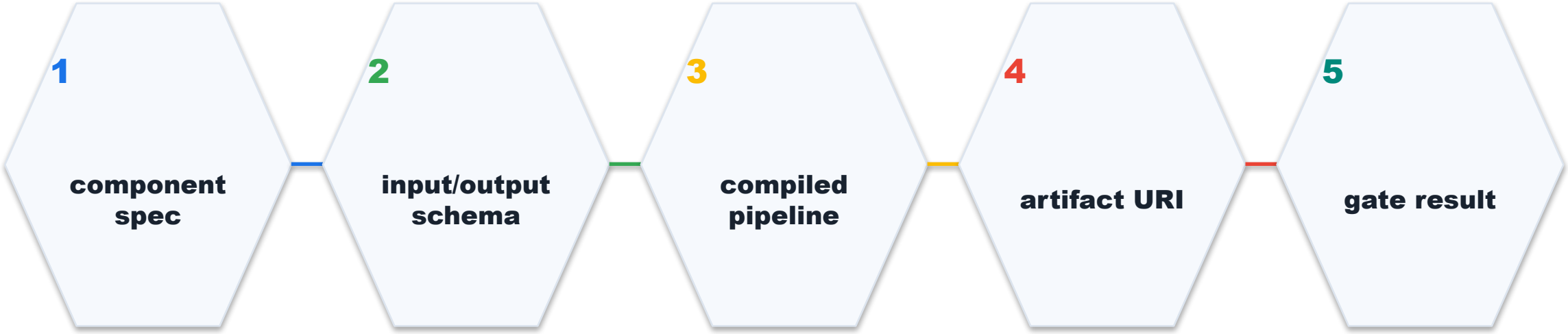
Pipeline component contracts — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Hidden dependency	component reads global path	declare input
Cache poison	mutable URI	immutable artifact
Wrong deploy	gate not wired	explicit conditional

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Pipeline component contracts — evidence chain



A valid production decision is reproducible from the complete chain.

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Pipeline component contracts — exam decision

SCENARIO

A pipeline retrains successfully but deploys a model evaluated on the previous run's dataset.

BEST DECISION

Make data/model/metrics explicit artifacts and wire deployment to the current evaluation output.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 14 evidence — Compile a Vertex AI Pipeline

ACTIVITY 14

Connect validate, transform, train, evaluate and conditional deployment components.

Evidence: activity artifacts and acceptance checks

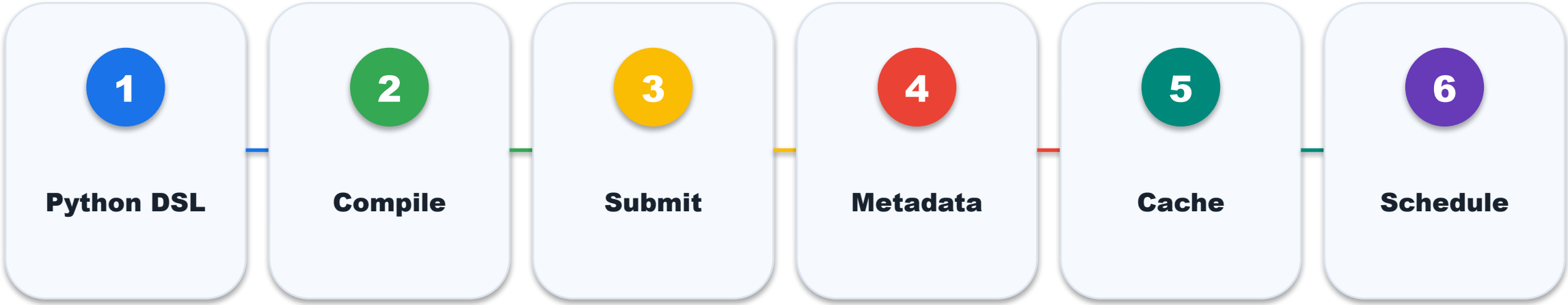
Detailed procedures: Learner Guide + activity README/PDF

- `pipeline.py`
- `pipeline_spec.json`
- `component_contracts.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Vertex AI Pipelines and Kubeflow — mechanism map



Control point Decision evidence must bind python dsl to schedule through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Vertex AI Pipelines and Kubeflow — service decision matrix

OPTION 1

Vertex AI Pipelines

Use when

managed KFP execution

OPTION 2

Kubeflow on GKE

Use when

cluster-level control

OPTION 3

Local runner

Use when

component unit testing only

Exam cue: select from workload evidence—not from product familiarity.

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Vertex AI Pipelines and Kubeflow — runnable configuration

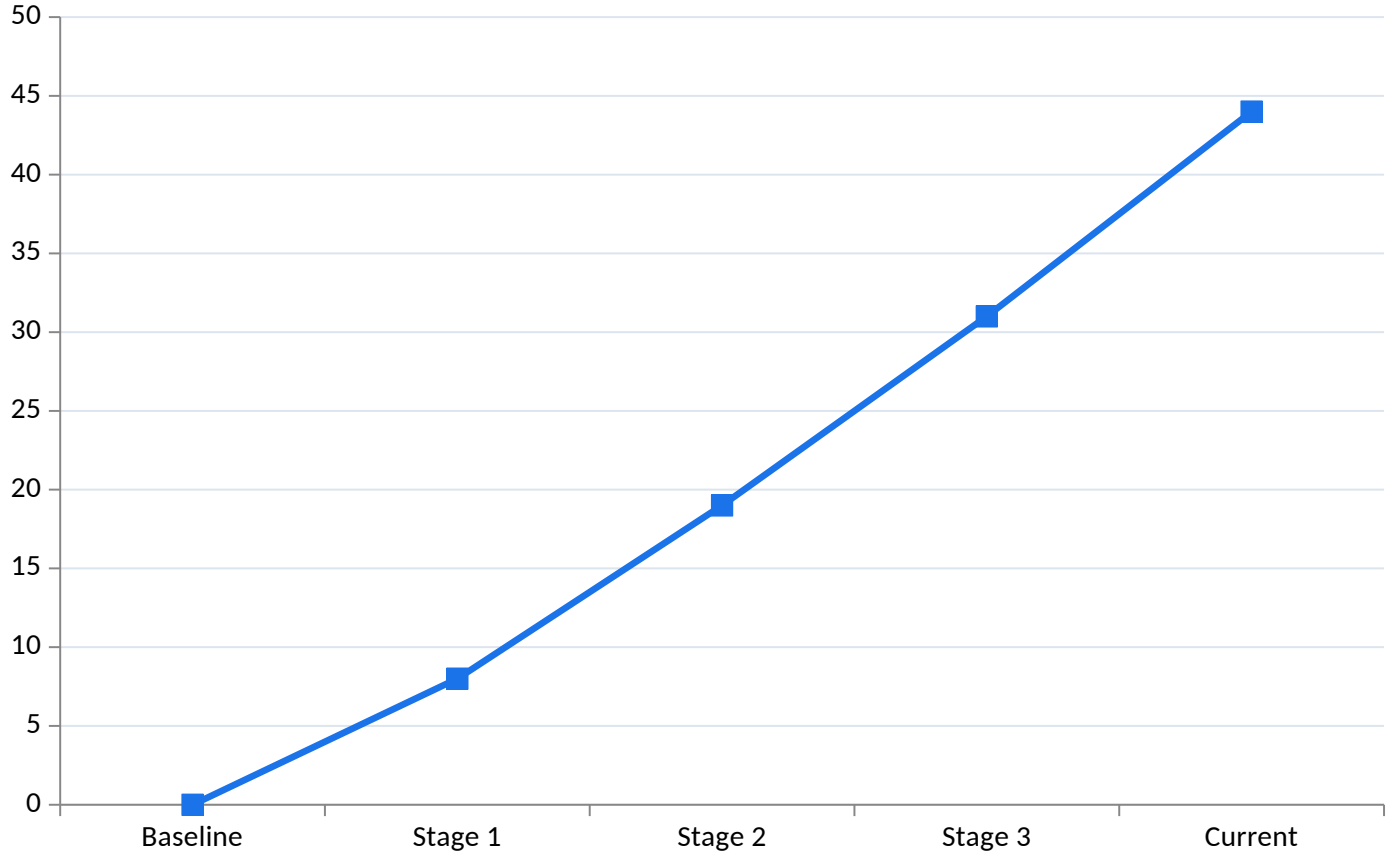
```
compiler.Compiler().compile(pipeline_func=training_pipeline
, package_path='pipeline.json')
aiplatform.PipelineJob(display_name='churn',
template_path='pipeline.json').run()
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Vertex AI Pipelines and Kubeflow — metric evidence



Cached step savings

Formula

```
saved =
uncached_duration -
cached_duration
```

Interpret trend, threshold and uncertainty before acting.

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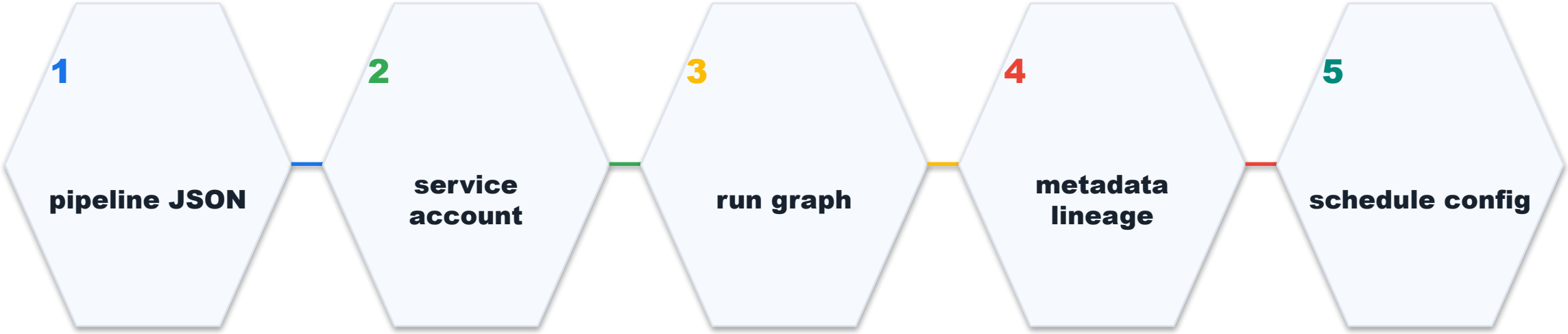
Vertex AI Pipelines and Kubeflow — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Compile error	type mismatch	typed component contracts
Permission failure	pipeline SA incomplete	role matrix
Unexpected cache	inputs unchanged but code changed	version component/image

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Vertex AI Pipelines and Kubeflow — evidence chain



A valid production decision is reproducible from the complete chain.

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Vertex AI Pipelines and Kubeflow — exam decision

SCENARIO

A weekly training pipeline must be reproducible, inspectable and rerun only changed components.

BEST DECISION

Use Vertex AI Pipelines with versioned components, managed metadata and deliberate caching.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 14 evidence — Compile a Vertex AI Pipeline

ACTIVITY 14

Connect validate, transform, train, evaluate and conditional deployment components.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `pipeline.py`
- `pipeline_spec.json`
- `component_contracts.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Cloud Composer and Airflow — mechanism map



Control point Decision evidence must bind dag to notification through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Cloud Composer and Airflow — service decision matrix

OPTION 1

Composer

Use when

cross-service scheduling and dependencies

OPTION 2

Vertex schedule

Use when

simple pipeline cadence

OPTION 3

Cloud Scheduler

Use when

single HTTP/event trigger

Exam cue: select from workload evidence—not from product familiarity.

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Cloud Composer and Airflow — runnable configuration

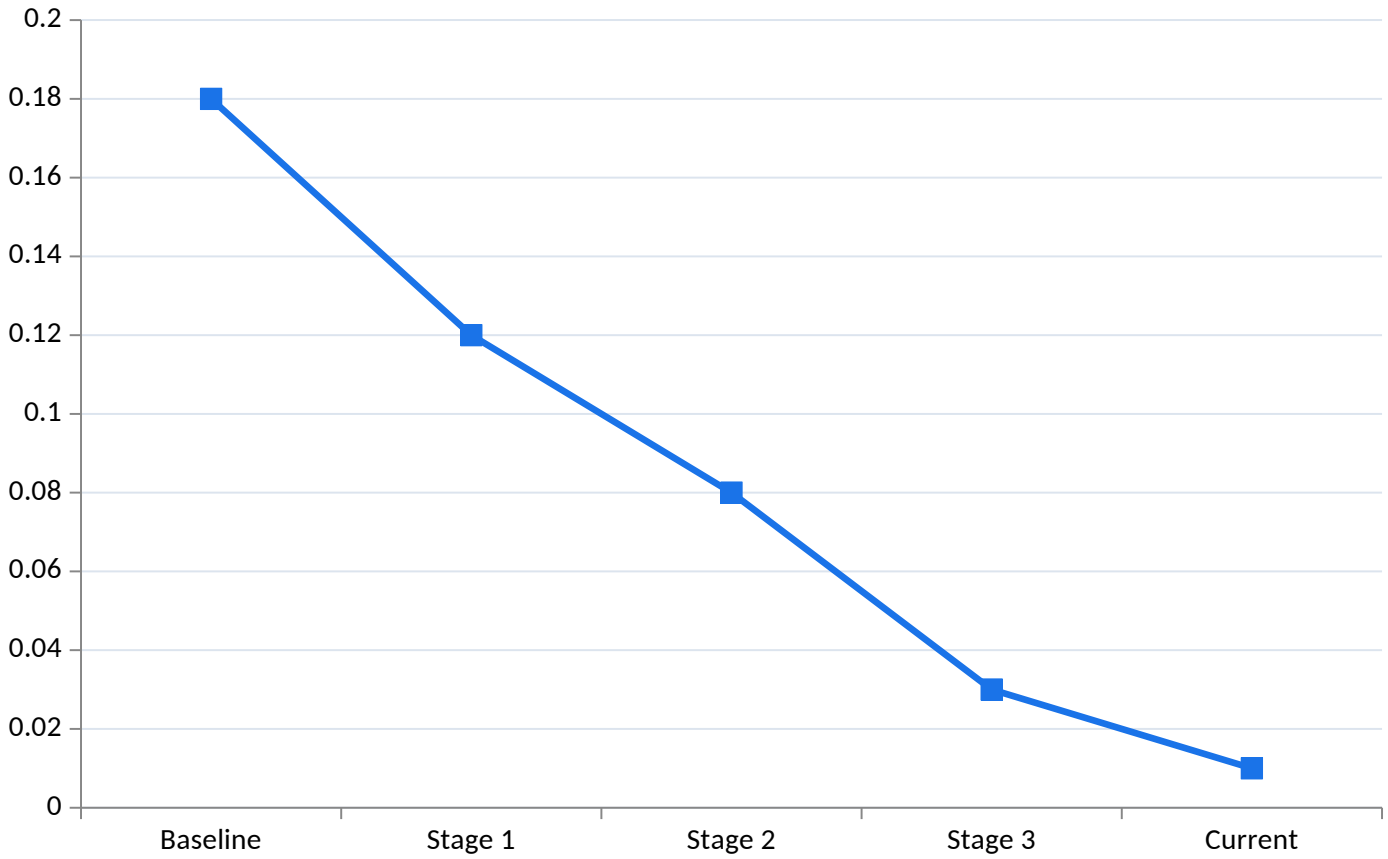
```
with DAG('weekly_ml', schedule='0 2 * * 1', catchup=False)
as dag:
    wait_data >> launch_dataflow >> run_vertex_pipeline >>
verify_slo
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Cloud Composer and Airflow — metric evidence



DAG SLA misses

Formula

```
miss_rate =
missed_runs /
scheduled_runs
```

Interpret trend, threshold and uncertainty before acting.

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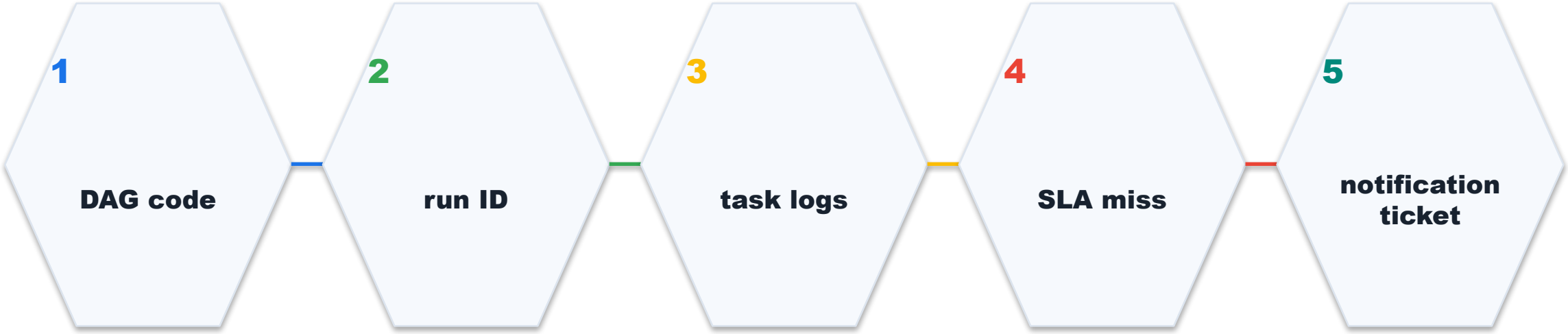
Cloud Composer and Airflow — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Backfill storm	catchup enabled	set catchup deliberately
Stuck sensor	polling resource	deferrable sensor
Duplicate run	non-idempotent task	run IDs and idempotency

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Cloud Composer and Airflow — evidence chain



A valid production decision is reproducible from the complete chain.

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Cloud Composer and Airflow — exam decision

SCENARIO

Training waits for three upstream data products and must notify owners when the freshness SLA fails.

BEST DECISION

Use Composer for cross-team dependencies; keep ML logic inside the versioned Vertex pipeline.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 14 evidence — Compile a Vertex AI Pipeline

ACTIVITY 14

Connect validate, transform, train, evaluate and conditional deployment components.

Evidence: activity artifacts and acceptance checks

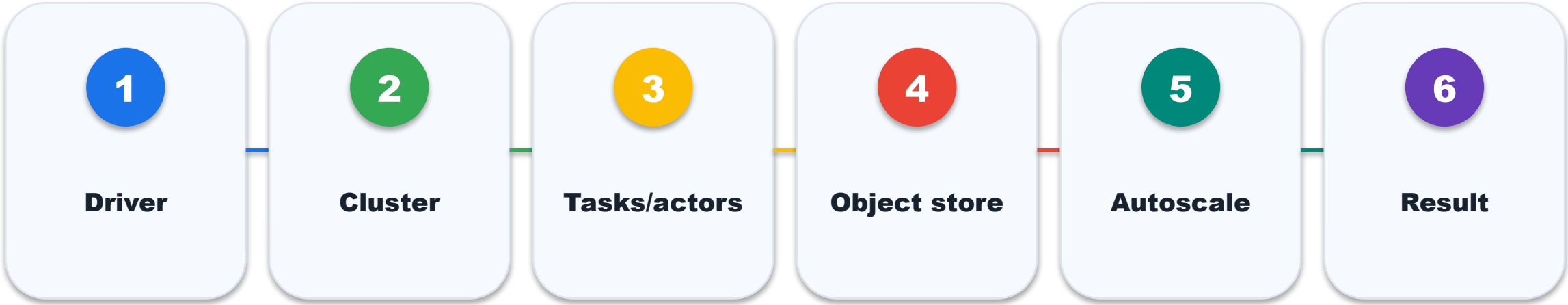
Detailed procedures: Learner Guide + activity README/PDF

- `pipeline.py`
- `pipeline_spec.json`
- `component_contracts.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Ray on Vertex AI — mechanism map



Control point Decision evidence must bind driver to result through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Ray on Vertex AI — service decision matrix

OPTION 1

Ray

Use when

distributed Python workloads and tuning

OPTION 2

Dataflow

Use when

Beam data processing

OPTION 3

CustomJob

Use when

bounded training job

Exam cue: select from workload evidence—not from product familiarity.

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Ray on Vertex AI — runnable configuration

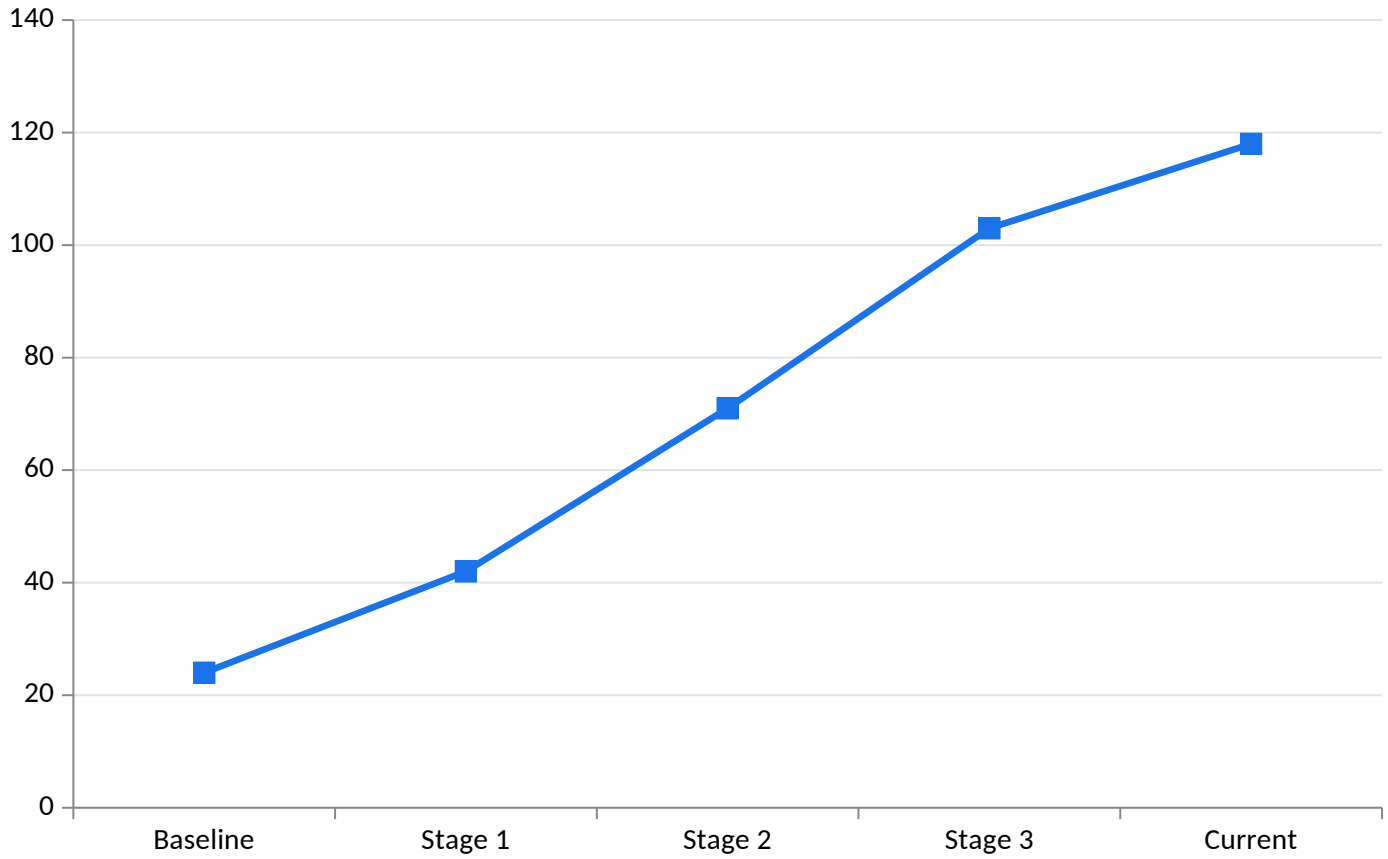
```
@ray.remote
def score_shard(uri):
    return score(load(uri))
results = ray.get([score_shard.remote(u) for u in
shard_uris])
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Ray on Vertex AI — metric evidence



Distributed task throughput

Formula

```
throughput =
completed_tasks / minute
```

Interpret trend, threshold and uncertainty before acting.

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Ray on Vertex AI — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Object-store pressure	large returned objects	write artifacts externally
Task retries	non-idempotent side effect	idempotent tasks
Cluster idle	no teardown	job-scoped cluster policy

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Ray on Vertex AI — evidence chain



A valid production decision is reproducible from the complete chain.

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Ray on Vertex AI — exam decision

SCENARIO

A Python-native embedding job fans out 20,000 independent shards and aggregates a manifest.

BEST DECISION

Use Ray tasks with external artifact writes, idempotent retries and explicit cluster teardown.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 14 evidence — Compile a Vertex AI Pipeline

ACTIVITY 14

Connect validate, transform, train, evaluate and conditional deployment components.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `pipeline.py`
- `pipeline_spec.json`
- `component_contracts.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Training-serving preprocessing parity — mechanism map



Control point Decision evidence must bind source schema to golden test through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Training-serving preprocessing parity — service decision matrix

OPTION 1

tf.Transform

Use when

TensorFlow preprocessing graph

OPTION 2

Sklearn Pipeline

Use when

classical Python models

OPTION 3

Feature Store

Use when

central shared features

Exam cue: select from workload evidence—not from product familiarity.

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Training-serving preprocessing parity — runnable configuration

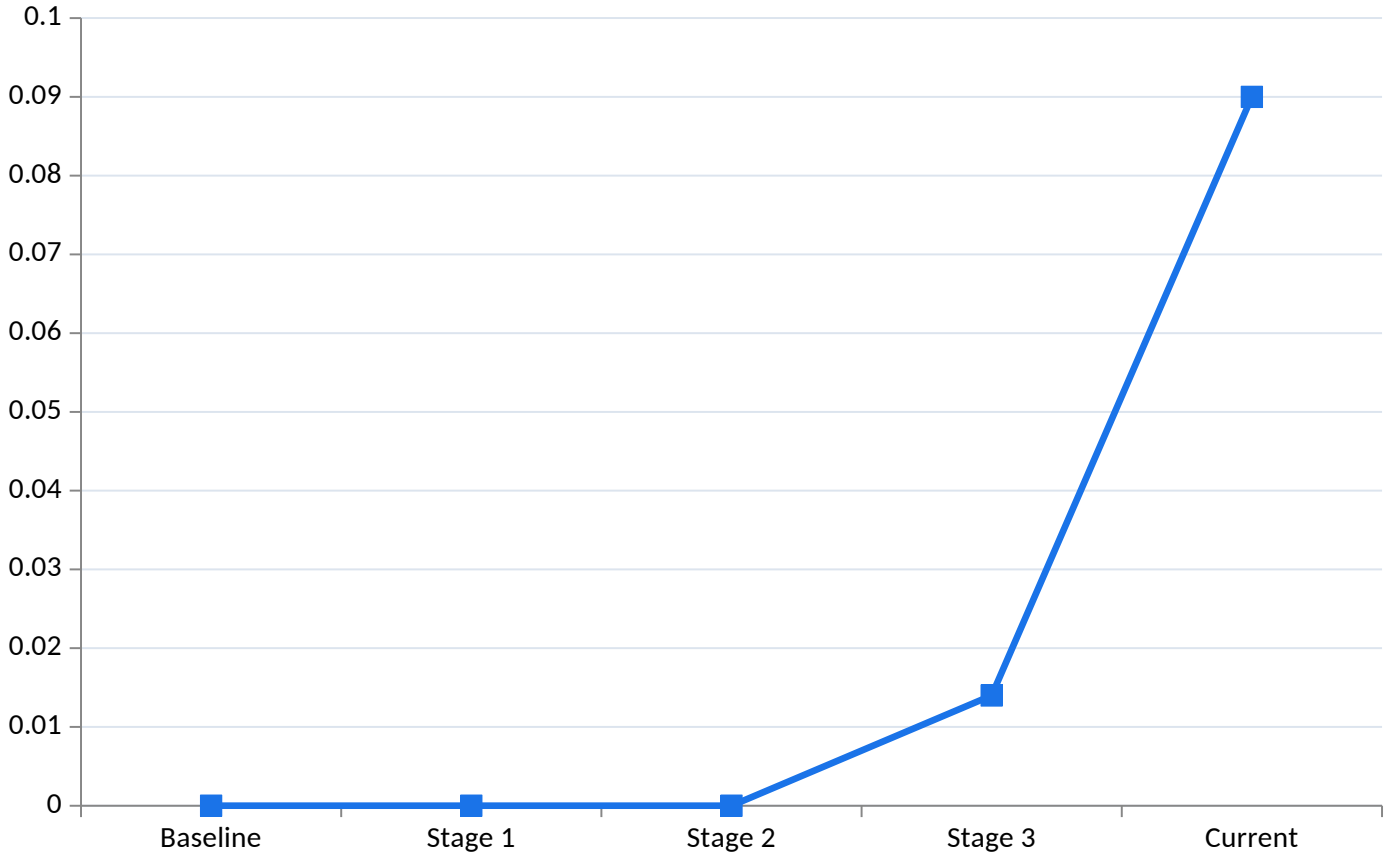
```
expected = offline_pipeline.transform(golden)
actual = requests.post(endpoint, json=golden).json()
['features']
np.testing.assert_allclose(actual, expected, rtol=1e-6)
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Training-serving preprocessing parity — metric evidence



Golden feature max error

Formula

`max_abs(actual-expected)`

Interpret trend, threshold and uncertainty before acting.

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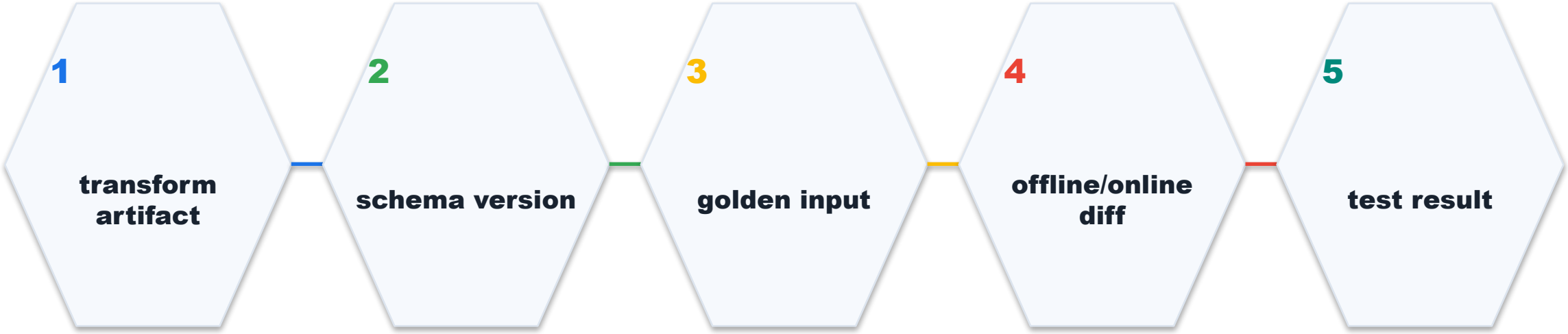
Training-serving preprocessing parity — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Skew	two transform implementations	one exported graph
Schema mismatch	independent contracts	versioned schema
Silent rounding	different dtypes	assert tolerances

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Training-serving preprocessing parity — evidence chain



A valid production decision is reproducible from the complete chain.

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Training-serving preprocessing parity — exam decision

SCENARIO

A category vocabulary updates in training, but the online service keeps last month's mapping.

BEST DECISION

Version and deploy the transformation with the model; block rollout on golden parity tests.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 11 evidence — Verify Feature Store Point-in-Time Parity

ACTIVITY 11

Model offline/online feature definitions and detect stale or leaked features.

Evidence: activity artifacts and acceptance checks

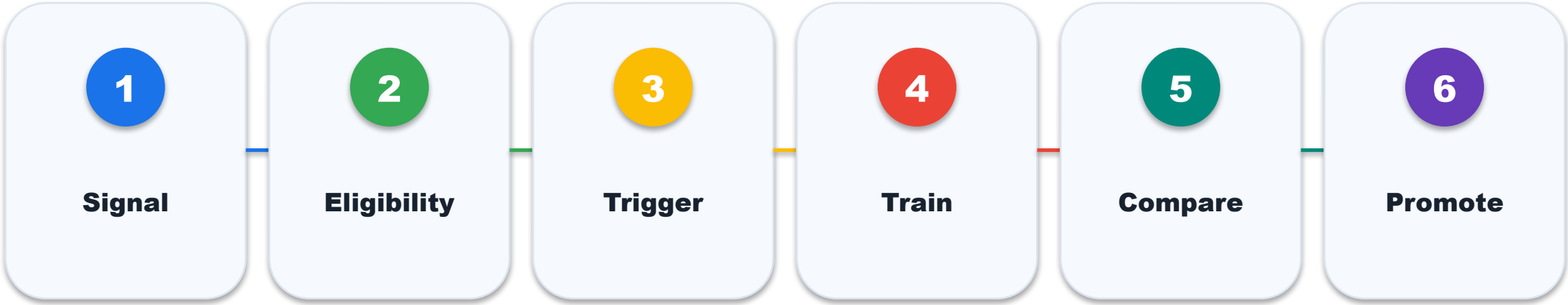
Detailed procedures: Learner Guide + activity README/PDF

- `offline_features.csv`
- `online_features.csv`
- `check_parity.py`
- `feature_view.yaml`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Retraining policy — mechanism map



Control point Decision evidence must bind signal to promote through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Retraining policy — service decision matrix

OPTION 1

Schedule

Use when

stable cadence and predictable data

OPTION 2

Drift trigger

Use when

distribution changes matter

OPTION 3

Performance trigger

Use when

labels arrive and business metric degrades

Exam cue: select from workload evidence—not from product familiarity.

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Retraining policy — runnable configuration

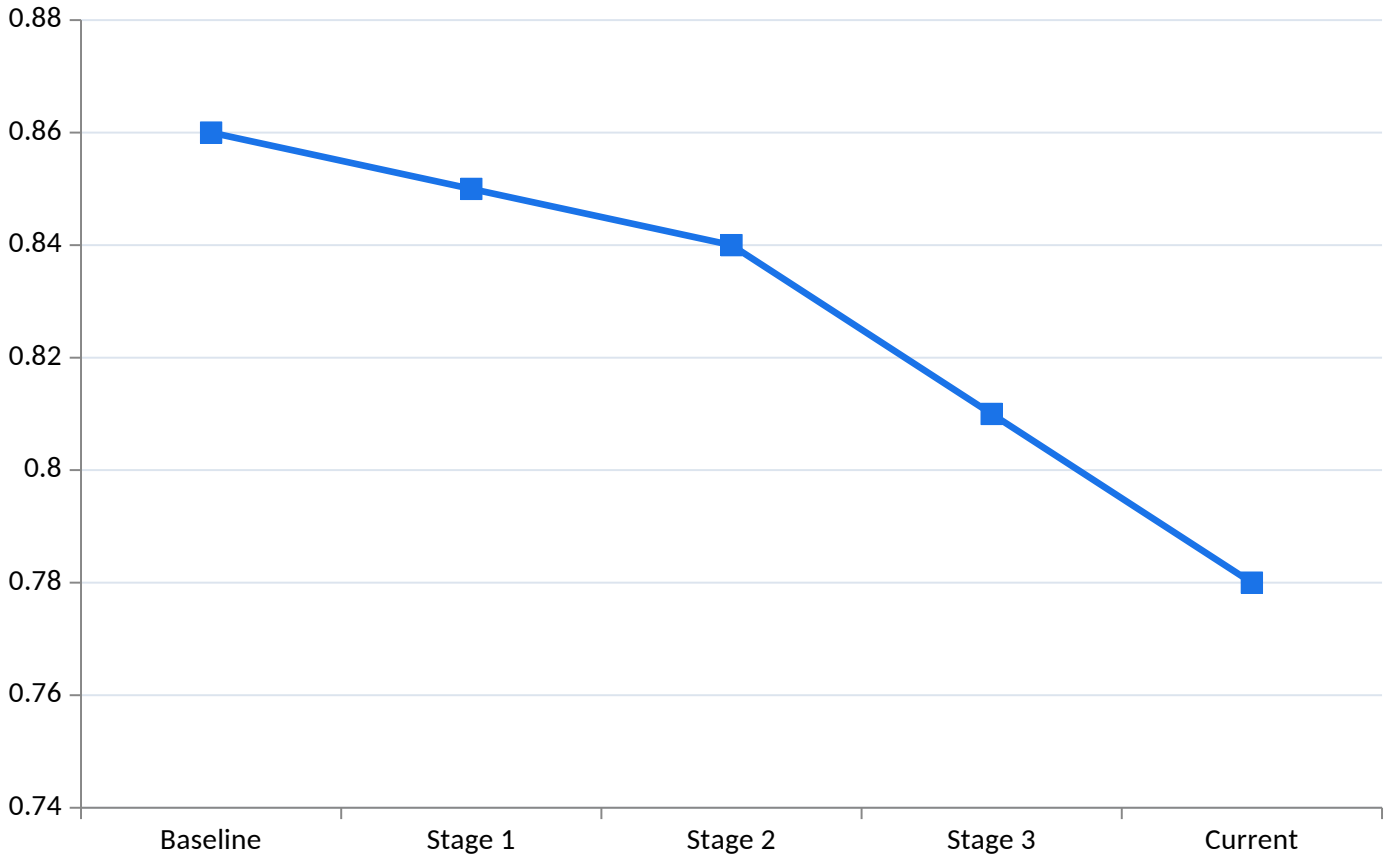
```
trigger_if =
{'psi_gt':0.2,'auc_drop_gt':0.03,'min_new_labels':10000}
cooldown_hours = 72
require_human_approval = True
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Retraining policy — metric evidence



AUC over time

Formula

```
drop = baseline_auc - current_auc
```

Interpret trend, threshold and uncertainty before acting.

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Retraining policy — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Retrain loop	no cooldown	minimum interval
Bad labels	freshness mistaken for quality	label validation
Auto-promote harm	single metric	multi-gate approval

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Retraining policy — evidence chain



A valid production decision is reproducible from the complete chain.

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Retraining policy — exam decision

SCENARIO

Labels arrive weekly; data drift is common but only some drift reduces business performance.

BEST DECISION

Trigger on performance with enough fresh labels; use drift as investigation, not automatic proof.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

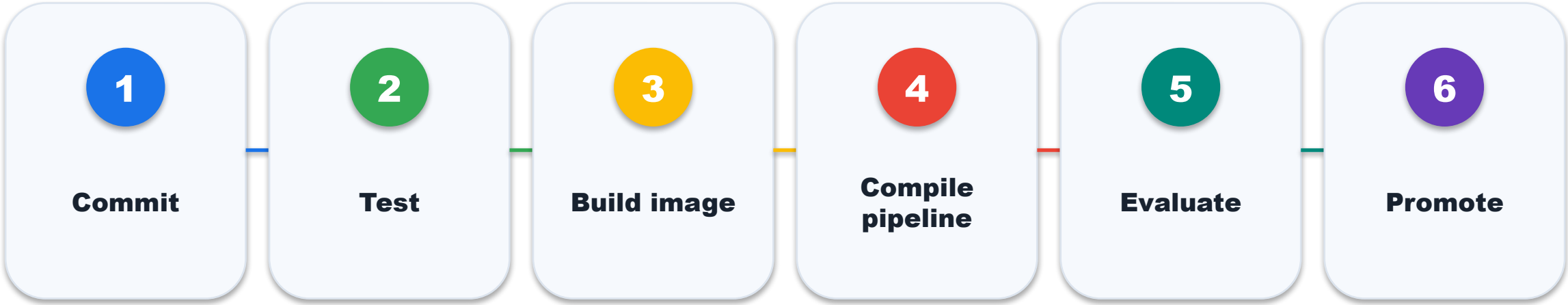
Detailed procedures: Learner Guide + activity README/PDF

- `baseline.csv`
- `current.csv`
- `monitor.py`
- `incident_template.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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CI/CD/CT with Cloud Build — mechanism map



Control point Decision evidence must bind commit to promote through versioned artifacts.

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DOMAIN 5 · DECISION RULE

CI/CD/CT with Cloud Build — service decision matrix

OPTION 1

CI

Use when

unit, contract and security tests

OPTION 2

CD

Use when

deploy validated artifacts

OPTION 3

CT

Use when

retrain from governed triggers

Exam cue: select from workload evidence—not from product familiarity.

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CI/CD/CT with Cloud Build — runnable configuration

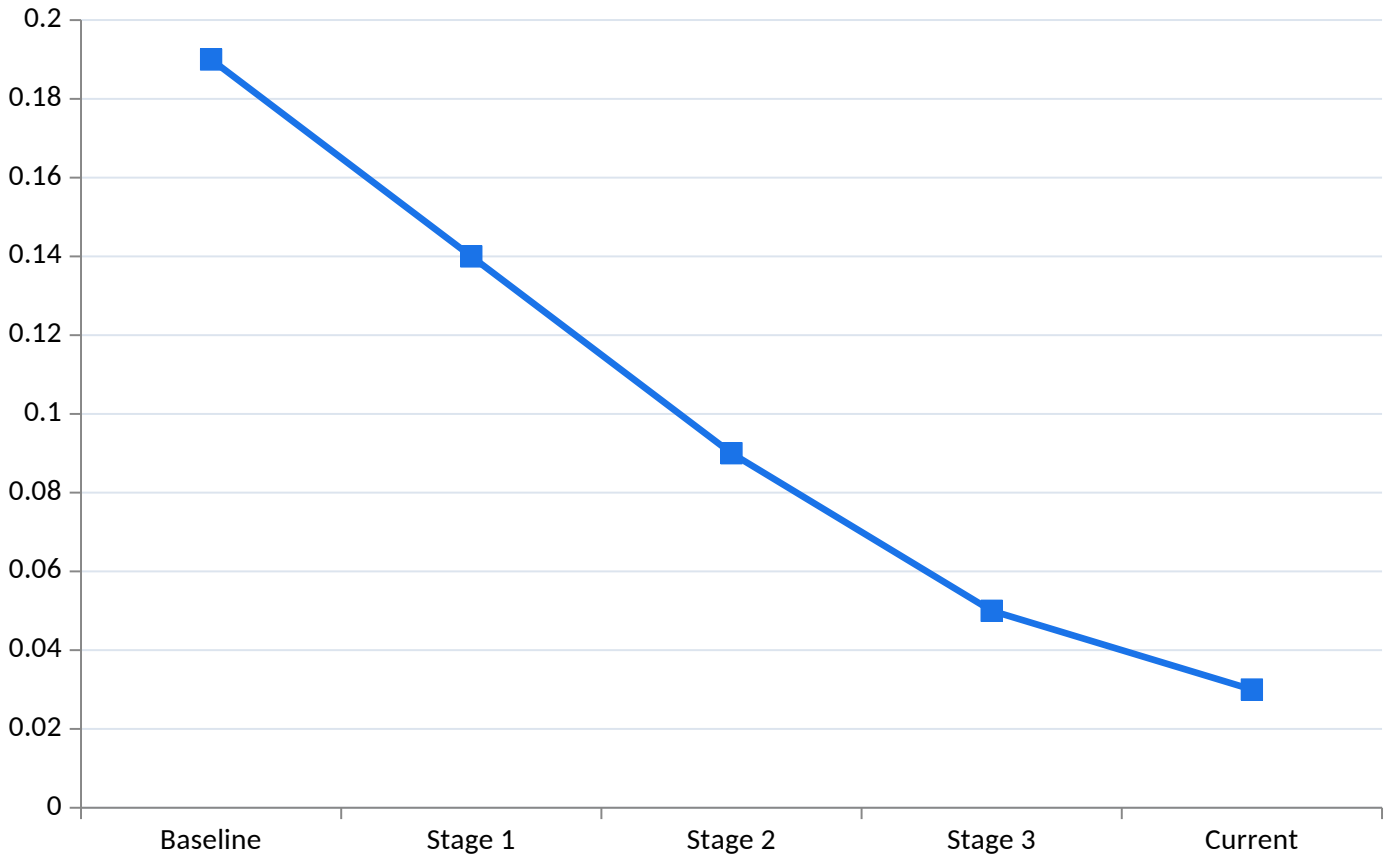
```
steps:
- name: python:3.11
  entrypoint: pytest
- name: gcr.io/cloud-builders/docker
  args: ['build','-t','$IMAGE','.']
images: ['$IMAGE']
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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CI/CD/CT with Cloud Build — metric evidence



Change failure rate

Formula

```
failed_deployments / deployments
```

Interpret trend, threshold and uncertainty before acting.

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CI/CD/CT with Cloud Build — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Mutable latest	tag reused	deploy digest
Test gap	only code unit tests	data/model/contract tests
Privilege escalation	broad build SA	least-privilege build identity

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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CI/CD/CT with Cloud Build — evidence chain



A valid production decision is reproducible from the complete chain.

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CI/CD/CT with Cloud Build — exam decision

SCENARIO

A training image passes local tests but production deploys a different dependency set.

BEST DECISION

Build once in Cloud Build, lock dependencies, attest the digest and deploy that exact artifact.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 15 evidence — Design an ML CI/CD/CT Release

ACTIVITY 15

Create Cloud Build gates, artifact provenance and a model card release record.

Evidence: activity artifacts and acceptance checks

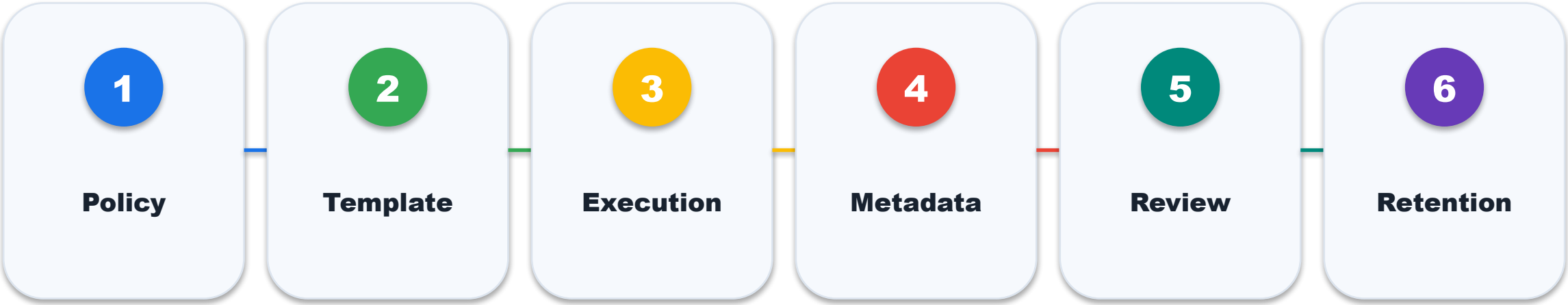
Detailed procedures: Learner Guide + activity README/PDF

- `cloudbuild.yaml`
- `release_gate.py`
- `model_card.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Pipeline governance and reproducibility — mechanism map



Control point Decision evidence must bind policy to retention through versioned artifacts.

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DOMAIN 5 · DECISION RULE

Pipeline governance and reproducibility — service decision matrix

OPTION 1

Org policy

Use when

prevent forbidden resource settings

OPTION 2

Template

Use when

approved repeatable architecture

OPTION 3

Policy-as-code

Use when

automated configuration gate

Exam cue: select from workload evidence—not from product familiarity.

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Pipeline governance and reproducibility — runnable configuration

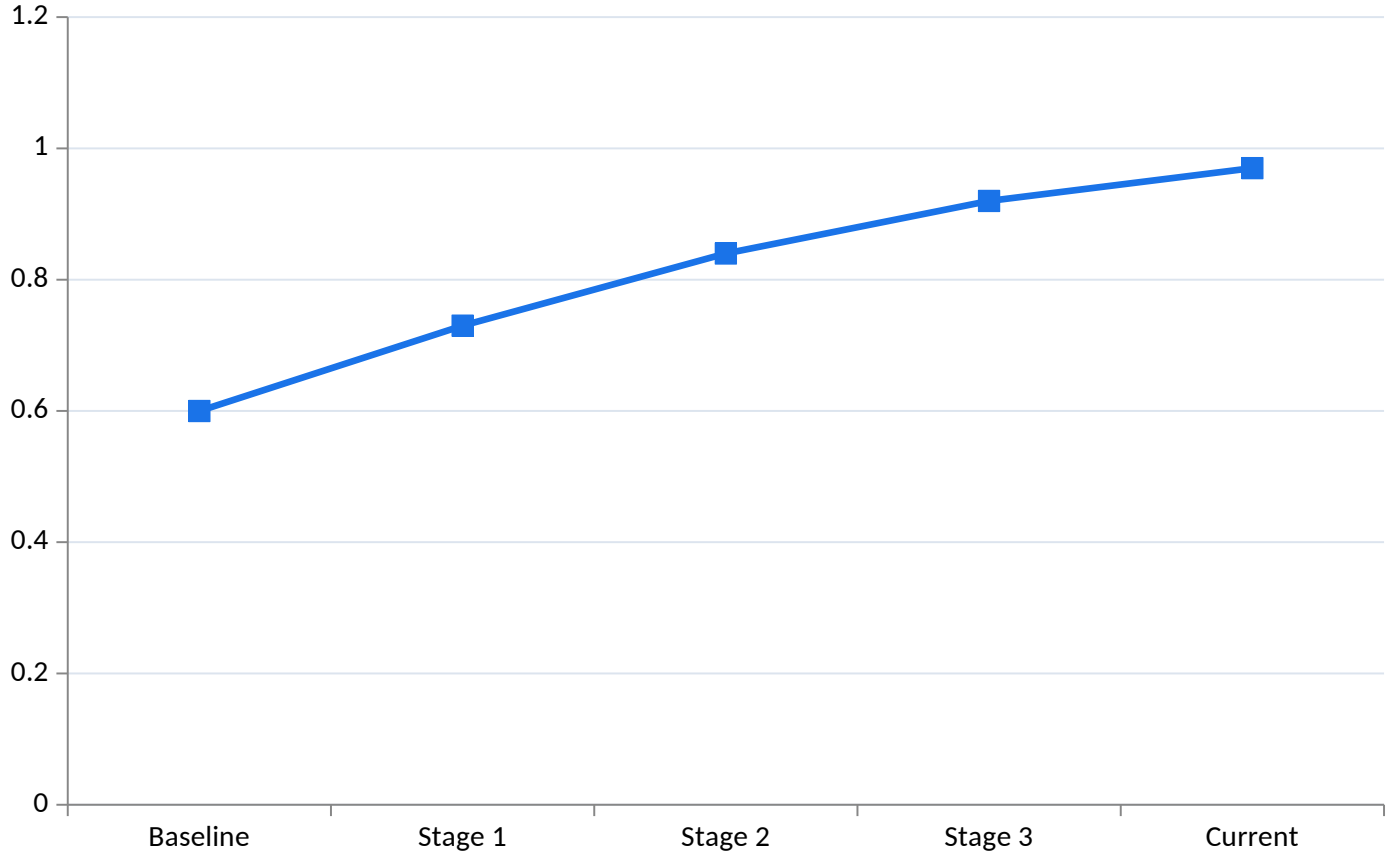
```
required_labels =
['owner','data_classification','model_risk']
forbidden = ['public_ip','default_service_account']
retention_days = 365
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Pipeline governance and reproducibility — metric evidence



Reproducible run rate

Formula

```
reproduced_within_tolerance / audited_runs
```

Interpret trend, threshold and uncertainty before acting.

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Pipeline governance and reproducibility — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Missing owner	labels optional	admission check
Data unavailable	retention too short	align lineage retention
Environment drift	manual runtime	versioned container

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Pipeline governance and reproducibility — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 5 · SCENARIO

Pipeline governance and reproducibility — exam decision

SCENARIO

An auditor selects a six-month-old prediction and asks which code, data and model produced it.

BEST DECISION

Retain end-to-end lineage, immutable artifacts and policy evidence long enough for replay.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 15 evidence — Design an ML CI/CD/CT Release

ACTIVITY 15

Create Cloud Build gates, artifact provenance and a model card release record.

Evidence: activity artifacts and acceptance checks

Detailed procedures: Learner Guide + activity README/PDF

- `cloudbuild.yaml`
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DOMAIN 6 · ~13%

Monitoring AI solutions

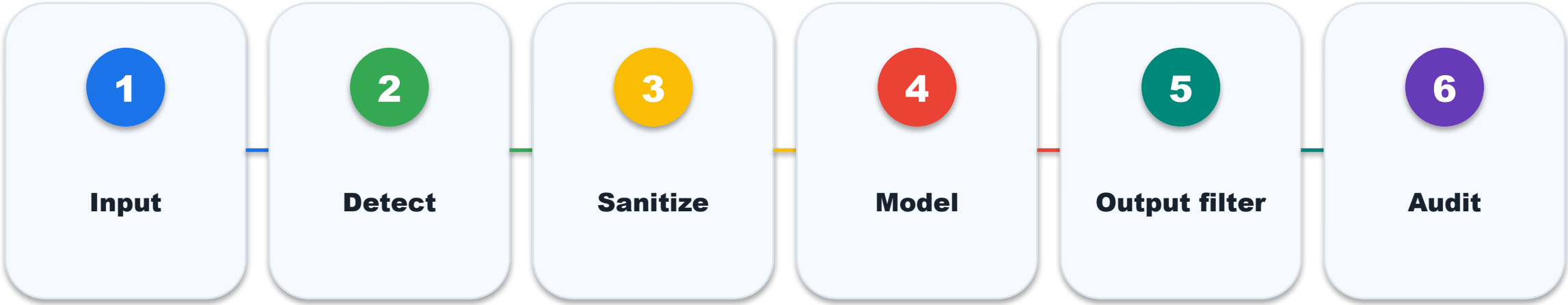
Topics 6–8: testing, monitoring, troubleshooting and MLOps

8 technical mechanisms · 64 anchored slides



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AI security and Model Armor — mechanism map



Control point Decision evidence must bind input to audit through versioned artifacts.

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DOMAIN 6 · DECISION RULE

AI security and Model Armor — service decision matrix

OPTION 1

Regex/DLP

Use when

known sensitive patterns

OPTION 2

Safety filters

Use when

harm categories

OPTION 3

Model Armor

Use when

managed prompt/response protection

Exam cue: select from workload evidence—not from product familiarity.

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AI security and Model Armor — runnable configuration

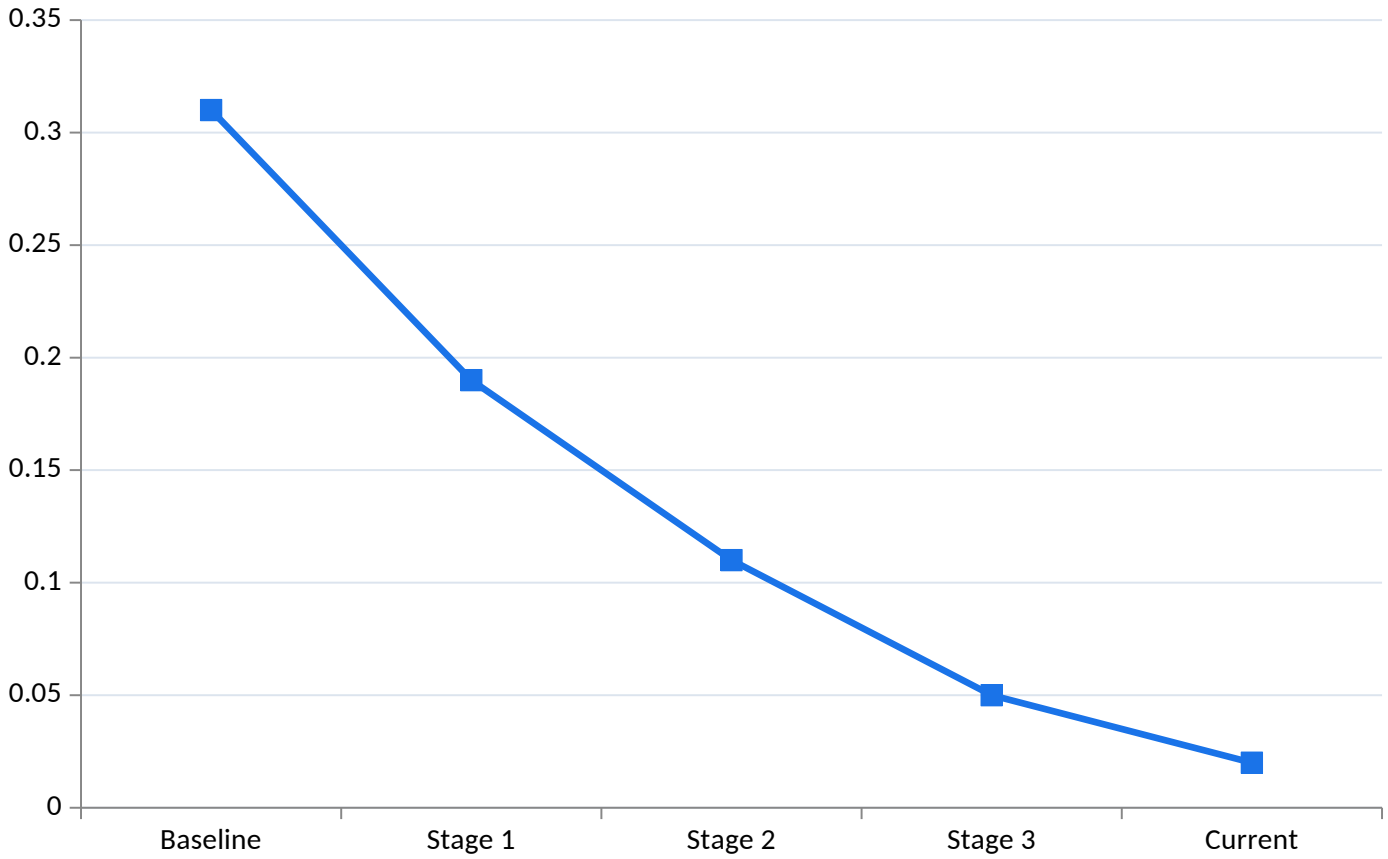
```
blocked_patterns = ['ignore previous instructions','BEGIN
PRIVATE KEY']
scan_prompt = True
scan_response = True
log_payload = False
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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AI security and Model Armor — metric evidence



Attack success rate

Formula

```
successful_attacks /
red_team_attempts
```

Interpret trend, threshold and uncertainty before acting.

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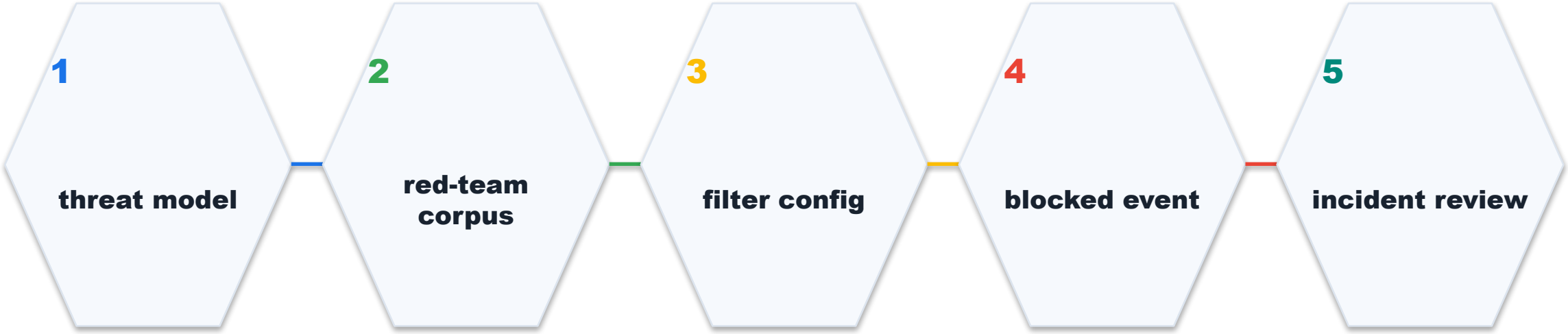
AI security and Model Armor — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Indirect injection	retrieved document contains commands	treat context as data
Exfiltration	model reveals secret	minimize context and output scan
Log exposure	payload retained	redaction and access control

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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AI security and Model Armor — evidence chain



A valid production decision is reproducible from the complete chain.

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DOMAIN 6 · SCENARIO

AI security and Model Armor — exam decision

SCENARIO

A grounded assistant retrieves a document that instructs the model to reveal system prompts.

BEST DECISION

Separate instructions from untrusted context, scan inputs/outputs and test indirect injections.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

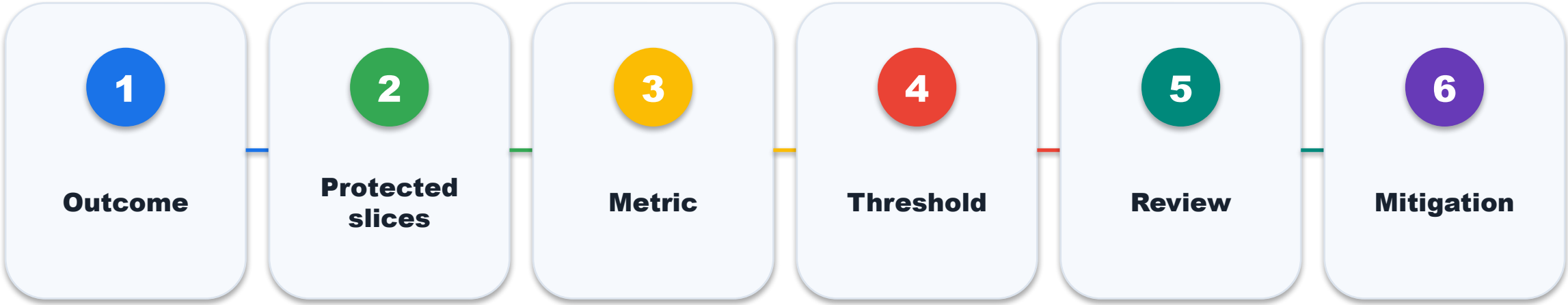
Detailed procedures: Learner Guide + activity README/PDF

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- `current.csv`
- `monitor.py`
- `incident_template.md`

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Responsible AI and bias monitoring — mechanism map



Control point Decision evidence must bind outcome to mitigation through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Responsible AI and bias monitoring — service decision matrix

OPTION 1

Demographic parity

Use when

selection rates where appropriate

OPTION 2

Equal opportunity

Use when

true-positive rates

OPTION 3

Calibration

Use when

probabilities match outcomes

Exam cue: select from workload evidence—not from product familiarity.

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Responsible AI and bias monitoring — runnable configuration

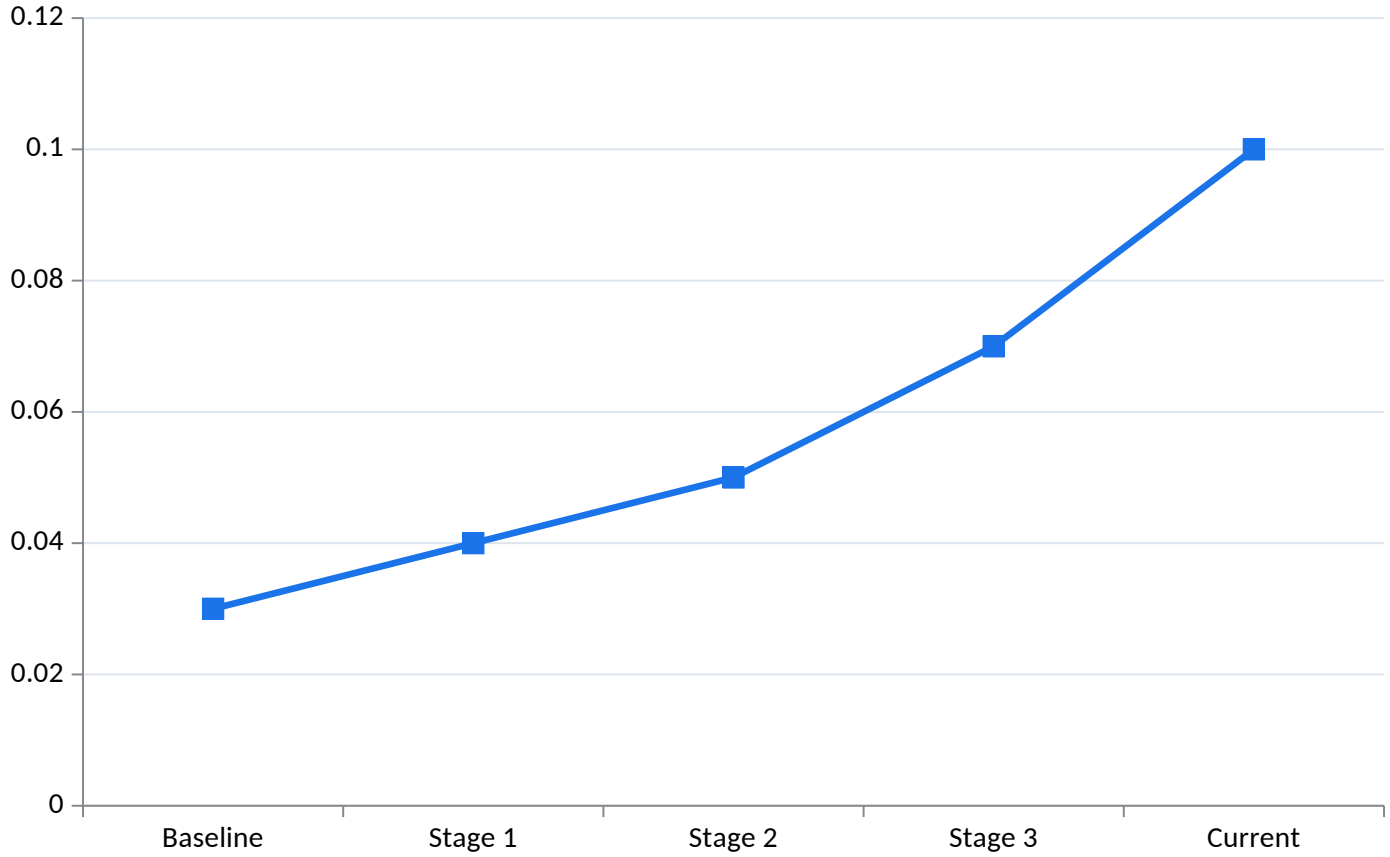
```
monitor_slices = ['region','age_band']
max_tpr_gap = 0.05
min_sample = 200
escalate_to = 'model-risk-committee'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Responsible AI and bias monitoring — metric evidence



True-positive-rate gap

Formula

`max(TPR)-min(TPR)`

Interpret trend, threshold and uncertainty before acting.

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Responsible AI and bias monitoring — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Small slice noise	insufficient N	confidence bounds/min sample
Proxy feature	correlated attribute	causal/domain review
Metric conflict	one fairness metric only	risk-specific metric set

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Responsible AI and bias monitoring — evidence chain



A valid production decision is reproducible from the complete chain.

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Responsible AI and bias monitoring — exam decision

SCENARIO

Approval rates are stable overall but qualified applicants in one region see falling true-positive rate.

BEST DECISION

Alert on slice TPR gap with sample safeguards and require model-risk review before threshold changes.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

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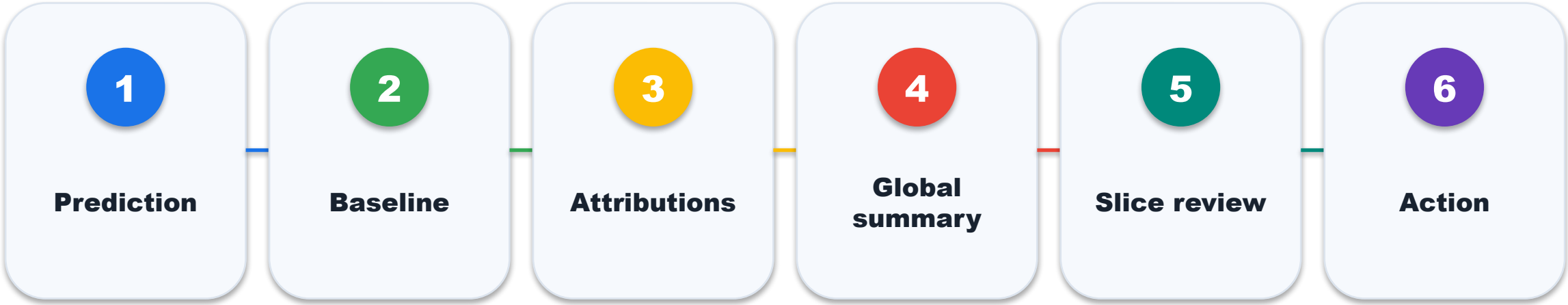
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Explainability and feature attribution — mechanism map



Control point Decision evidence must bind prediction to action through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Explainability and feature attribution — service decision matrix

OPTION 1

Local attribution

Use when

explain one prediction

OPTION 2

Global importance

Use when

aggregate model behavior

OPTION 3

Counterfactual

Use when

what feasible change alters outcome

Exam cue: select from workload evidence—not from product familiarity.

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Explainability and feature attribution — runnable configuration

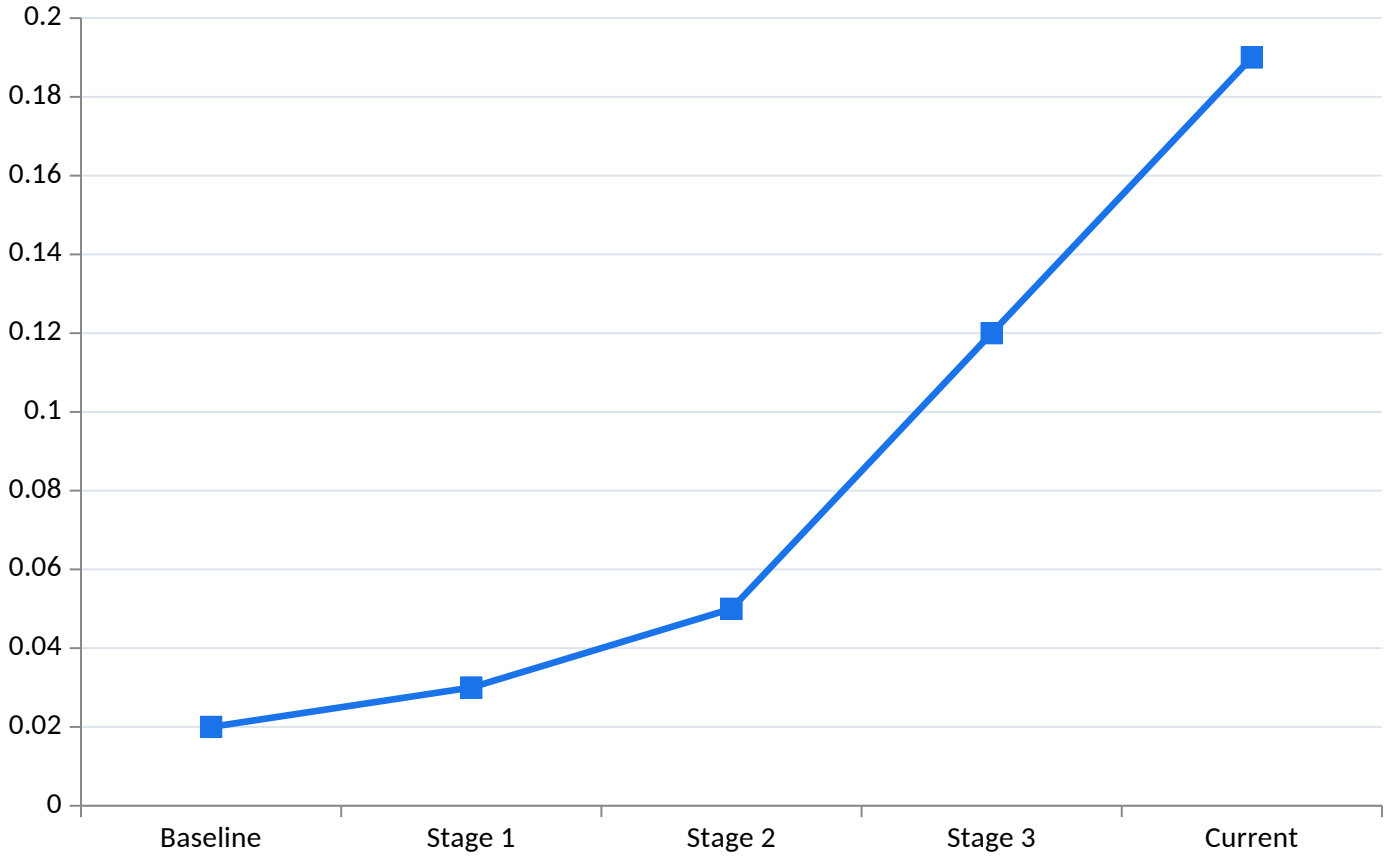
```
explanation_spec =
{'method':'sampled_shapley','sampled_shapley_attribution':
{'path_count':20}}
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Explainability and feature attribution — metric evidence



Attribution drift

Formula

```
distance(current_attributions,
baseline_attributions)
```

Interpret trend, threshold and uncertainty before acting.

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Explainability and feature attribution — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Causal claim	attribution mistaken for cause	state limitation
Unstable rank	correlated features	group and test stability
Sensitive proxy	unexpected importance	investigate governance

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Explainability and feature attribution — evidence chain



A valid production decision is reproducible from the complete chain.

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Explainability and feature attribution — exam decision

SCENARIO

A credit model keeps AUC, but postcode suddenly becomes the top feature after a data release.

BEST DECISION

Investigate attribution drift and proxy risk; do not accept stable accuracy as proof of safety.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

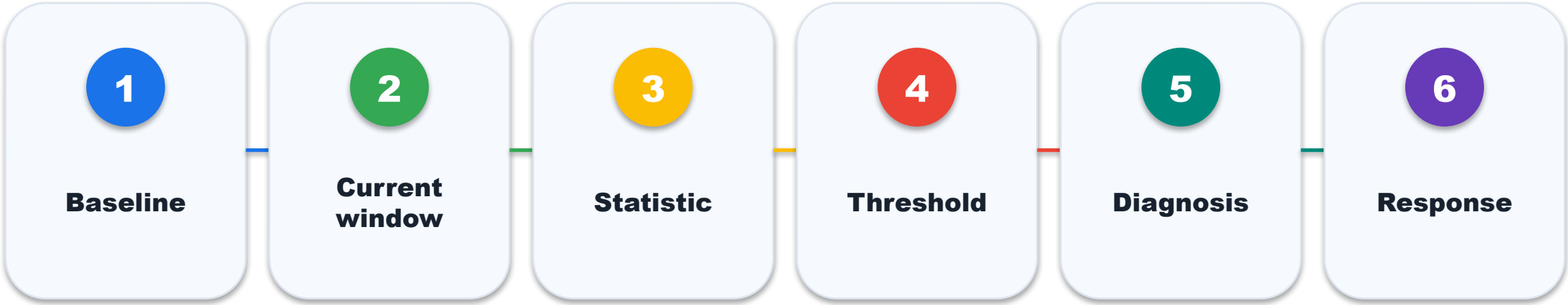
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Data drift, skew and concept drift — mechanism map



Control point Decision evidence must bind baseline to response through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Data drift, skew and concept drift — service decision matrix

OPTION 1

Training-serving skew

Use when

training vs serving pipeline mismatch

OPTION 2

Data drift

Use when

input distribution changes

OPTION 3

Concept drift

Use when

relationship between input and outcome changes

Exam cue: select from workload evidence—not from product familiarity.

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Data drift, skew and concept drift — runnable configuration

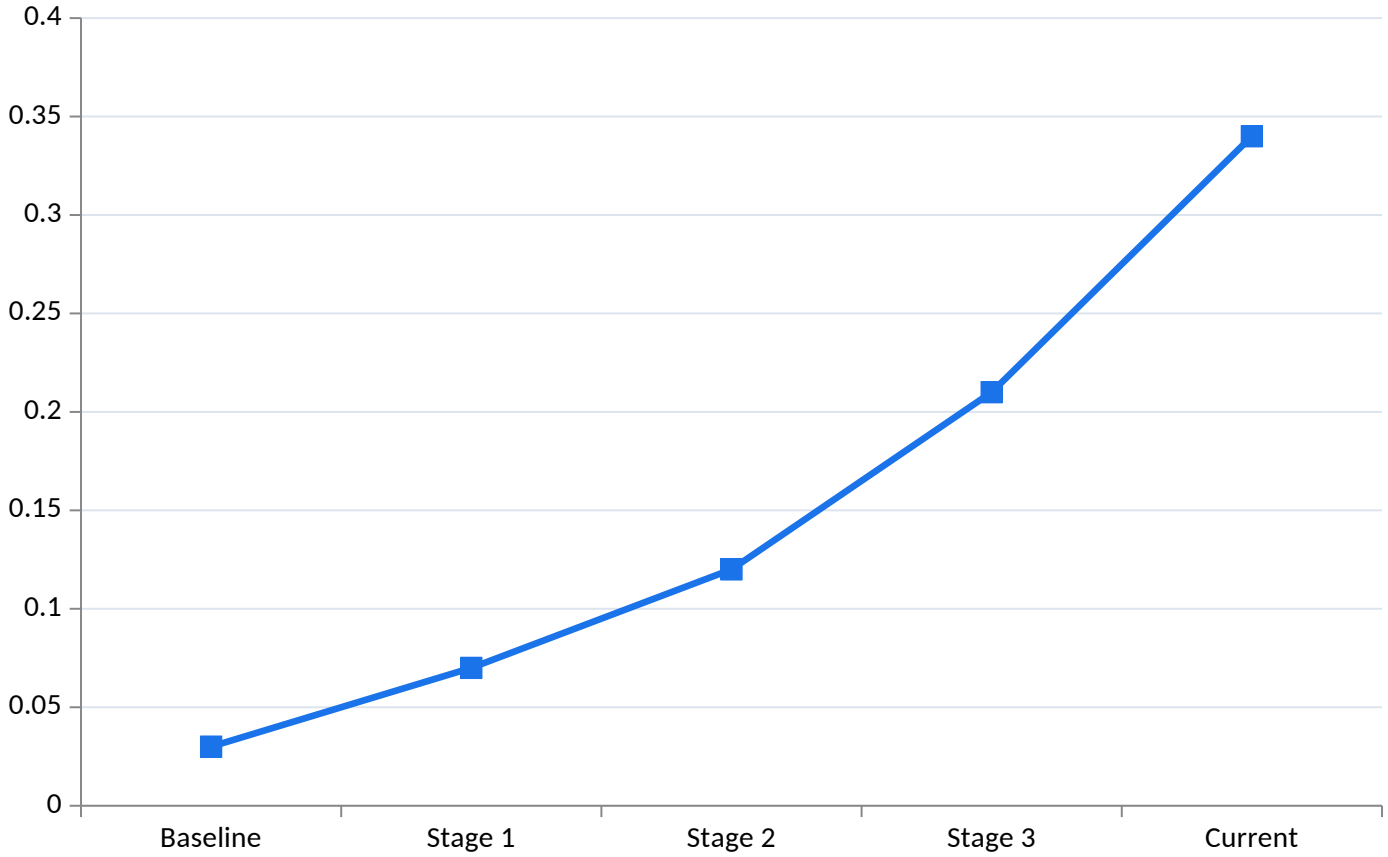
```
psi = sum((current_pct - base_pct) *
log(current_pct/base_pct))
alert_if = 'psi > 0.2 for 2 windows'
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Data drift, skew and concept drift — metric evidence



Population stability index

Formula

```
PSI = sum((A-E)*ln(A/E))
```

Interpret trend, threshold and uncertainty before acting.

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Data drift, skew and concept drift — failure diagnosis

Observable symptom	Likely mechanism	Control or test
False drift	seasonality	seasonal baseline
Skew	transform version differs	parity test
Concept drift	inputs stable but errors rise	label-based performance monitor

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Data drift, skew and concept drift — evidence chain



A valid production decision is reproducible from the complete chain.

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Data drift, skew and concept drift — exam decision

SCENARIO

A fraud feature distribution shifts every holiday, but loss rises only after a new payment channel launch.

BEST DECISION

Use seasonal baselines; distinguish benign data drift from labeled concept drift before retraining.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

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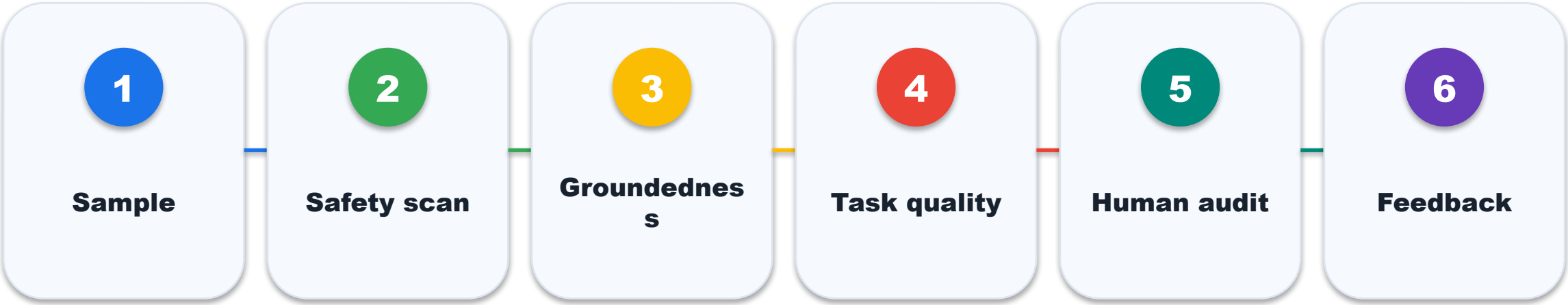
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Gen AI evaluation in production — mechanism map



Control point Decision evidence must bind sample to feedback through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Gen AI evaluation in production — service decision matrix

OPTION 1

Continuous eval

Use when

sample production traffic

OPTION 2

Synthetic red team

Use when

probe rare attacks

OPTION 3

Human audit

Use when

calibrate nuanced risk

Exam cue: select from workload evidence—not from product familiarity.

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Gen AI evaluation in production — runnable configuration

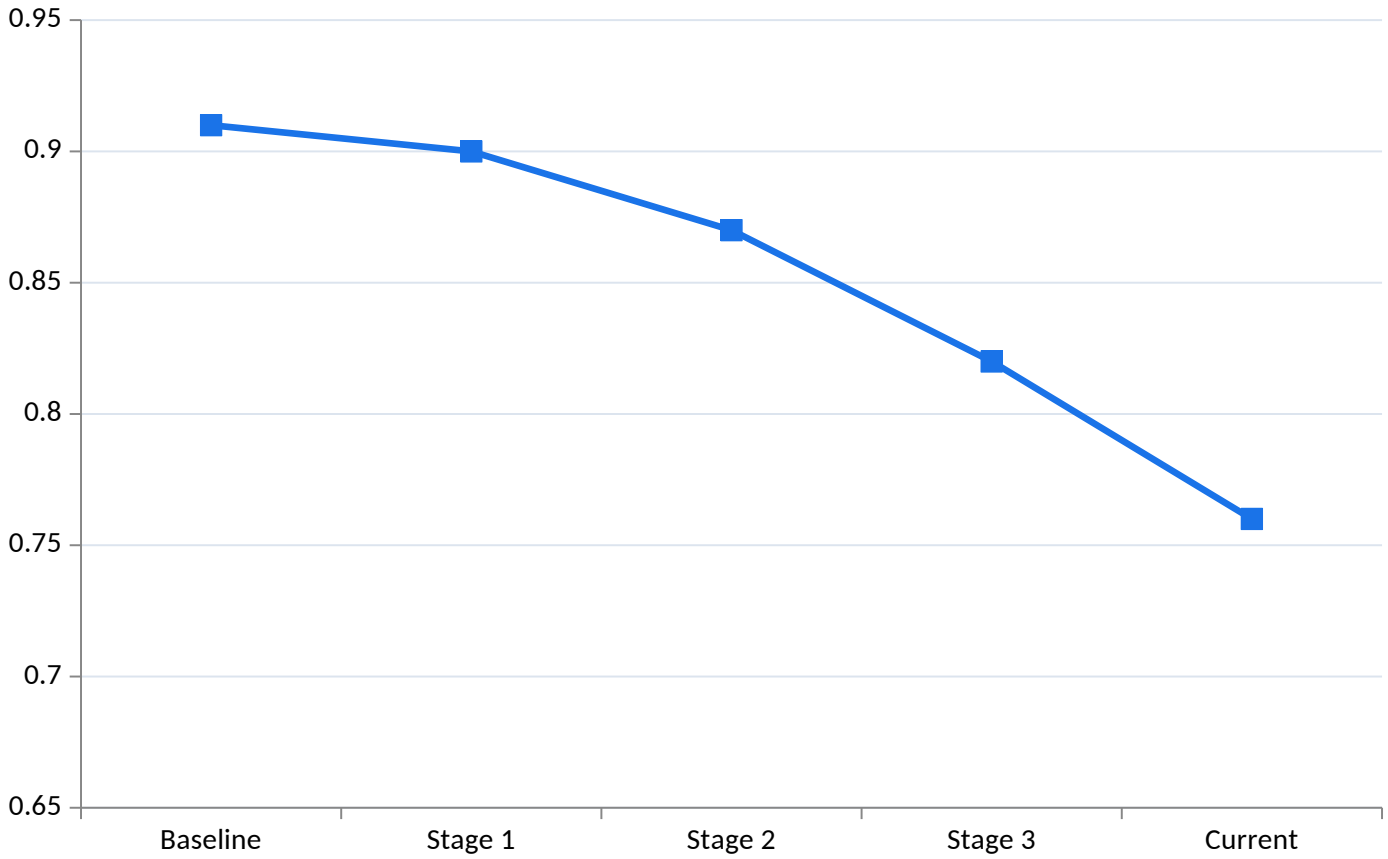
```
sample_rate = 0.02
metrics = ['groundedness','instruction_following','safety']
critical_violation_budget = 0
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Gen AI evaluation in production — metric evidence



Grounded answer rate

Formula

```
grounded_answers /
evaluated_answers
```

Interpret trend, threshold and uncertainty before acting.

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Gen AI evaluation in production — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Privacy leak	raw prompts in eval set	redact and restrict
Judge drift	judge model changed	version and calibrate
Silent regression	no release baseline	champion comparison

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Gen AI evaluation in production — evidence chain



A valid production decision is reproducible from the complete chain.

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Gen AI evaluation in production — exam decision

SCENARIO

A support assistant remains fluent but citations increasingly fail to support its claims.

BEST DECISION

Continuously score groundedness, audit samples and block releases that regress against the champion.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

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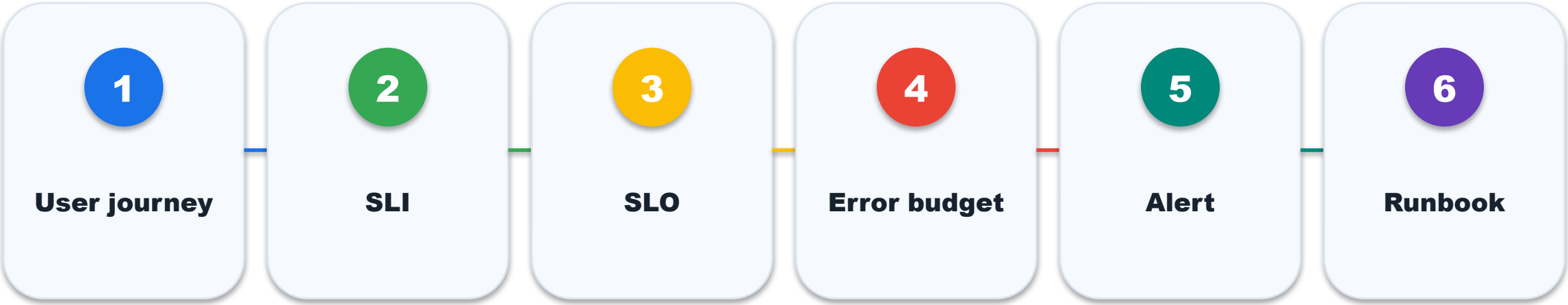
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Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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SLOs, logs and alerts — mechanism map



Control point Decision evidence must bind user journey to runbook through versioned artifacts.

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DOMAIN 6 · DECISION RULE

SLOs, logs and alerts — service decision matrix

OPTION 1

Availability

Use when

successful valid responses

OPTION 2

Latency

Use when

p95/p99 per request class

OPTION 3

Quality

Use when

validated prediction or grounded response

Exam cue: select from workload evidence—not from product familiarity.

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SLOs, logs and alerts — runnable configuration

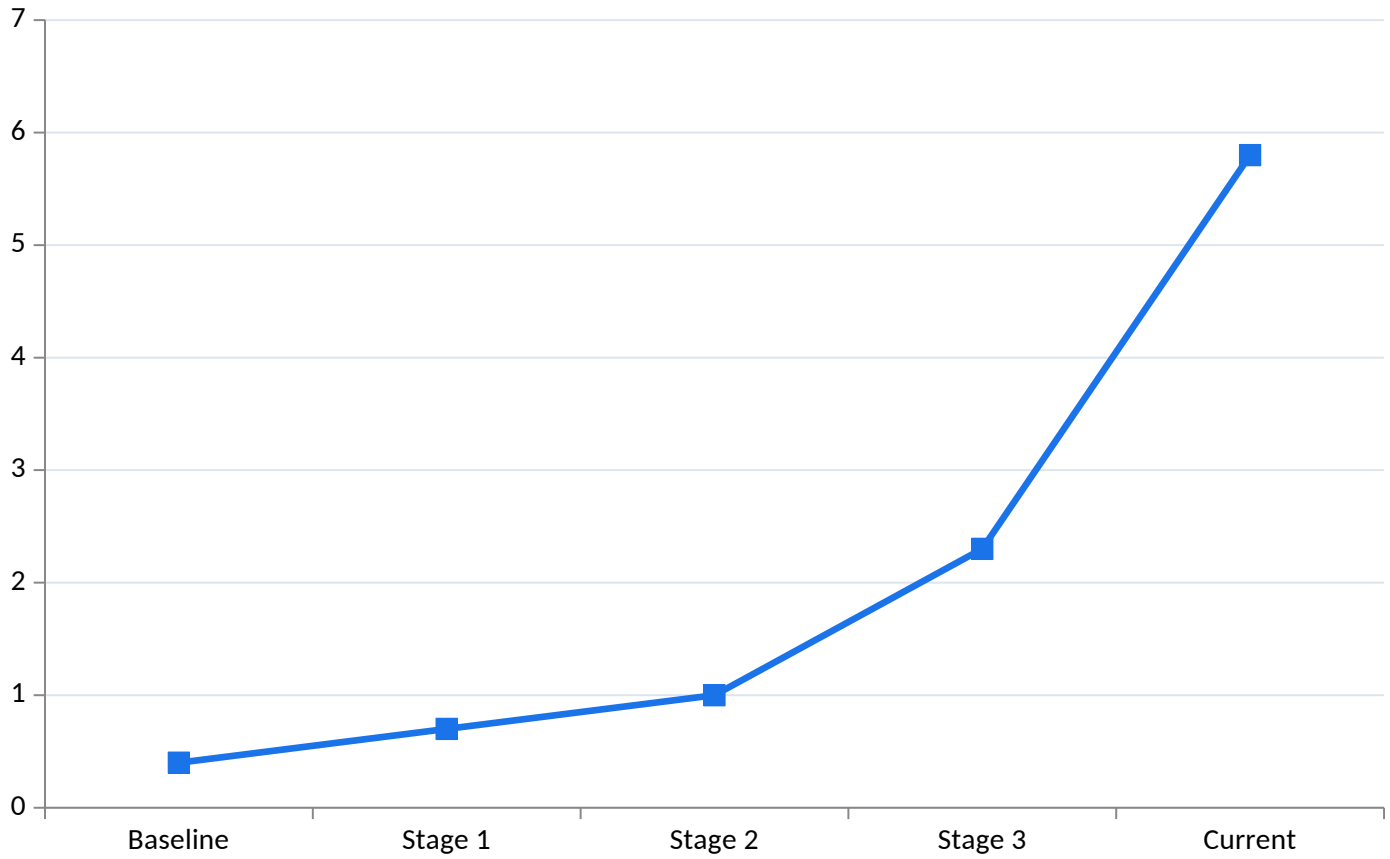
```
slo:
  availability: 0.999
  p95_latency_ms: 300
  quality_floor: 0.82
alert_on_burn_rate: true
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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SLOs, logs and alerts — metric evidence



Error-budget burn

Formula

```
burn =
observed_error_rate /
allowed_error_rate
```

Interpret trend, threshold and uncertainty before acting.

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SLOs, logs and alerts — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Alert storm	symptom alerts duplicate	burn-rate aggregation
No context	logs lack version	structured version fields
Green infra/red users	only CPU monitored	user-centric quality SLI

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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SLOs, logs and alerts — evidence chain



A valid production decision is reproducible from the complete chain.

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SLOs, logs and alerts — exam decision

SCENARIO

Infrastructure is healthy, yet users receive stale predictions for six hours after a failed batch.

BEST DECISION

Add freshness and quality SLIs, not just infrastructure metrics; page on error-budget burn.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

ACTIVITY 16

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Evidence: activity artifacts and acceptance checks

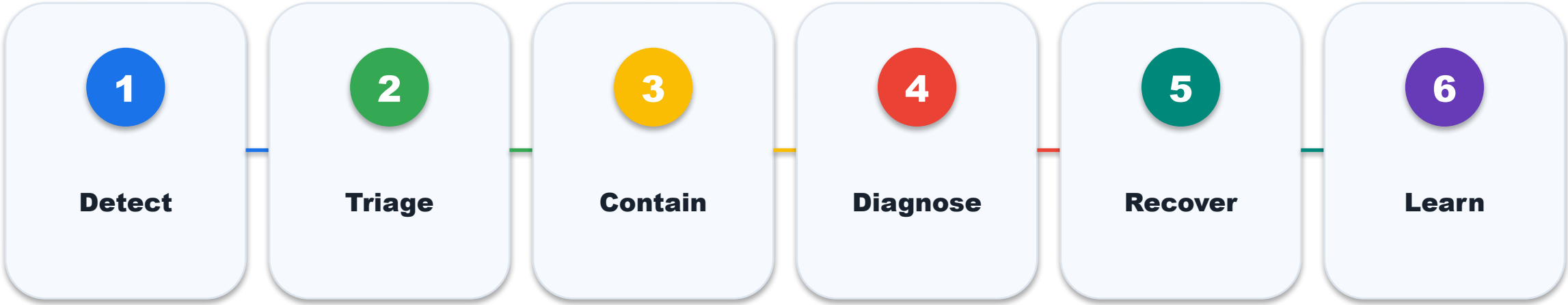
Detailed procedures: Learner Guide + activity README/PDF

- `baseline.csv`
- `current.csv`
- `monitor.py`
- `incident_template.md`

Slide shows artifacts and acceptance evidence only—no click-by-click procedure.

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Incident troubleshooting — mechanism map



Control point Decision evidence must bind detect to learn through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Incident troubleshooting — service decision matrix

OPTION 1

Rollback

Use when

new version caused regression

OPTION 2

Scale

Use when

capacity saturation

OPTION 3

Disable feature

Use when

upstream corruption

Exam cue: select from workload evidence—not from product familiarity.

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Incident troubleshooting — runnable configuration

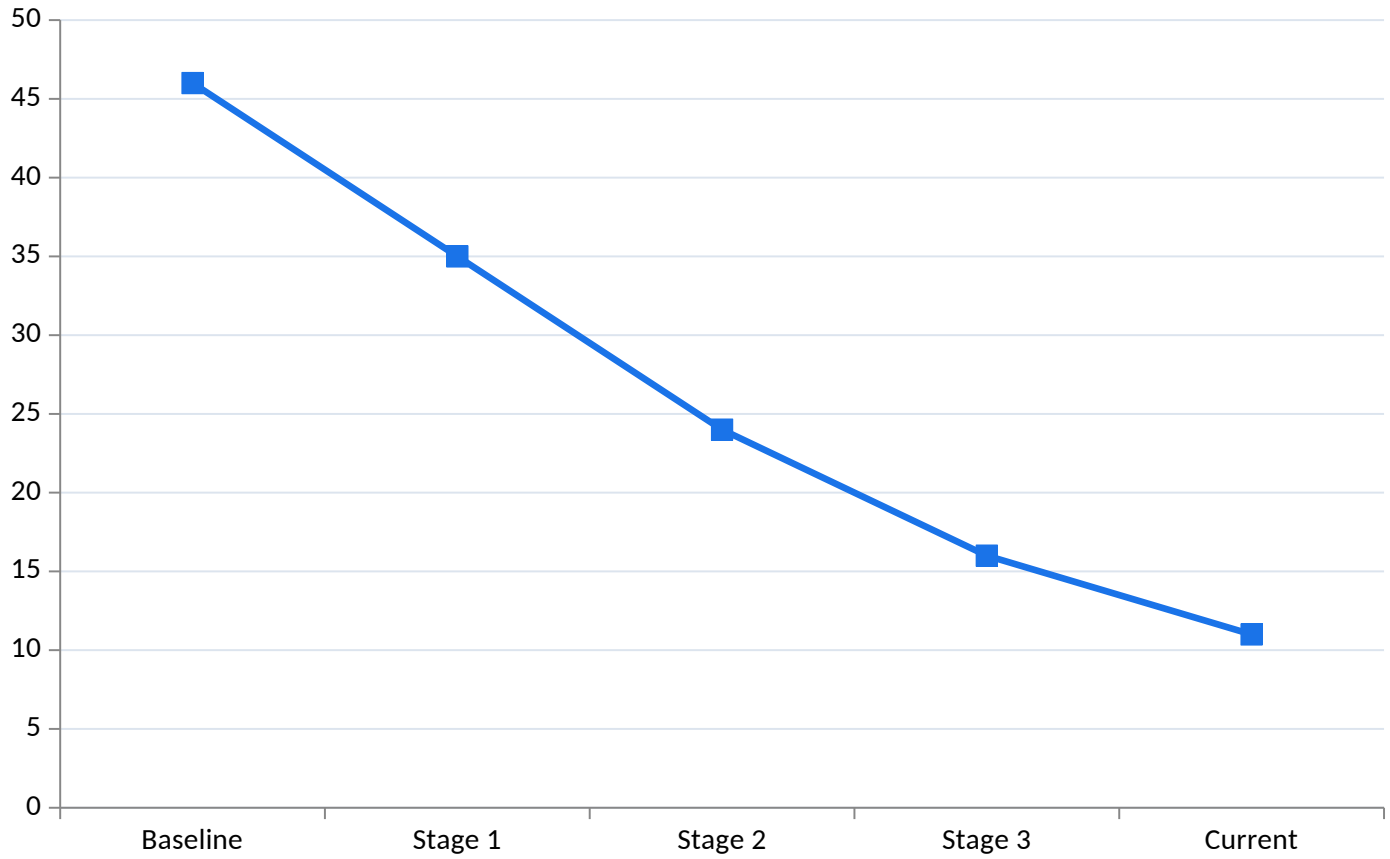
```
incident_id: ML-2026-0912
model_version: candidate-17
start: 14:05Z
containment: rollback_to_champion
owner: ml-platform
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Incident troubleshooting — metric evidence



Mean time to recovery

Formula

```
MTTR =
sum(recovery_duration)/i
ncident_count
```

Interpret trend, threshold and uncertainty before acting.

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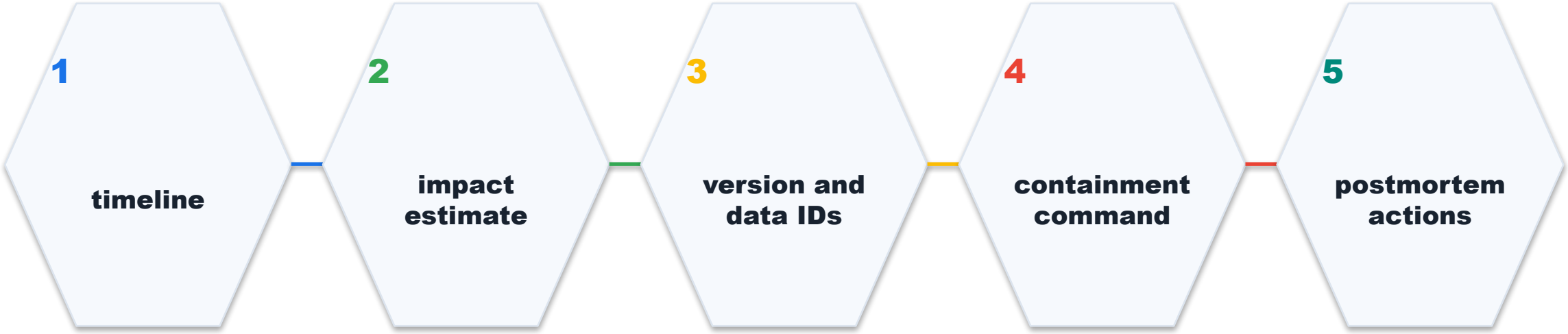
Incident troubleshooting — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Slow triage	no version tags	structured logs
Repeat issue	no action owner	postmortem with due dates
Evidence loss	ephemeral logs	central retention

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Incident troubleshooting — evidence chain



A valid production decision is reproducible from the complete chain.

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Incident troubleshooting — exam decision

SCENARIO

p95 latency triples immediately after a new model version receives 20% traffic.

BEST DECISION

Compare per-version metrics, roll back the candidate, preserve traces, then reproduce under load.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

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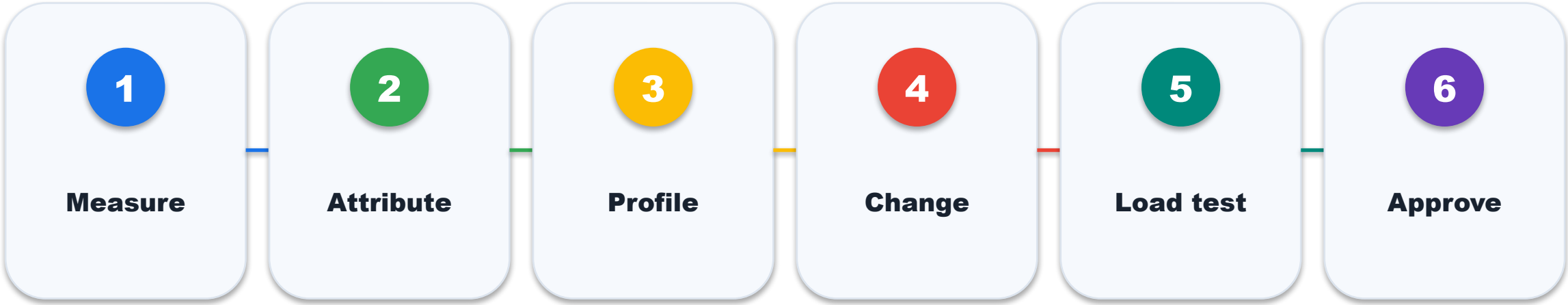
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Cost and performance optimization — mechanism map



Control point Decision evidence must bind measure to approve through versioned artifacts.

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DOMAIN 6 · DECISION RULE

Cost and performance optimization — service decision matrix

OPTION 1

Right-size

Use when

remove unused CPU/memory/accelerator

OPTION 2

Batching

Use when

amortize inference overhead

OPTION 3

Quantization

Use when

trade small quality loss for speed/cost

Exam cue: select from workload evidence—not from product familiarity.

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Cost and performance optimization — runnable configuration

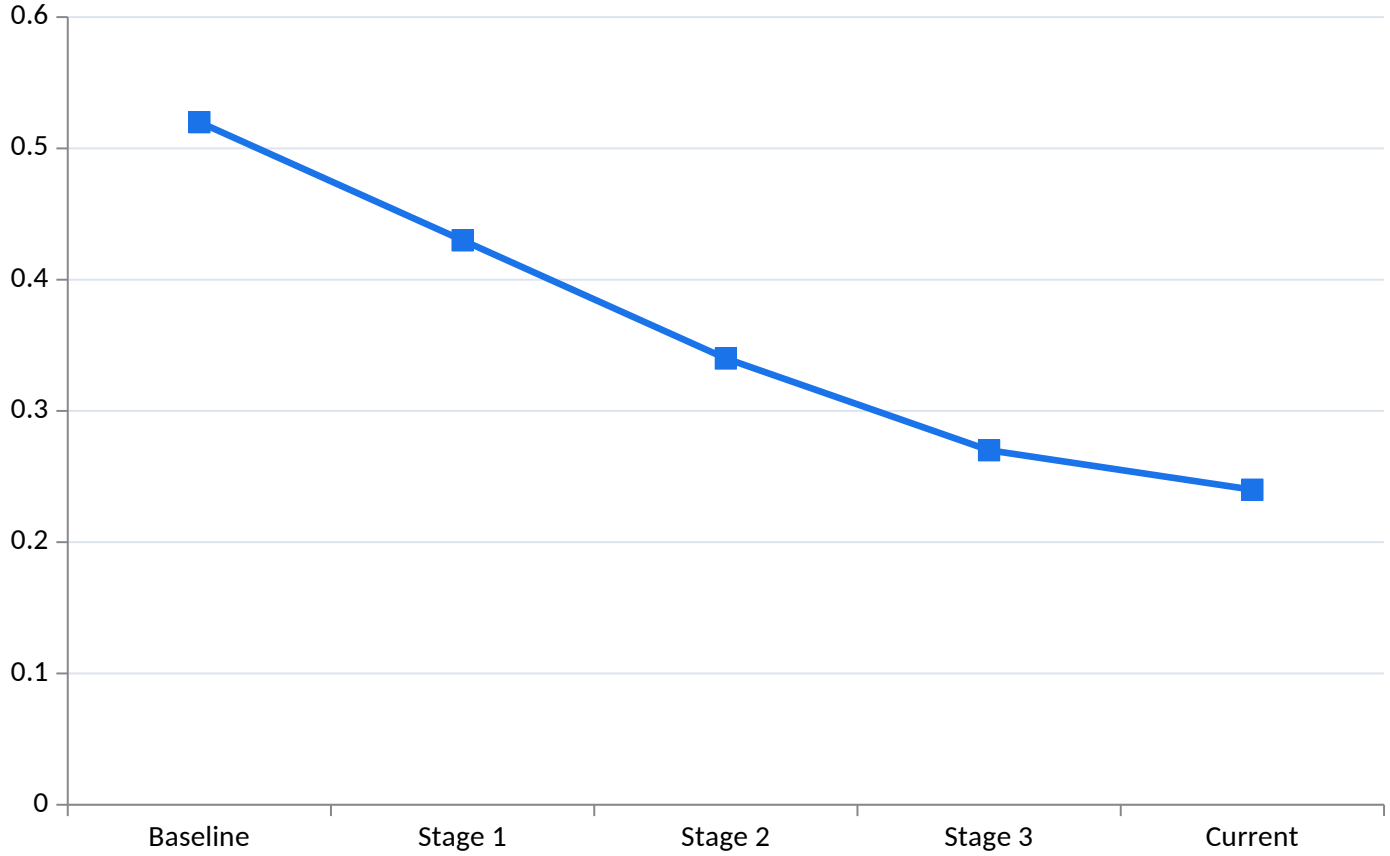
```
baseline = {'p95_ms':280,'cost_1k':0.42,'auc':0.846}
candidate = {'p95_ms':170,'cost_1k':0.24,'auc':0.842}
max_auc_loss = 0.005
```

Read it like an engineer

- Identity and region are explicit
- Inputs and outputs are versionable
- Metrics or health can be verified
- No secret is embedded

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Cost and performance optimization — metric evidence



Cost per 1,000 predictions

Formula

```
total_cost /
prediction_count * 1000
```

Interpret trend, threshold and uncertainty before acting.

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Cost and performance optimization — failure diagnosis

Observable symptom	Likely mechanism	Control or test
Cheap but worse	quality not gated	multi-objective gate
Benchmark bias	toy load	production traffic shape
Hidden egress	cross-region path	co-locate services

Escalation requires evidence: job/revision ID · version · timestamp · affected slice · attempted control

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Cost and performance optimization — evidence chain



A valid production decision is reproducible from the complete chain.

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Cost and performance optimization — exam decision

SCENARIO

Quantization halves cost but reduces AUC by 0.004 and improves p95 latency by 110 ms.

BEST DECISION

Accept only if the documented quality tolerance is 0.005 and slice/edge-case gates also pass.

Decision rule: satisfy explicit constraints with the least operational complexity and verifiable controls.

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Activity 16 evidence — Investigate Drift, Bias and an ML Incident

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Exam Decision Checklist

Clarify workload

1

Data modality, volume, freshness and business loss.

Constrain architecture

2

Quality, latency, cost, availability, privacy and explainability.

Prefer managed fit

3

Choose the least complex service that satisfies measured requirements.

Demand evidence

4

Versioned artifacts, metrics, lineage, monitoring and rollback.

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SOURCE COVERAGE

Legacy v7 Coverage Preserved

Topics 1–3

1

GCP foundations, ML workflow, algorithms and managed AI APIs.

Topics 4–5

2

TensorFlow/Keras, tf.data and feature engineering.

Topics 6–7

3

AutoML/custom training, batch/online serving and production systems.

Topic 8

4

MLOps, CI/CD/CT, pipelines, monitoring and troubleshooting.

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LEARNING OUTCOMES

Course Achievement Evidence

Architecture

1

Decision records tied to documented business constraints.

Implementation

2

Runnable scripts, containers and pipeline specifications.

Operations

3

Metrics, canary controls and production monitoring.

Responsible AI

4

Privacy, fairness, security, explainability and incident records.

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COURSE CLOSING

Course Feedback

Feedback form

1

Complete the course feedback form shared by the trainer.

Certificate

2

Your certificate of completion is emailed after the course.

Questions

3

Raise any outstanding questions with the trainer before you leave.

Keep learning

4

Revisit the activities and Learner Guide to consolidate your evidence.

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Thank You

Build evidence. Make the decision. Operate responsibly.